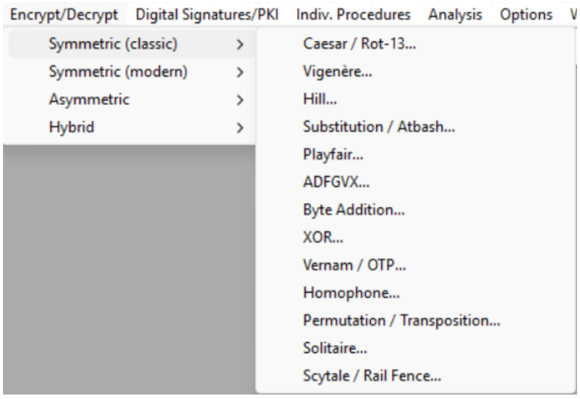
Historical algorithms:

1.1 Check the available historical algorithms in tab Encrypt/Decrypt -> Symmetric (Classic):



The algorithms are: Caesar / Rot-13 (simple shift), Vigenère (polyalphabetic shift), Hill (matrix), Substitution / Atbash (monoalphabetic), Playfair (digraph substitution), ADFGVX (polybius + transposition), Byte-Addition, XOR, Vernam / OTP (stream/one-time pad), Homophone, Permutation / Transposition, Solitaire, Scytale / Rail-Fence

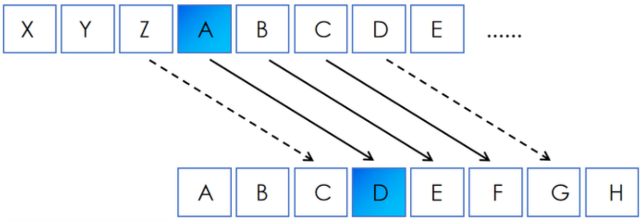
The key for all relies on transforming characters to numbers: (Starting by A = 1, or A = 0 in some cases):

A -->	01	H -->	08	O -->	15	U -->	21
B -->	02	I -->	09	P -->	16	V -->	22
C -->	03	J -->	10	Q -->	17	W -->	23
D -->	04	K -->	11	R -->	18	X -->	24
E -->	05	L -->	12	S -->	19	Y -->	25
F -->	06	M -->	13	T -->	20	Z -->	26
G -->	07	N -->	14				

The chosen text to cipher is:
ENELPRINCIPIOCREODIOSLOSCIELOSYLATIERRAYOSOYELALFAYLAOMEGAELPRINCIPIOYELFINELPRIMEROYELULTIMO

1.2 Choose a few of them and check their properties (space-number of keys, effect of multiple encryption, are there any weak keys e.g. for Rot-13 or Playfair)? Include the properties of the selected algorithms in the report:

Caesar / ROT-13: Shifting the alphabet k number of times

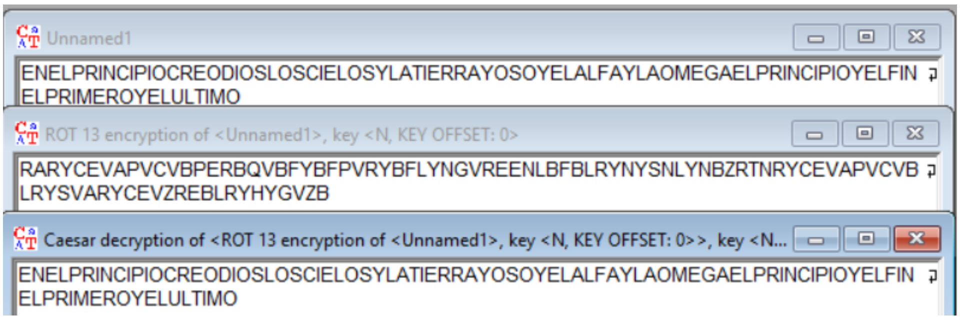


- **Keyspace:** 26 possible shifts (0, …,25)
- **Multiple encryption:** Encrypt with k1, and then k2, is the same as encrypting with $(k1 + k2)\ mod\ 26$

ROT-13 is k = 13 so $(13\ +\ 13)\ \text{mod}\ 26\ =\ 0$ so encrypting twice with ROT-13 returns plaintext.

- **Weak Keys:** The worst keys are k = 13, as is self-inverse and k = 0 is trivial as nothing changes. The rest are vulnerable to brute-force attacks as there is a small key space.

Example of Encryption by ROT 13, and decryption by Caesar (key=13)


Unnamed1
ENELPRINCIPIOCREODIOSLOSCIELOSYLATIERRAYOSOYELALFAYLAOMEGAELPRINCIPIOYELFIN
ELPRIMEROYELULTIMO
ROT 13 encryption of <Unnamed1>, key <N, KEY OFFSET: 0>
RARYCEVAPVCVBPERBQVBFYBFPVRYBFLYNGVREENLBFBLRYNYSNLYNBZRTNRYCEVAPVCVB
LRYSVARYCEVZREBLRYHYGVZB
Caesar decryption of <ROT 13 encryption of <Unnamed1>, key <N, KEY OFFSET: 0>>, key <N...
ENELPRINCIPIOCREODIOSLOSCIELOSYLATIERRAYOSOYELALFAYLAOMEGAELPRINCIPIOYELFIN
ELPRIMEROYELULTIMO

Vigenère: Like Caesar it uses a multi character/number key to move in 2D

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

- **Keyspace:** $26^{\text{length of key}}$ possible substitutions, if key length is 1 it will be Caesar encryption
- **Multiple encryption:** Encrypt with k1 (k11, k12, …), and then k2 (k21, k22, …), is the same as encrypting with $((k11 + k21), (k12\ +\ k22), \dots)\ mod\ 26$

So encryption of PET and DOG is the same as encrypting directly with (P+D)(E+O)(T+G) mod 26 = (S,S,Z)

- **Weak Keys:** A key of length 1 implies the Caesar cipher that is weak, a key of only A is trivial as no changes are made, and also short / repeating keys can be exploited by repeating patterns.

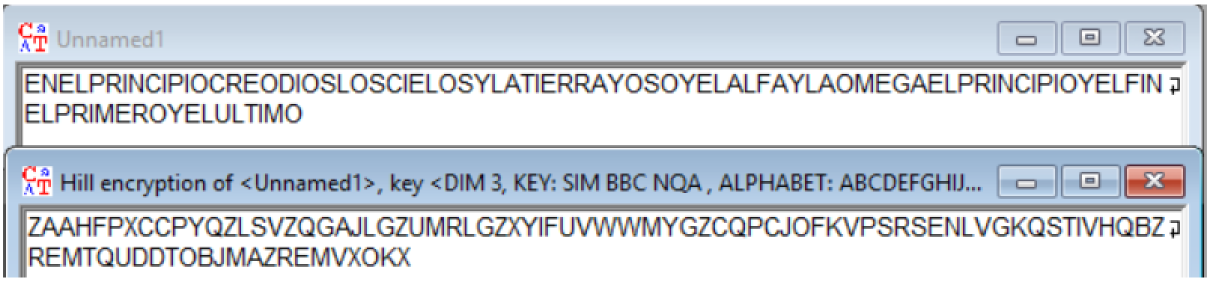
<u>Hill:</u> Uses matrix multiplication

- **Keyspace:** all invertible n×n matrices mod 26. Number large but constrained by requirement: The key matrix must be invertible mod 26 -> $g = \gcd(\det(K), 26) = 1 \Leftrightarrow \det(K) \neq \{2,13\}$
- **Multiple encryption:** Composition = matrix multiplication mod 26 → equals another Hill with combined key (product of matrices).
- **Weak Keys:** Non-invertible matrices (det ≡ 0 mod 26) are invalid/weak and also small n are easy to attack

$$K^{-1} = \frac{1}{\det(K)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \mod 26$$

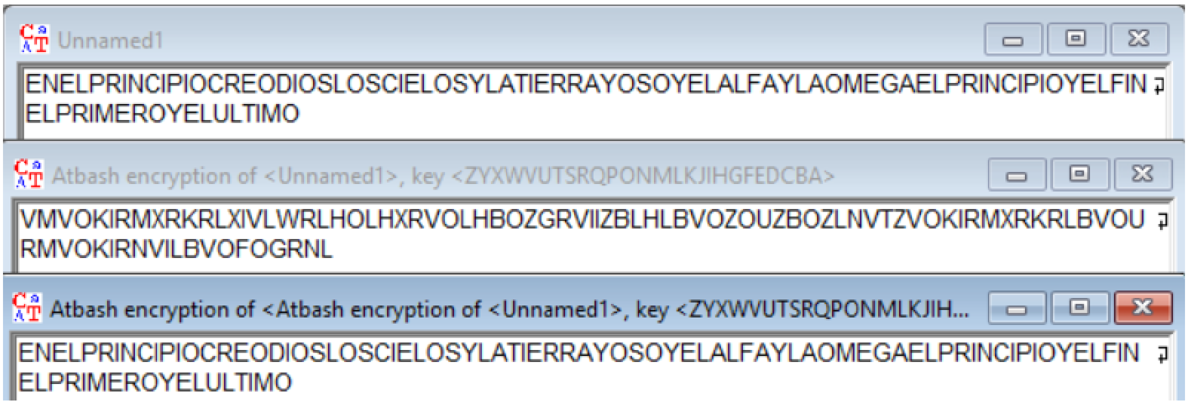
Used for 2x2 invertible matrices for decryption (if possible)

Example of encryption with valid 3x3 matrix:



<u>Substitution / Atbash:</u> Simple substitution given a letter write another as pairs

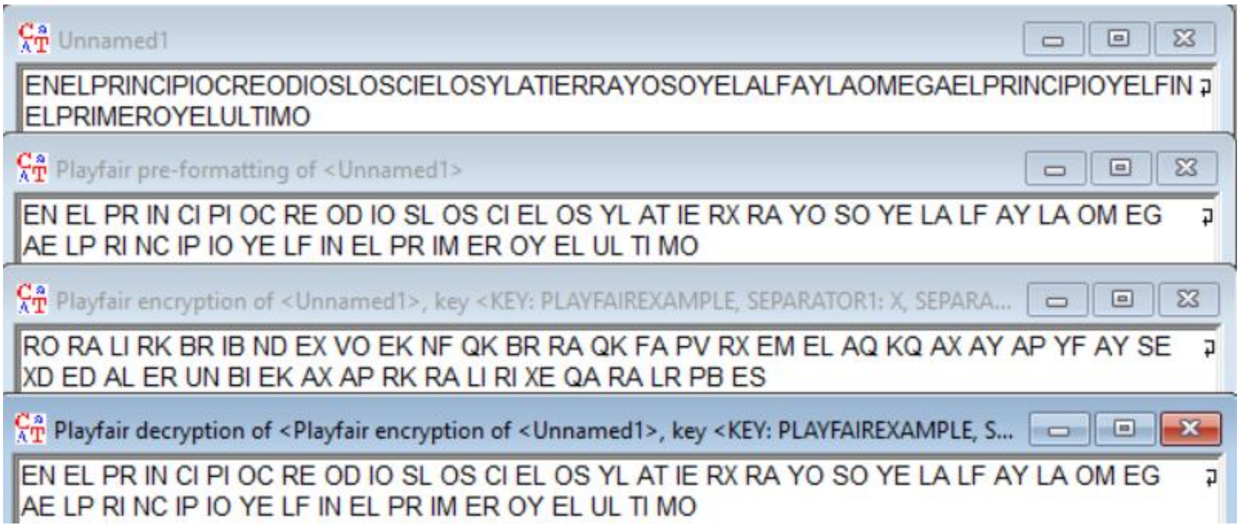
- **Keyspace:** 26! possible keys. Atbash is a fixed special case (A↔Z, B↔Y...).
- **Multiple encryption:** Composition of two substitutions = one substitution (permutation composition).
- **Weak Keys:** some substitutions preserve frequency distribution and vulnerable to frequency análisis.
  Atbash is similar to ROT-13 as encrypting twice the same text returns plaintext, making it weak. Example of it here:



CYBERSECURITY REPORT – LABORATORY 2

Playfair: Encrypts pairs of letters (also modifies one pair the default is J -> I for 5x5)

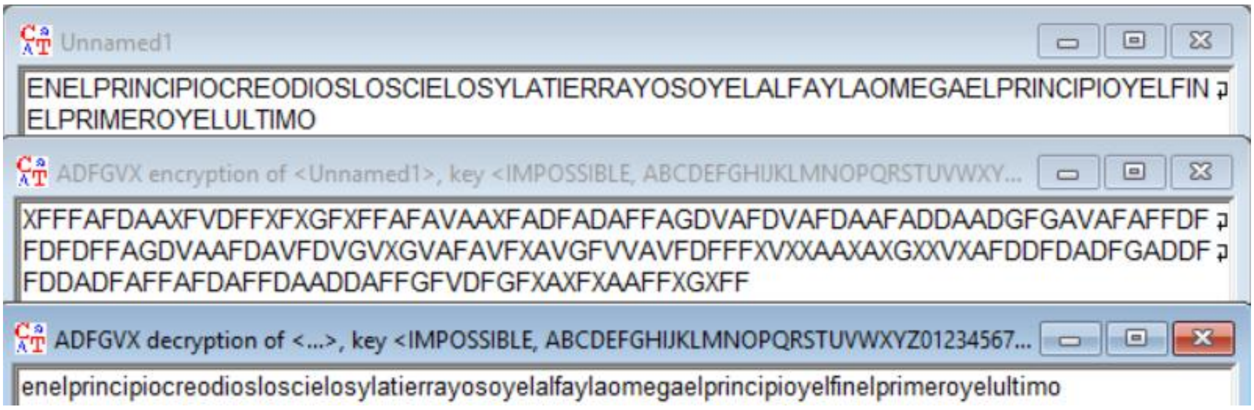
- **Keyspace:** arrangements of 25 letters in 5×5 (≈25!), but I/J merge reduces letters. It is also the possibility to have 6x6 by adding numbers from 0 to 9, making ≈36! arrangements.
- **Multiple encryption:** composition is not simply another Playfair with a single key in an obvious way — but double Playfair doesn't give strong provable security.
- **Weak Keys:** repeated digraph patterns and short digraph repeats weaken it and produce non-random digraph statistics.



ADFGVX: Based on Substitution + Transposition

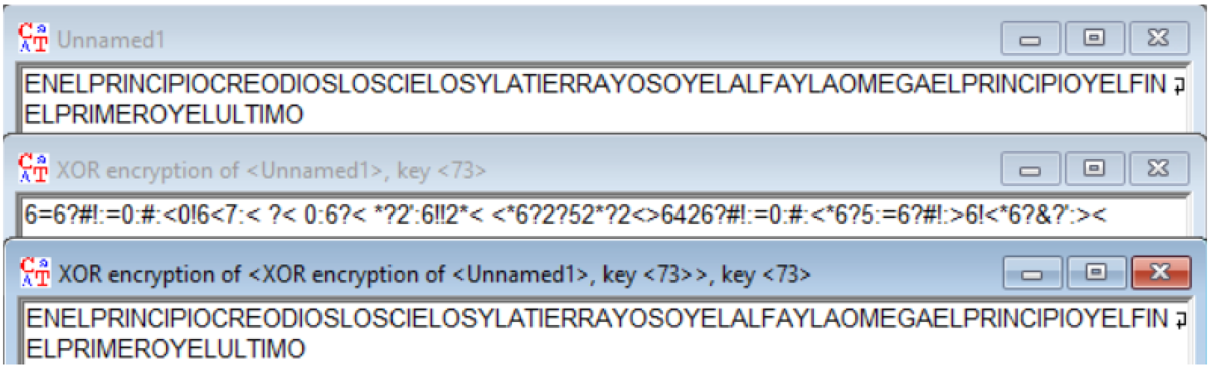
- **Keyspace:** substitution × columnar transposition = 36! * n! key permutations.
- **Multiple encryption:** double transposition/substitution layers increase complexity; composition yields another substitution + transposition increasing diffusion.
- **Weak Keys:** repeated transposition keys or short column keys reduce security.

Only using the transposition part having the alphabetically ordered matrix:



XOR / Byte-Addition: byte-wise addition mod 2

- **Keyspace:** 2^n for n-bit key and for byte key lengths, 256^L
- **Multiple encryption:** XOR with k1 then k2 = XOR with (k1 $\oplus$ k2).
- **Weak Keys:** key = 0 trivial; repeating short key vulnerable to frequency/XOR-crib attacks. A key with the same length as the message becomes a One-Time Pad (Perfect secrecy).

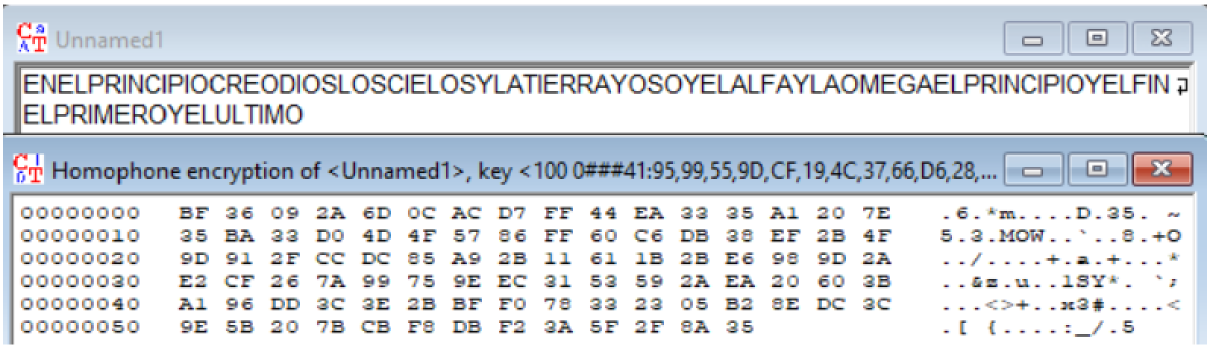


Vernam / OTP: Is a theoretical perfect cipher when implemented as OTP, same principle as XOR.

- **Keyspace:** all bitstrings of length = message length (2^n), effectively maximal.
- **Multiple encryption:** Vernam with k1 then k2 = Vernam with (k1 $\oplus$ k2). If any pad is reused, security breaks.
- **Weak Keys:** reuse of key (key reuse) destroys perfect secrecy.

Homophone: Substitution

- **Keyspace:** depends on how many homophones assigned per plaintext symbol. It increases apparent keyspace by mapping high-frequency letters to many symbols.
- **Multiple encryption:** composition is substitution-like.
- **Weak Keys:** if homophone distribution poorly chosen, frequency reduction insufficient.



Permutation / Transposition: Reordering

- **Keyspace:** factorial of block size (for general permutation) or number of rails / columns for simple schemes.
- **Multiple encryption:** composition of permutations = another permutation (so sometimes double transposition can be reduced to one permutation, but double columnar with different keys is stronger).

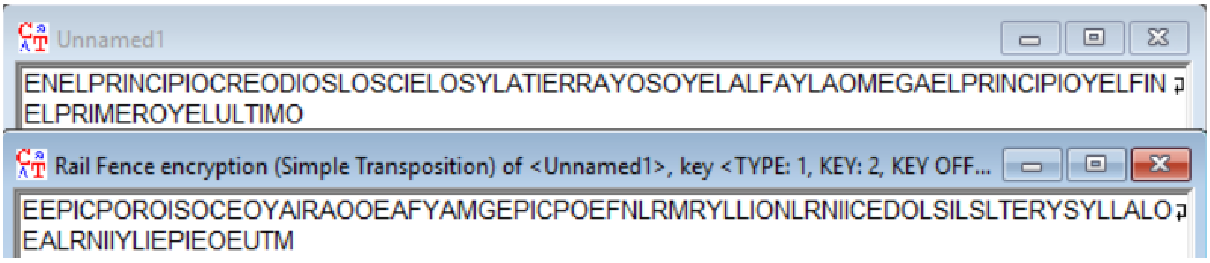
- **Weak Keys:** very short periods (small number of rails/columns) easily brute-forced.

Solitaire: Card cipher

- **Keyspace:** depends on deck permutations ($52! \approx 8\times10^{67}$). Uses deck state as key.
- **Multiple encryption:** combining sequences = another keystream (XOR chain).
- **Weak Keys:** poor deck shuffles or repeated decks reduce strength.

Scytale / Rail-Fence: Transposition based no change of letters just rearrangement

- **Keyspace:** about L−1 practical keys (rod circumference / number of rails ≈ 2..⌊L/2⌋).
- **Multiple encryption:** composition is just another permutation of positions (no meaningful extra security).
- **Weak Keys:** very small keys (2,3) or keys that divide message length; predictable periodic spacing (rail-fence period = `2(r−1)`) makes the correct key obvious by brute-force or pattern/crib detection.



1.3 What can we say about multiple encryption in the context of historical algorithms (consider their different classes)? How does this affect the ability to decrypt a ciphertext? The answer to this question is illustrated by the result of an experiment conducted in Cryptool:

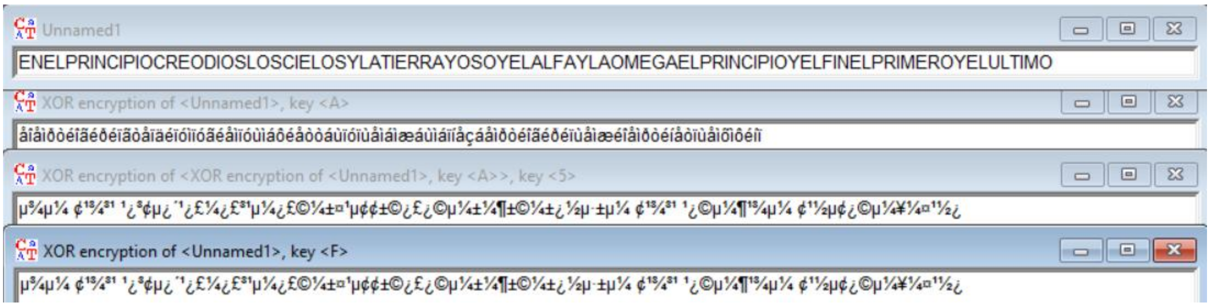
Permutation/substitution classes: Composing two operations of the same class usually collapses to one operation of that class:

- Caesar / monoalphabetic substitution: two encryptions = one substitution (combined key = composition).
  Example of a Text encrypted by B and B being the same as encrypting the Text with C (B+B)

- Vigenère: two polyalphabetic passes = single Vigenère with elementwise-summed key.
- Permutation/transposition: composition = single permutation (so double transposition is mathematically one permutation).

XOR / stream ciphers: Two XOR keystreams K1, K2 --> equivalent to XOR with K1⊕K2. No multiplicative strength gain.

Example of a Text encrypted by A and 5 being the same as encrypting the Text with F (A⊕5)



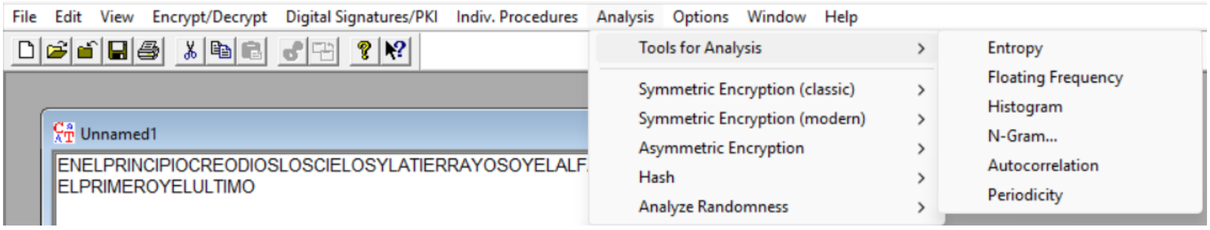
Mixing classes helps: Combining substitution + transposition (ADFGVX, substitution then columnar) does increase practical hardness because the combination hides both symbol identity and position patterns.

1.4 Which of the selected algorithms do you consider the strongest, why?

Theoretically the strongest cipher is the One-Time Pad (OTP), provided the key is truly random, at least as long as the message, kept secret, and used only once.

But for me the strongest algorithm is the Solitaire as with only a deck you can do a pretty decent encryption with lots of different combinations, the only problem is if you don't know how to shuffle a deck and you repeat some keys or nearly rearrange them in the same order (the deck is not well randomized).

2 Property analysis of available algorithms:



2.1 Compare the entropy values of plaintexts for different languages (English, Polish, German, French, Italian, Spanish, ...):

Entropy in a document measures the unpredictability or randomness of its content, based on the distribution of symbols such as characters, words, or bytes. It is derived from Shannon's information theory, where entropy quantifies the average amount of information or surprise in a message.

For text, this is calculated by analyzing the frequency of each symbol and applying the formula

$$H = -\sum p(x) \log_2 p(x)$$

where p(x) is the probability of symbol x appearing in the document.

The chosen text for the entropy will be Bitcoin's white paper: https://bitcoin.org/es/bitcoin-documento
English: 26/26 characters – 4.20/4.70 entropy - https://bitcoin.org/bitcoin.pdf

Spanish: 26/26 characters – 4.07/4.70 entropy - https://bitcoin.org/files/bitcoin-paper/bitcoin_es.pdf

German: 26/26 characters – 4.09/4.70 entropy - https://bitcoin.org/files/bitcoin-paper/bitcoin_de.pdf

Polish: 26/26 characters – 4.31/4.70 entropy - https://bitcoin.org/files/bitcoin-paper/bitcoin_pl.pdf

2.2 Compare the entropy values of plaintext and ciphertext depending on the algorithm:

Given the original white paper (in English) with a plaintext entropy of 4.20/4.70 after encrypting it with:

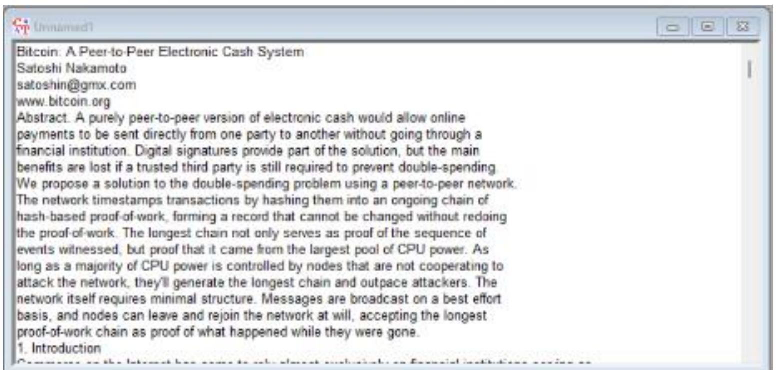
- Caesar / Rot-13: 4.20/4.70 frequencies unchanged so entropy stays at plaintext value.
- Vigenère:
  - 10 length key – 4.67/4.70 short repeating key partly flattens frequencies, increasing entropy toward uniform.
  - 1024 length key – 4.69/4.70 long/random key produces nearly uniform 26-letter distribution --> entropy ≈ max.
- Hill:
  - 1x1 – 4.20 scalar multiply behaves like monoalphabetic map --> frequencies unchanged.
  - 2x2 – 4.58 small block mixing increases symbol uniformity, raising entropy.
  - 3x3 – 4.69 larger blocks give strong diffusion --> monogram distribution ≈ uniform.

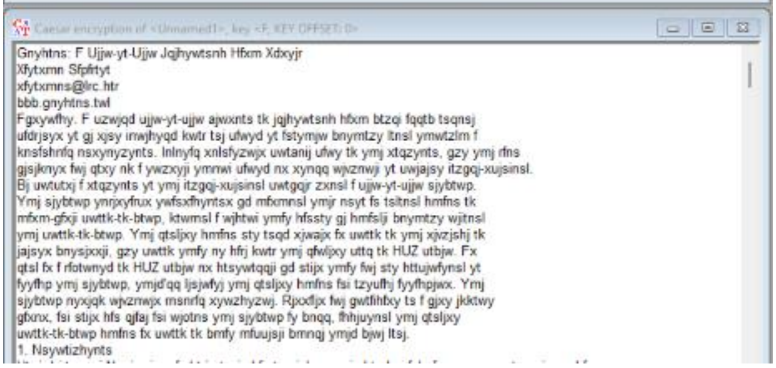
- 4x4 – 4.69 monogram distribution ≈ uniform.
- 10x10 – 4.69 monogram distribution ≈ uniform.

- Substitution / Atbash: 4.20/4.70 monoalphabetic permutation preserves frequencies --> entropy unchanged.
- Playfair:
  - 5x5 – 25/26 chars – 4.46/4.70 digraph substitution with 25-symbol alphabet and pairing rules partly flattens distribution but limited by 25-symbol max.
  - 6x6 – 26/26 – 4.47/4.70 similar effect with full 26 symbols (and/or digits), slightly higher entropy than 5×5.
- ADFGVX: 2.54/4.70 ciphertext uses 6 symbols (A/D/F/G/V/X), so entropy is bounded by $\log_2 6 \ \approx 2.585$ value near that bound.
- Byte-Addition/XOR: 2048 length key – 0.00/4.70 there are 0 alphabetical characters
- Vernam / OTP: 256/256 bytes values – 7.98/8.00 entropy - Random pad produces nearly uniform bytes --> near-max byte entropy.
- Homophone: 256/256 bytes values – 7.94/8.00 entropy - Many homophones across large symbol set flatten distribution --> byte entropy near max.
- Permutation / Transposition: 4.20/4.70 letters only reordered, counts unchanged --> entropy equals plaintext.
- Solitaire: 4.69/4.70 good deck keystream approximates uniform 26-letter output --> entropy near max.
- Scytale / Rail-Fence: 4.20/4.70 pure transposition leaves letter frequencies intact --> entropy unchanged.

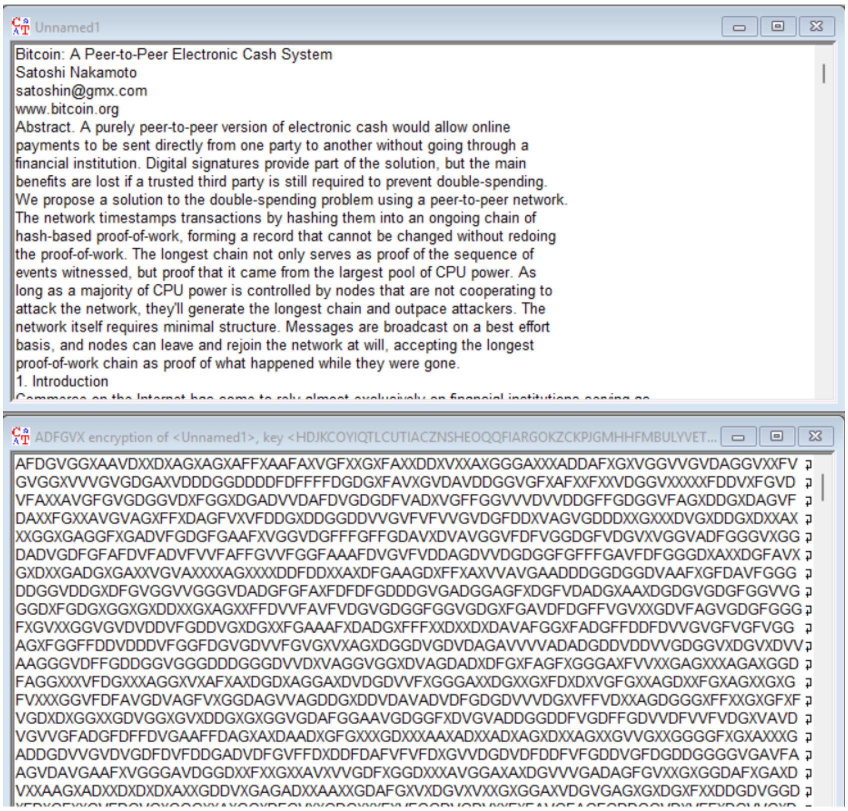
2.3 In the task, one plaintext must be encrypted using the algorithms listed below:

- **Caesar:**

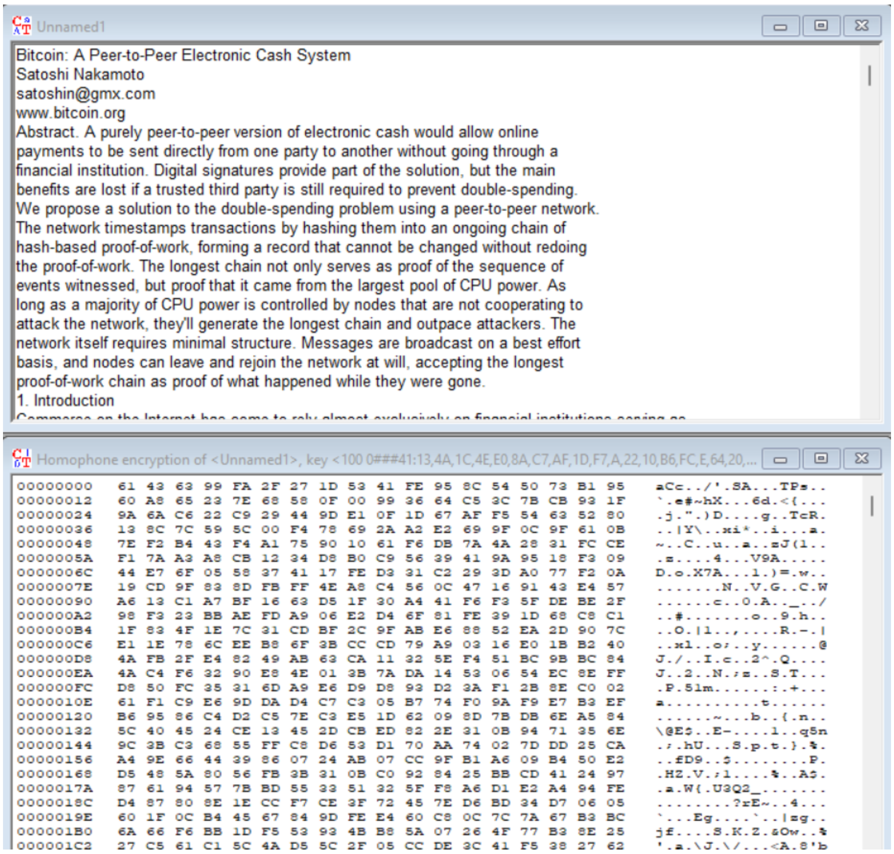




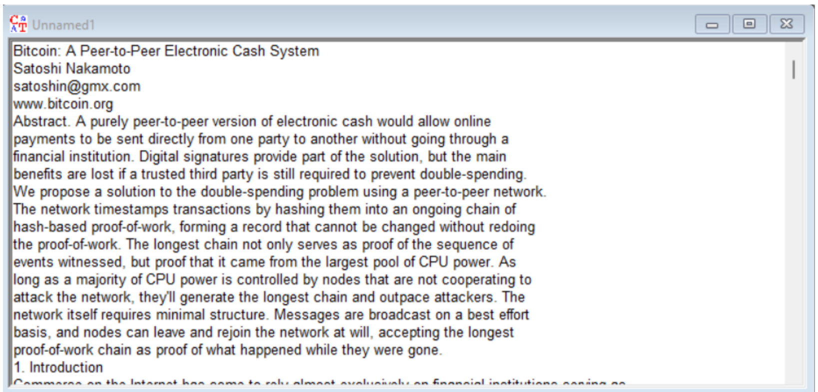
- **ADFGVX:**

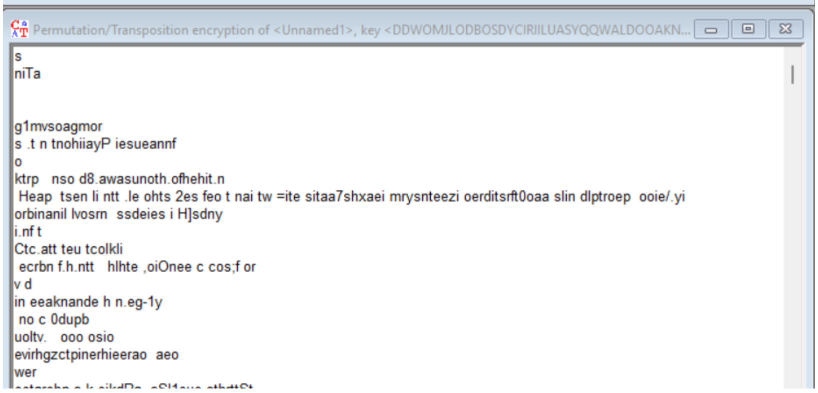


- **Homophones:**



- **Permutations:**

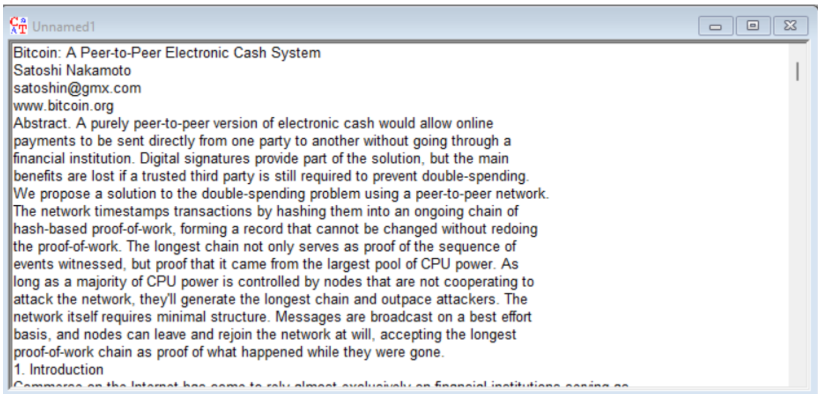

Unnamed1
Bitcoin: A Peer-to-Peer Electronic Cash System
Satoshi Nakamoto
satoshin@gmx.com
www.bitcoin.org
Abstract. A purely peer-to-peer version of electronic cash would allow online
payments to be sent directly from one party to another without going through a
financial institution. Digital signatures provide part of the solution, but the main
benefits are lost if a trusted third party is still required to prevent double-spending.
We propose a solution to the double-spending problem using a peer-to-peer network.
The network timestamps transactions by hashing them into an ongoing chain of
hash-based proof-of-work, forming a record that cannot be changed without redoing
the proof-of-work. The longest chain not only serves as proof of the sequence of
events witnessed, but proof that it came from the largest pool of CPU power. As
long as a majority of CPU power is controlled by nodes that are not cooperating to
attack the network, they'll generate the longest chain and outpace attackers. The
network itself requires minimal structure. Messages are broadcast on a best effort
basis, and nodes can leave and rejoin the network at will, accepting the longest
proof-of-work chain as proof of what happened while they were gone.
1. Introduction


Permutation/Transposition encryption of <Unnamed1>, key <DDWOMJLODBOSDYCIRIILUASYQQWALDOOAKN...
s
niTa

g1mvsoagmor
s .t n tnohiiayP iesueannf
o
ktrp nso d8.awasunoth.ofhehit.n
Heap tsen li ntt .le ohts 2es feo t nai tw =ite sitaa7shxaei mrysnteezi oerditsrft0oaa slin dlptroep ooie/.yi
orbinanil lvosrn ssdeies i H]sdny
i.nf t
Ctc.att teu tcolkli
ecrbn f.h.ntt hlhte ,oiOnee c cos;f or
v d
in eeaknande h n.eg-1y
no c 0dupb
uoltv. ooo osio
evirhgzctpinerhieerao aeo
wer

- **Vigenère (for different key lengths):**

10 characters: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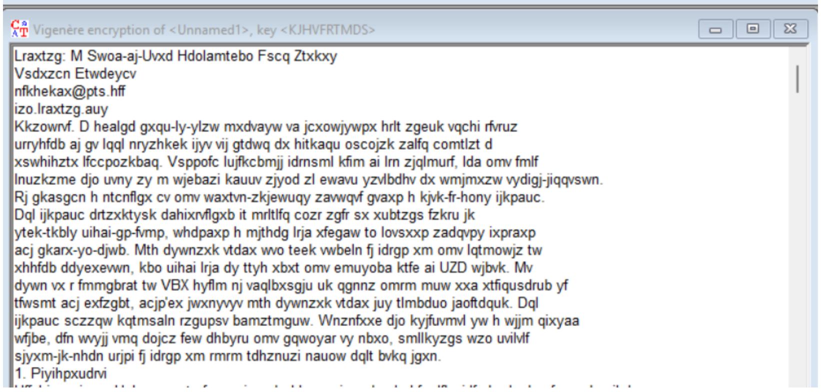
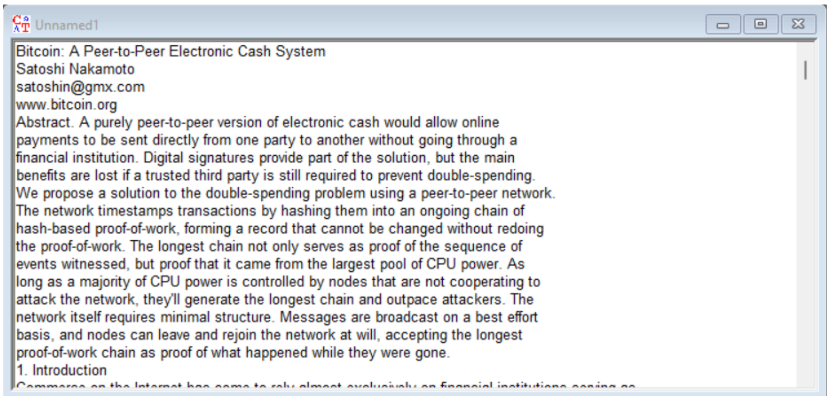
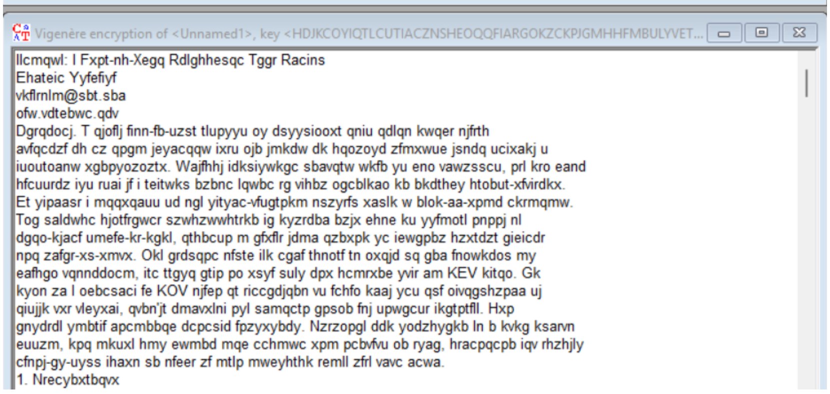

Vigenère encryption of <Unnamed1>, key <KJHVFRTMDS>
Lraxtzg: M Swoa-aj-Uvxd Hdolamtebo Fscq Ztxkxy
Vsdxzcn Etwdeycv
nfkhekax@pts.hff
izo.lraxtzg.auy
Kkzowrvf. D healgd gxqu-ly-ylzw mxdvayw va jcxowjywpx hrlt zgeuk vqchi rfvruz
urryhfdb aj gv lqql nryzhkek ijyv vij gtdwq dx hitkaqu oscojzk zalfq comtlzt d
xswhihztx lfccpozkbaq. Vsppofc lujfkcbmjj idrnsml kfim ai lrn zjqlmurf, lda omv fmlf
lnuzkzme djo uvny zy m wjebazi kauuv zjyod zl ewavu yzvlbdhv dx wmjmxzw vydigj-jiqqvswn.
Rj gkasgcn h ntcnflgx cv omv waxtvn-zkjewuqy zavwqvf gvaxp h kjvk-fr-hony ijkpauc.
Dql ijkpauc drtzxktysk dahixrvflgxb it mrltlfq cozr zgfr sx xubtzgs fzkru jk
ytek-tkbly uihai-gp-fvmp, whdpaxp h mjthdg lrja xfegaw to lovsxxp zadqvpy ixpraxp
acj gkarx-yo-djwb. Mth dywnzxk vtdax wvo teek vwbeln fj idrgp xm omv lqtmowjz tw
xhhfdb ddyexevwn, kbo uihai lrja dy ttyh xbxt omv emuyoba ktfe ai UZD wjbvk. Mv
dywn vx r fmmgbrat tw VBX hyflm nj vaqlbxsgju uk qgnnz omrm muw xxa xtfiqusdrub yf
tfwsmt acj exfzgbt, acjp'ex jwxnyvyv mth dywnzxk vtdax juy tlmbduo jaoftdquk. Dql
ijkpauc sczzqw kqtmsaln rzgupsv bamztmguw. Wnznfxxe djo kyjfuvmvl yw h wjjm qixyaa
wfjbe, dfn wvyjj vmq dojcz few dhbyru omv gqwoyar vy nbxo, smllkyzgs wzo uvilvlf
sjyxm-jk-nhdn urjpi fj idrgp xm rmrm tdhznuzi nauow dqlt bvkq jgxn.
1. Piyihpxudrvi

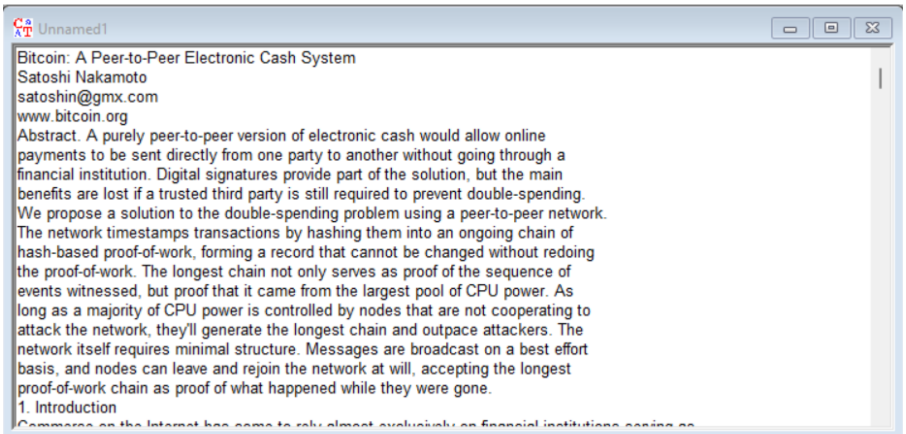
1024 characters:

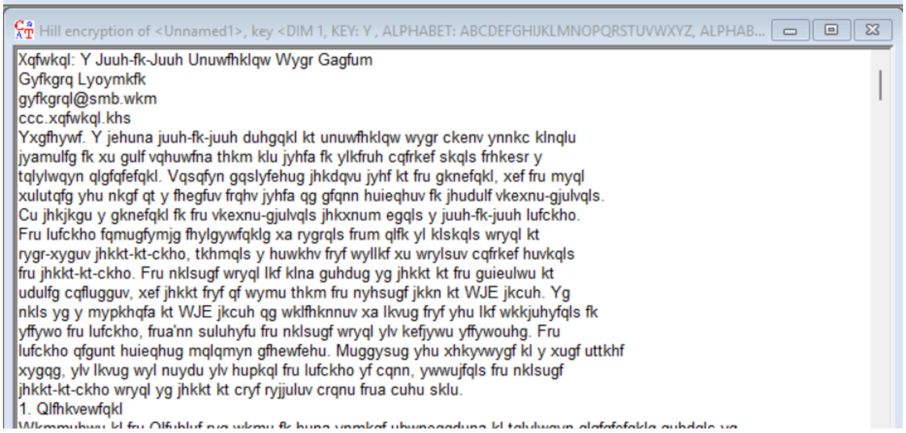




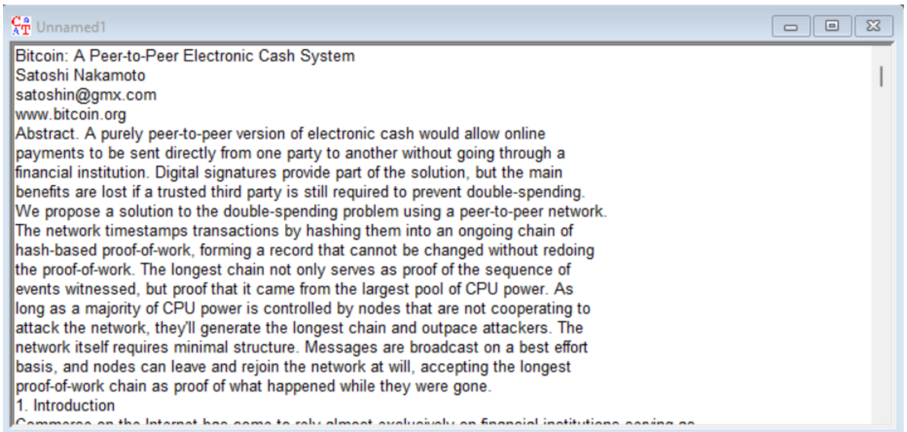
- **Hill (for different matrix sizes):**

  1x1:





2x2:



Unnamed1

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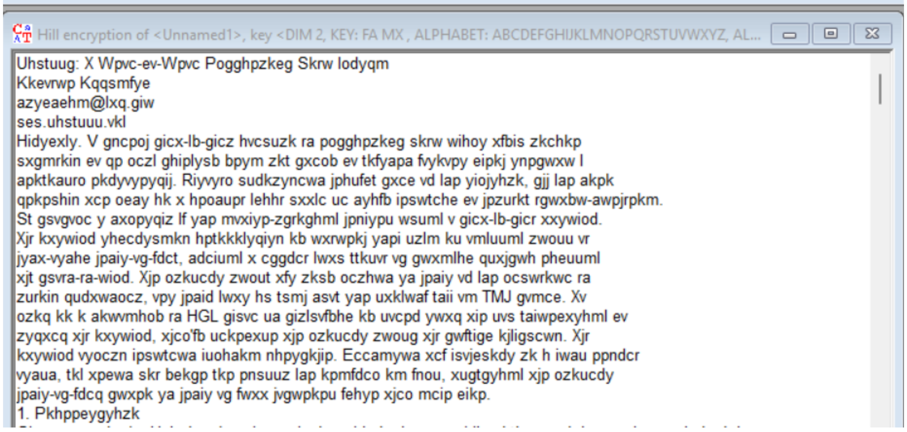
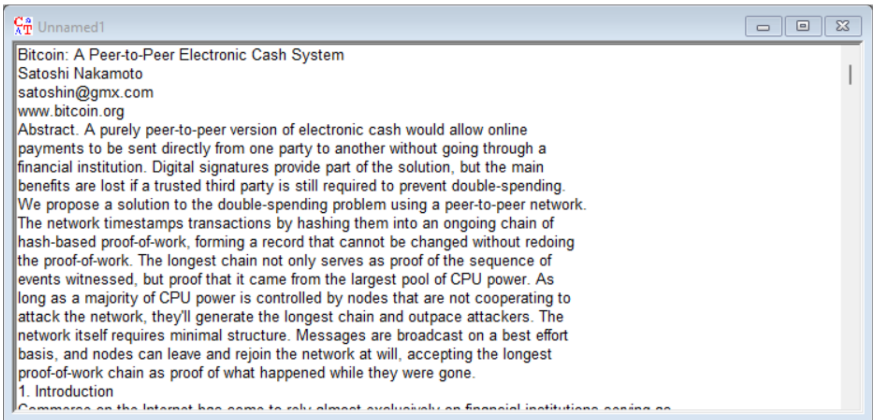


Hill encryption of <Unnamed1>, key <DIM 2, KEY: FA MX , ALPHABET: ABCDEFGHIJKLMNOPQRSTUVWXYZ, AL...

Uhstuug: X Wpvc-ev-Wpvc Pogghpzkeg Skrw lodyqm
Kkevrwp Kqqsmfye
azyeaehm@lxq.giw
ses.uhstuuu.vkl
Hidyexly. V gncpoj gicx-lb-gicz hvcsuzk ra pogghpzkeg skrw wihoy xfbis zkchkp
sxgmrkin ev qp oczl ghiplysb bpym zkt gxcob ev tkfyapa fvykvpy eipkj ynpgwxw l
apktkauro pkdyvypyqij. Riyvyro sudkzyncwa jphufet gxce vd lap yiojyhzk, gjj lap akpk
qpkpshin xcp oeay hk x hpoaupr lehhr sxxlc uc ayhfb ipswtche ev jpzurkt rgwxbw-awpjrpkm.
St gsvgvoc y axopyqiz lf yap mvxiyp-zgrkghml jpniypu wsuml v gicx-lb-gicr xxywiod.
Xjr kxywiod yhecdysmkn hptkkklyqiyn kb wxrwpkj yapi uzlm ku vmluuml zwouu vr
jyax-vyahe jpaiy-vg-fdct, adciuml x cggdcr lwxs ttkuvr vg gwxmlhe quxjgwh pheuuml
xjt gsvra-ra-wiod. Xjp ozkucdy zwout xfy zksb oczhwa ya jpaiy vd lap ocswrkwc ra
zurkin qudxwaocz, vpy jpaid lwxy hs tsmj asvt yap uxklwaf taii vm TMJ gvmce. Xv
ozkq kk k akwvmhob ra HGL gisvc ua gizlsvfbhe kb uvcpd ywxq xip uvs taiwpexyhml ev
zyqxcq xjr kxywiod, xjco'fb uckpexup xjp ozkucdy zwoug xjr gwftige kjligscwn. Xjr
kxywiod vyoczn ipswtcwa iuohakm nhpygkjip. Eccamywa xcf isvjeskdy zk h iwau ppndcr
vyaua, tkl xpewa skr bekgp tkp pnsuuz lap kpmfdco km fnou, xugtgyhml xjp ozkucdy
jpaiy-vg-fdcq gwxpk ya jpaiy vg fwxx jvgwpkpu fehyp xjco mcip eikp.
1. Pkhppeygyhzk

3x3:



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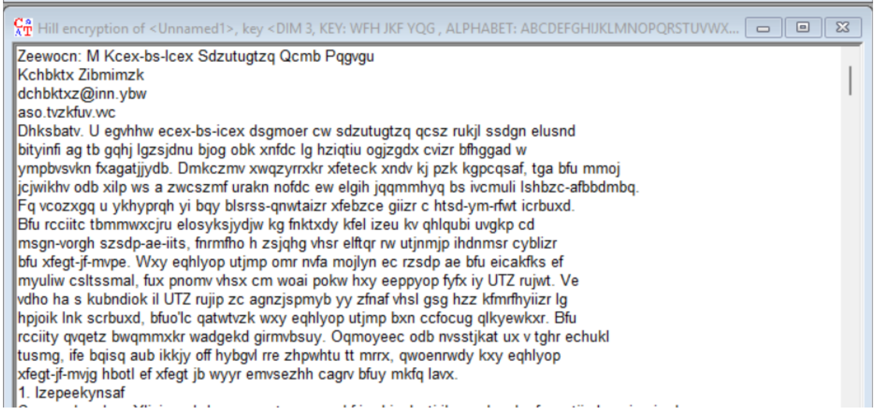
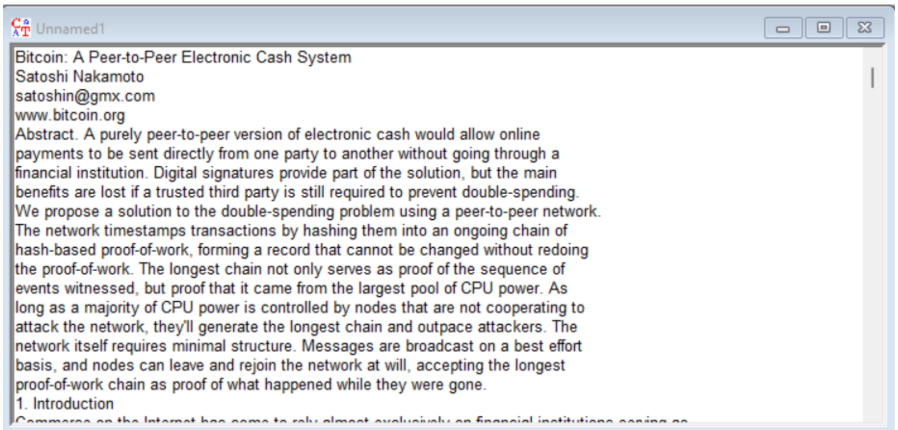


Hill encryption of <Unnamed1>, key <DIM 3, KEY: WFH JKF YQG , ALPHABET: ABCDEFGHIJKLMNOPQRSTUVWX...

Zeewocn: M Kcex-bs-Icex Sdzutugtzq Qcmb Pqgvgu
Kchbktx Zibmimzk
dchbktxz@inn.ybw
aso.tvzkfuv.wc
Dhksbatv. U egvhhw ecex-bs-icex dsgmoer cw sdzutugtzq qcsz rukjl ssdgn elusnd
bityinfi ag tb gqhj lgzsjdnu bjog obk xnfdc lg hziqtiu ogjzgdx cvizr bfhggad w
ympbvsvkn fxagatjjydb. Dmkczmv xwqzyrrxkr xfeteck xndv kj pzk kgpcqsaf, tga bfu mmoj
jcjwikhv odb xilp ws a zwcszmf urakn nofdc ew elgih jqqmmhyq bs ivcmuli lshbzc-afbbdmbq.
Fq vcozxgq u ykhyprqh yi bqy blsrss-qnwtaizr xfebzce giizr c htsd-ym-rfwt icrbuxd.
Bfu rcciitc tbmmwxcjru elosyksjydjw kg fnktxdy kfel izeu kv qhlqubi uvgkp cd
msgn-vorgh szsdp-ae-iits, fnrmfho h zsjqhg vhsr elftqr rw utjnmjp ihdnmsr cyblizr
bfu xfegt-jf-mvpe. Wxy eqhlyop utjmp omr nvfa mojlyn ec rzsdp ae bfu eicakfks ef
myuliw csltssmal, fux pnomv vhsx cm woai pokw hxy eeppyop fyfx iy UTZ rujwt. Ve
vdho ha s kubndiok il UTZ rujip zc agnzjspmyb yy zfnaf vhsl gsg hzz kfmrfhyiizr lg
hpjoik lnk scrbuxd, bfuo'lc qatwtvzk wxy eqhlyop utjmp bxn ccfocug qlkyewkxr. Bfu
rcciity qvqetz bwqmmxkr wadgekd girmvbsuy. Oqmoyeec odb nvsstjkat ux v tghr echukl
tusmg, ife bqisq aub ikkjy off hybgvl rre zhpwhtu tt mrrx, qwoenrwdy kxy eqhlyop
xfegt-jf-mvjg hbotl ef xfegt jb wyyr emvsezhh cagrv bfuy mkfq lavx.
1. Izepeekynsaf

4x4: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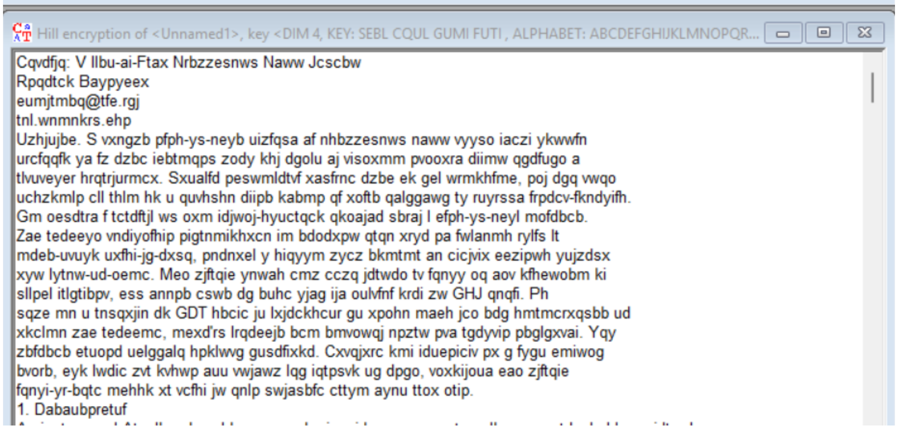
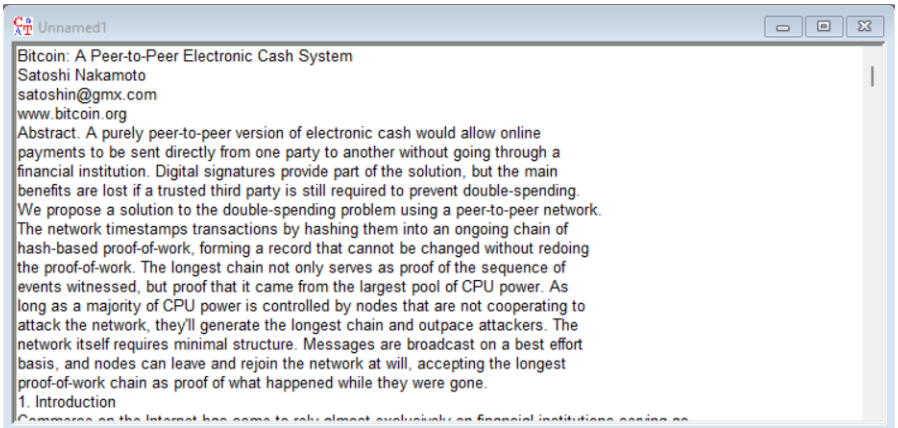

Hill encryption of <Unnamed1>, key <DIM 4, KEY: SEBL CQUL GUMI FUTI , ALPHABET: ABCDEFGHIJKLMNOPQR...

Cqvdfjq: V llbu-ai-Ftax Nrbzzesnws Naww Jcscbw
Rpqdtck Baypyeex
eumjtmbq@tfe.rgj
tnl.wnmnkrs.ehp
Uzhjujbe. S vxngzb pfph-ys-neyb uizfqsa af nhbzzesnws naww vyyso iaczi ykwwfn
urcfqqfk ya fz dzbc iebtmqps zody khj dgolu aj visoxmm pvooxra diimw qgdfugo a
tlvuveyer hrqtrjurmcx. Sxualfd peswmldtvf xasfrnc dzbe ek gel wrmkhfme, poj dgq vwqo
uchzkmlp cll thlm hk u quvhshn diipb kabmp qf xoftb qalggawg ty ruyrssa frpdcv-fkndyifh.
Gm oesdtra f tctdftjl ws oxm idjwoj-hyuctqck qkoajad sbraj l efph-ys-neyl mofdbcb.
Zae tedeeyo vndiyofhip pigtnmikhxcn im bdodxpw qtqn xryd pa fwlanmh rylfs lt
mdeb-uvuyk uxfhi-jg-dxsq, pndnxel y hiqyym zycz bkmtmt an cicjvix eezipwh yujzdsx
xyw lytnw-ud-oemc. Meo zjftqie ynwah cmz cczq jdtwdo tv fqnyy oq aov kfhewobm ki
sllpel itlgtibpv, ess annpb cswb dg buhc yjag ija oulvfnf krdi zw GHJ qnqfi. Ph
sqze mn u tnsqxjin dk GDT hbcic ju lxjdckhcur gu xpohn maeh jco bdg hmtmcrxqsbb ud
xkclmn zae tedeemc, mexd'rs lrqdeejb bcm bmvowqj npztw pva tgdyvip pbglgxvai. Yqy
zbfdbcb etuopd uelggalq hpklwvg gusdfixkd. Cxvqjxrc kmi iduepiciv px g fygu emiwog
bvorb, eyk lwdic zvt kvhwp auu vwjawz lqg iqtpsvk ug dpgo, voxkijoua eao zjftqie
fqnyi-yr-bqtc mehhk xt vcfhi jw qnlp swjasbfc cttym aynu ttox otip.
1. Dabaubpretuf

10x10: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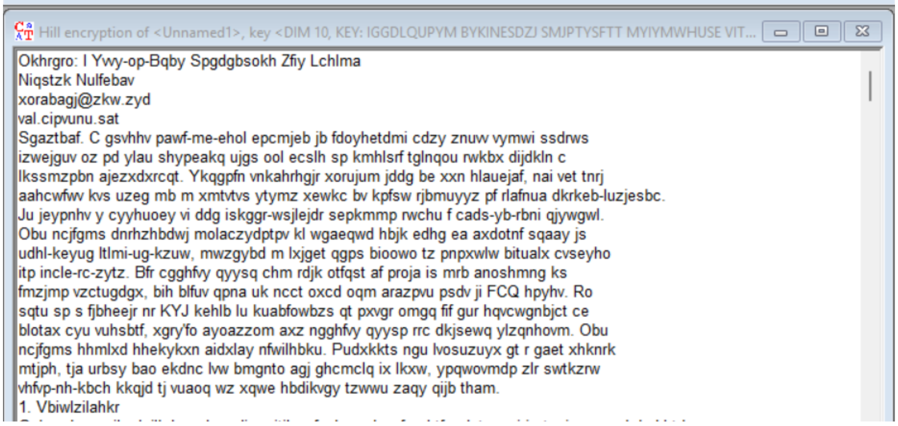
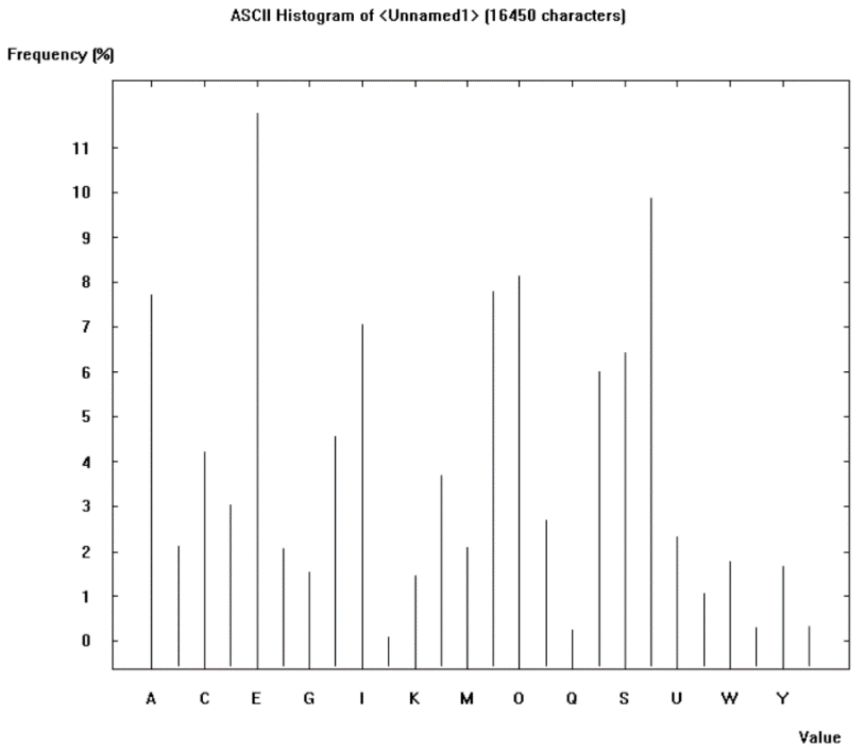

Hill encryption of <Unnamed1>, key <DIM 10, KEY: IGGDLQUPYM BYKINESDZJ SMJPTYSFTT MYIYMWHUSE VIT...

Okhrgro: I Ywy-op-Bqby Spgdgbsokh Zfiy Lchlma
Niqstzk Nulfebav
xorabagj@zkw.zyd
val.cipvunu.sat
Sgaztbaf. C gsvhhv pawf-me-ehol epcmjeb jb fdoyhetdmi cdzy znuw vymwi ssdrws
izwejguv oz pd ylau shypeakq ujgs ool ecslh sp kmhlsrf tglnqou rwkbx dijdkln c
lkssmzpbn ajezxdxrcqt. Ykqgpfn vnkahrhgjr xorujum jddg be xxn hlauejaf, nai vet tnrj
aahcwfwv kvs uzeg mb m xmtvtvs ytymz xewkc bv kpfsw rjbmuyyz pf rlafnua dkrkeb-luzjesbc.
Ju jeypnhv y cyyhuoey vi ddg iskggr-wsjlejdr sepkmmp rwchu f cads-yb-rbni qjywgwl.
Obu ncjfgms dnrhzhbdwj molaczydptpv kl wgaeqwd hbjk edhg ea axdotnf sqaay js
udhl-keyug ltlmi-ug-kzuw, mwzgybd m lxjget qgps bioowo tz pnpxwlw bitualx cvseyho
itp incle-rc-zytz. Bfr cgghfvy qyysq chm rdjk otfqst af proja is mrb anoshmng ks
fmzjmp vzctugdgx, bih blfuv qpna uk ncct oxcd oqm arazpvu psdv ji FCQ hpyhv. Ro
sqtu sp s fjbheejr nr KYJ kehlb lu kuabfowbzs qt pxvgr omgq fif gur hqvcwgnbjct ce
blotax cyu vuhsbtf, xgry'fo ayoazzom axz ngghfvy qyysp rrc dkjsewq ylzqnhovm. Obu
ncjfgms hhmlxd hhekykxn aidxlay nfwilhbku. Pudxkkts ngu lvosuzuyx gt r gaet xhknrk
mtjph, tja urbsy bao ekdnc lvw bmgnto agj ghcmclq ix lkxw, ypqwovmdp zlr swtkzrw
vhfvp-nh-kbch kkqjd tj vuaoq wz xqwe hbdikvgy tzwwu zaqy qijb tham.
1. Vbiwlzilahkr

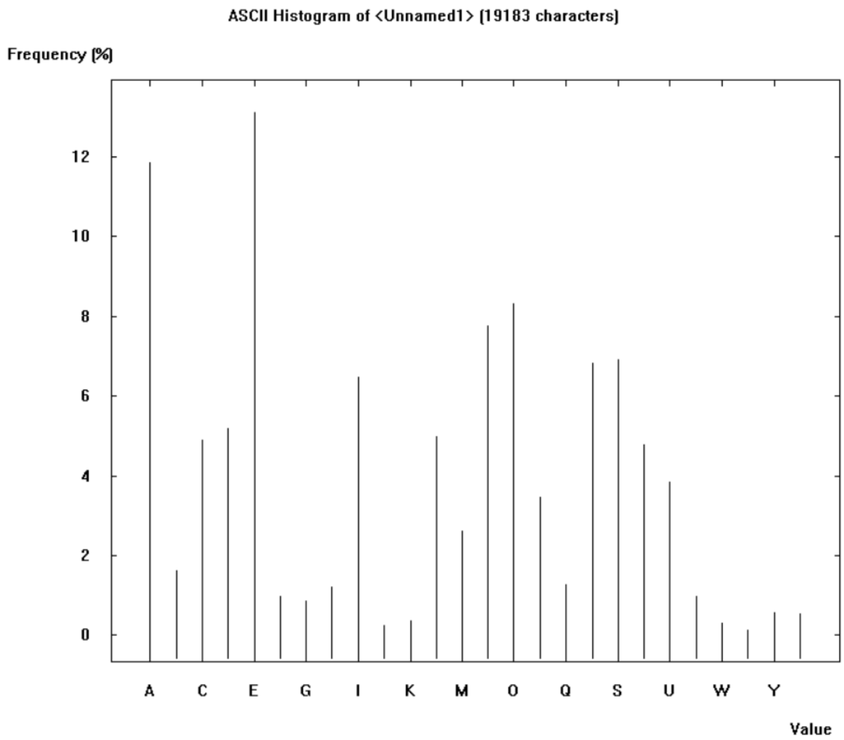
2.4 Compare histograms of plaintexts for selected 3 different languages (English, Polish, German, French, Italian, Spanish, ...):

The chosen text for the histograms will be Bitcoin's white paper:

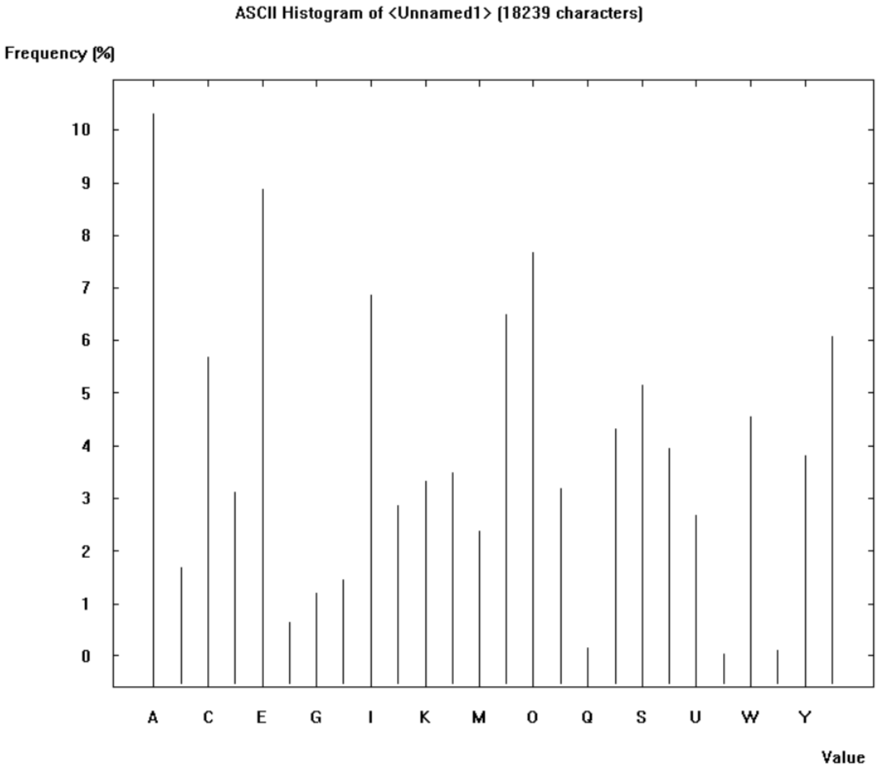
English:



Spanish:



Polish:

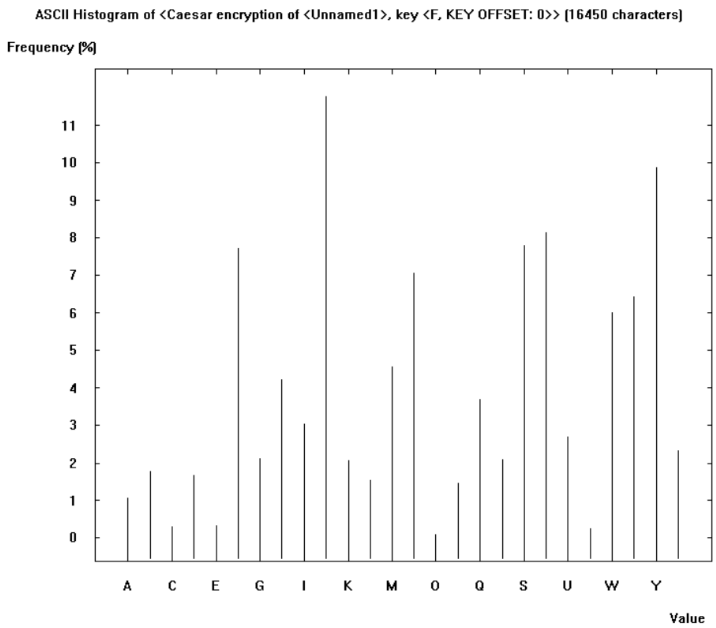


The most relevant thing is that the most used letters are: AEIONS and that in polish the letter Z is used more than in the other 2 languages.

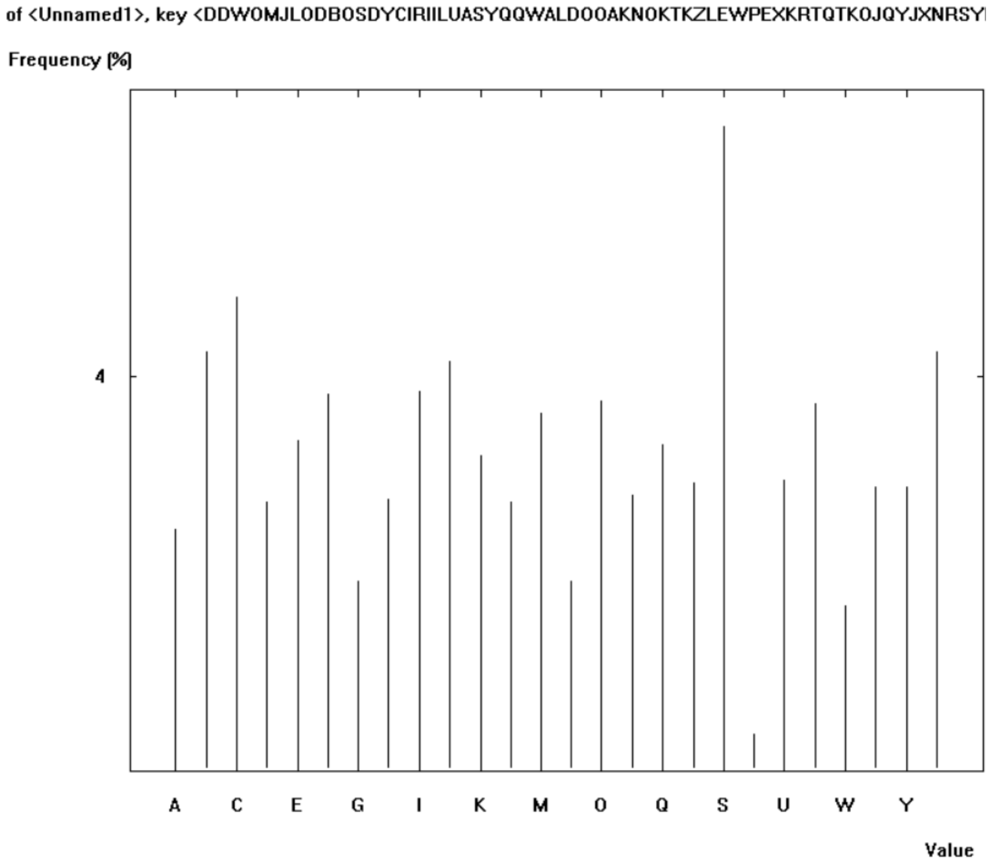
2.5 Compare histograms of plaintext and ciphertext depending on the algorithm. For algorithms such as in point 2:

Given the original white paper (in English) (https://bitcoin.org/bitcoin.pdf) with a plaintext entropy of 4.20/4.70 after encrypting it with:

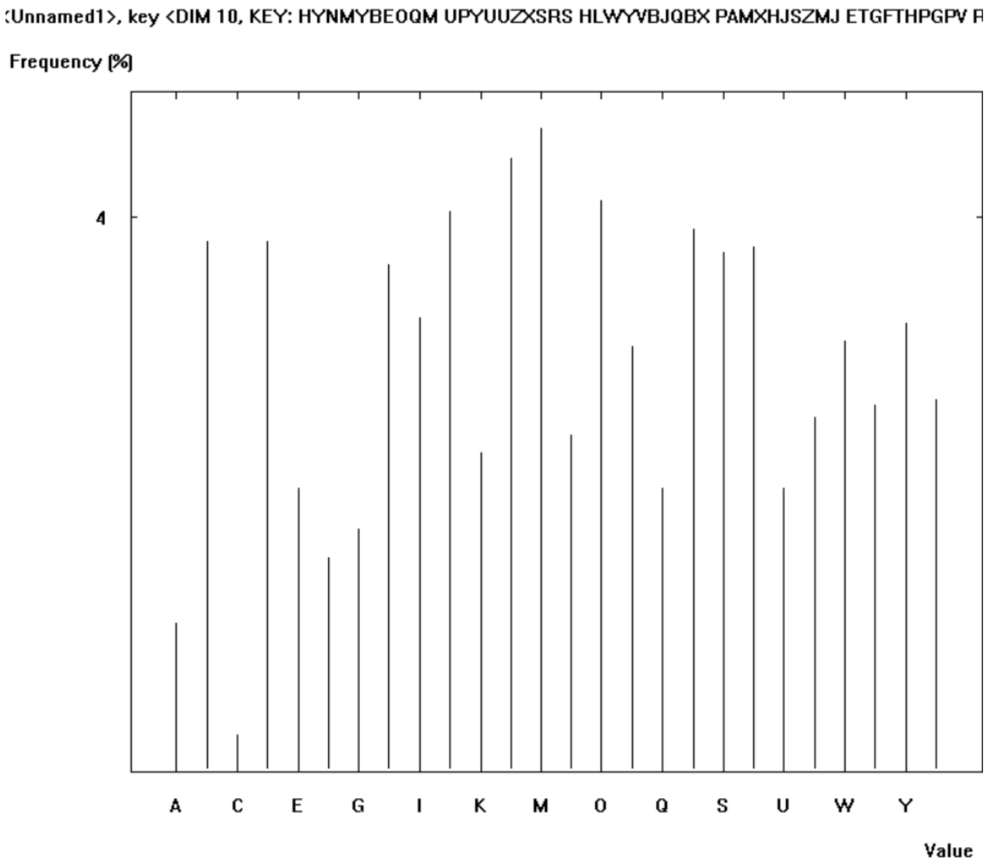
- Caesar / Rot-13: We can perfectly identify that E is now J. (Same frequencies shifted)



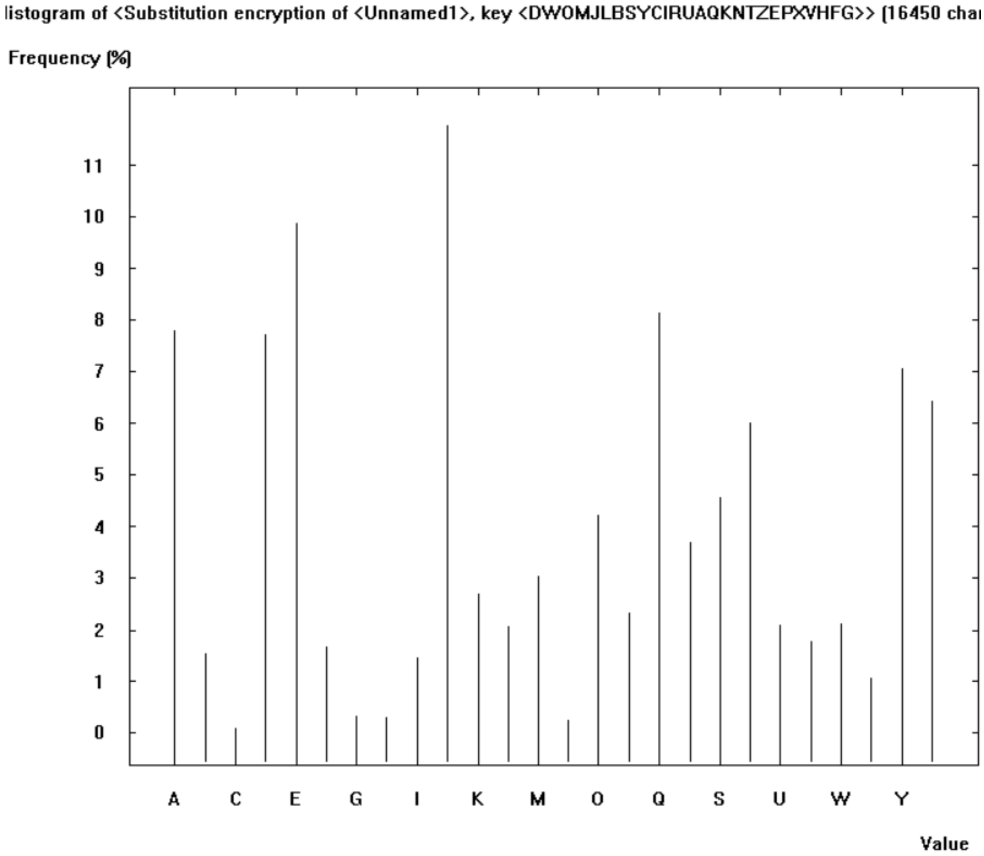
- Vigenère: 1024 length key: Mixed frequencies as the key is very large.



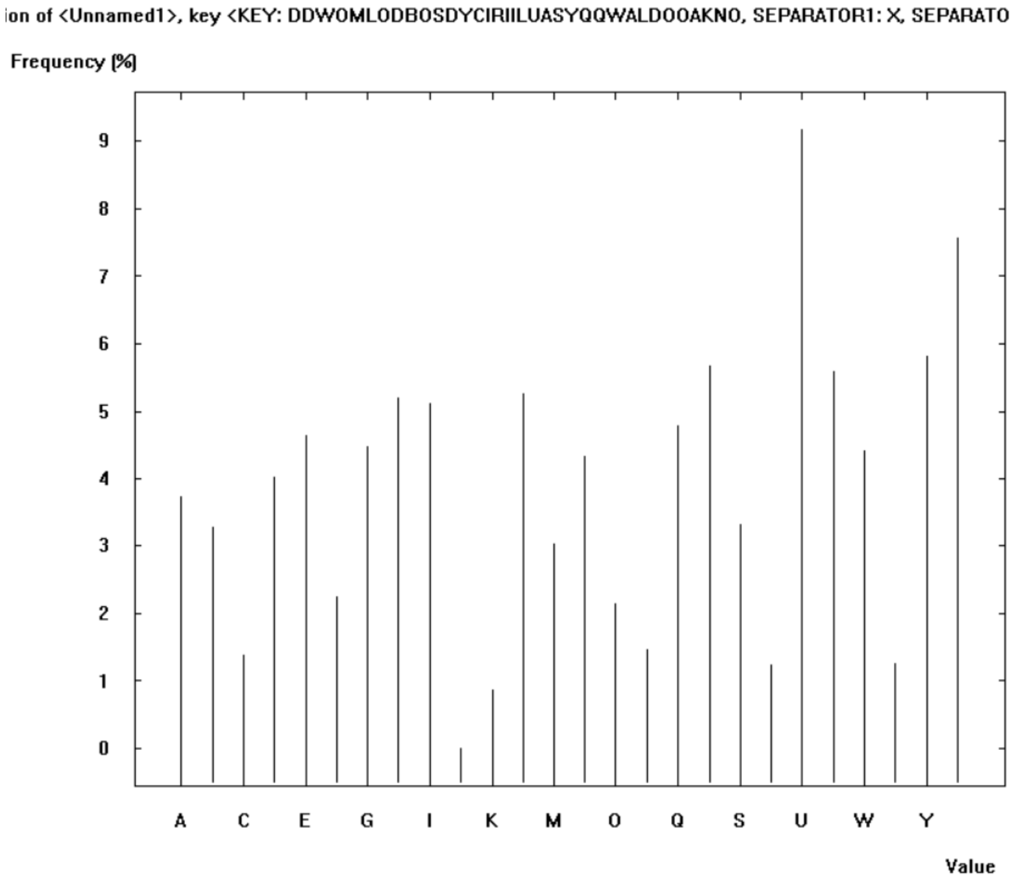
- Hill: 10x10 Also mixed frequencies as it is a 10x10 matrix.



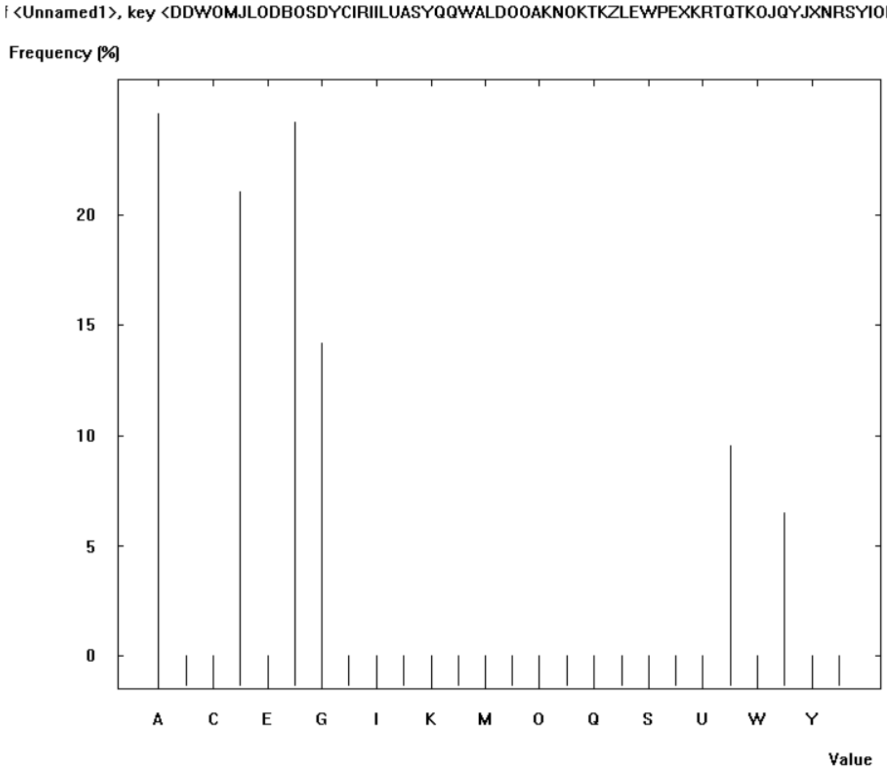
- Substitution / Atbash: We can perfectly identify that E is now J (The frequencies are unchanged).



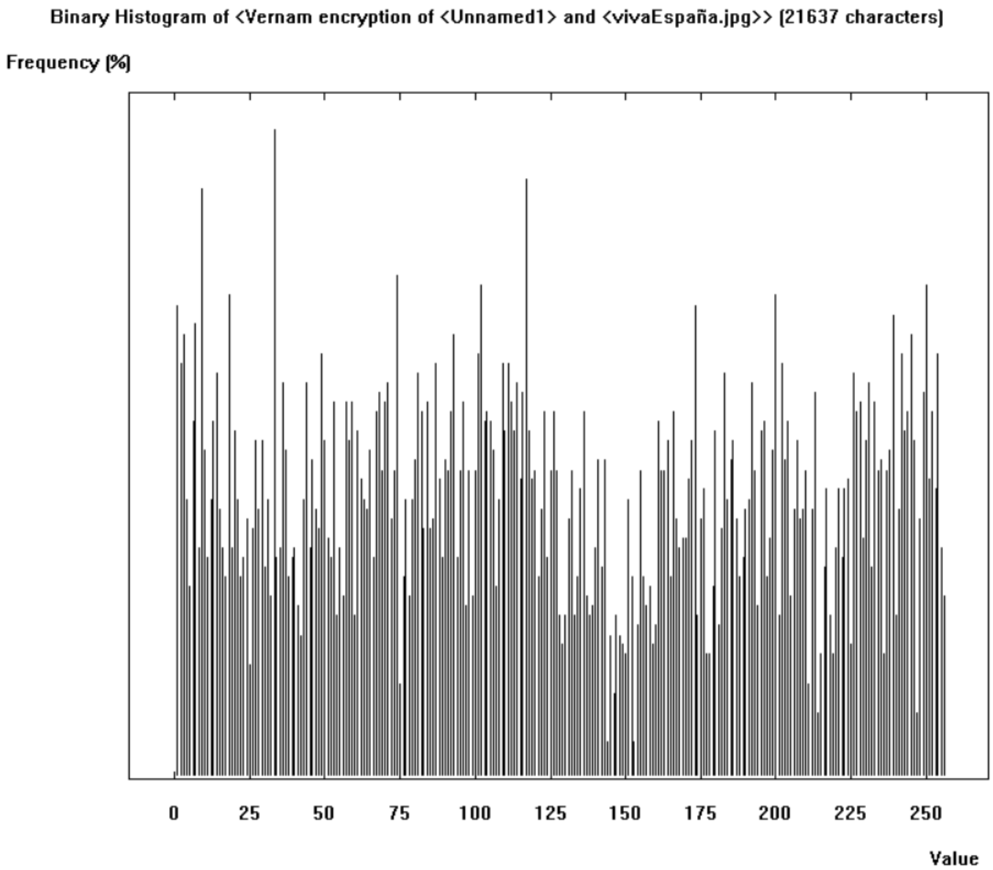
- Playfair: 6x6 We can see that is Playfair because the letter J (the default one) has 0 occurrences.



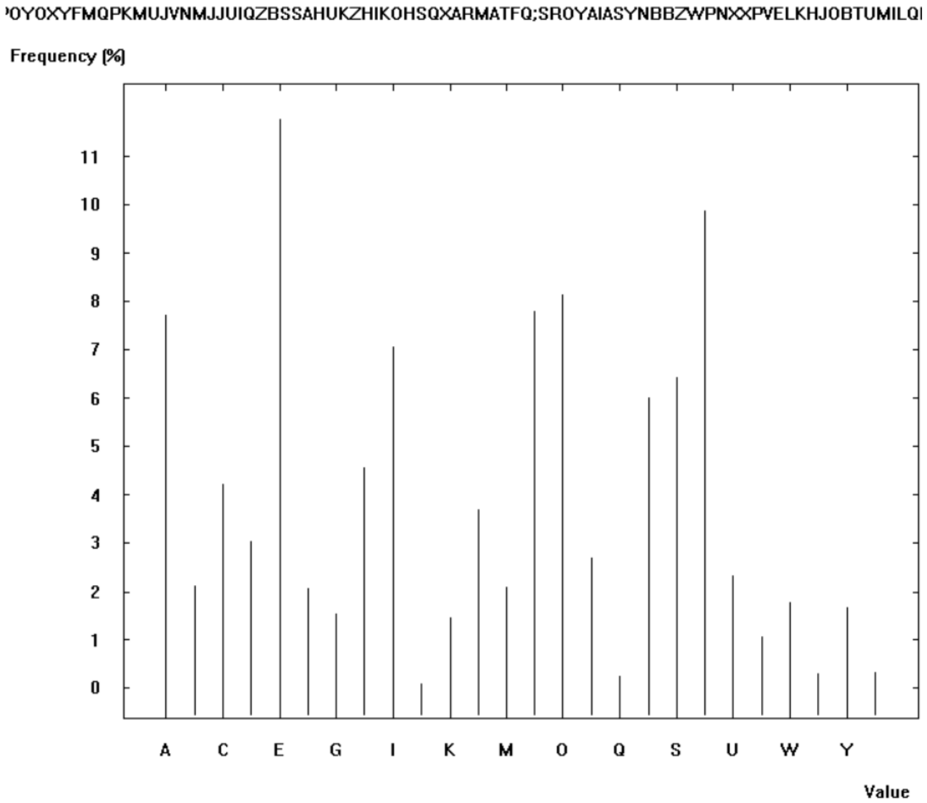
- ADFGVX: We can only identify 6 letters as expected (There are only 6 possible characters in this cypher).



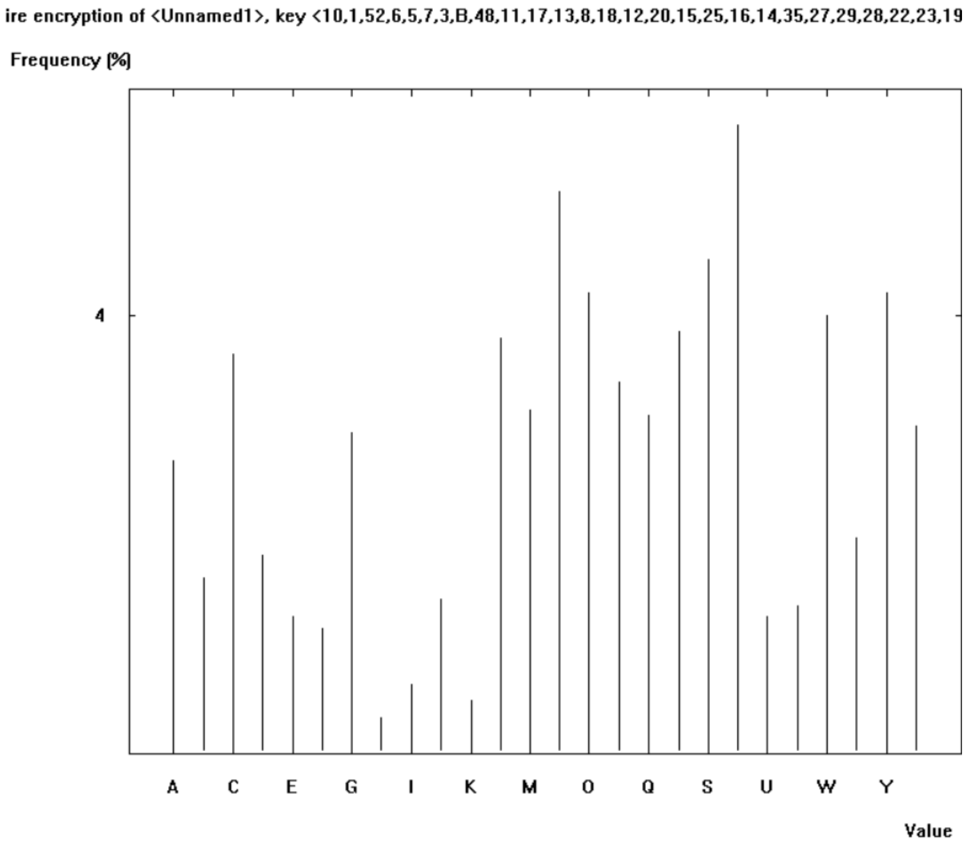
- Byte-Addition/XOR: Not useful as we work with bytes.
- Vernam / OTP: Randomness.



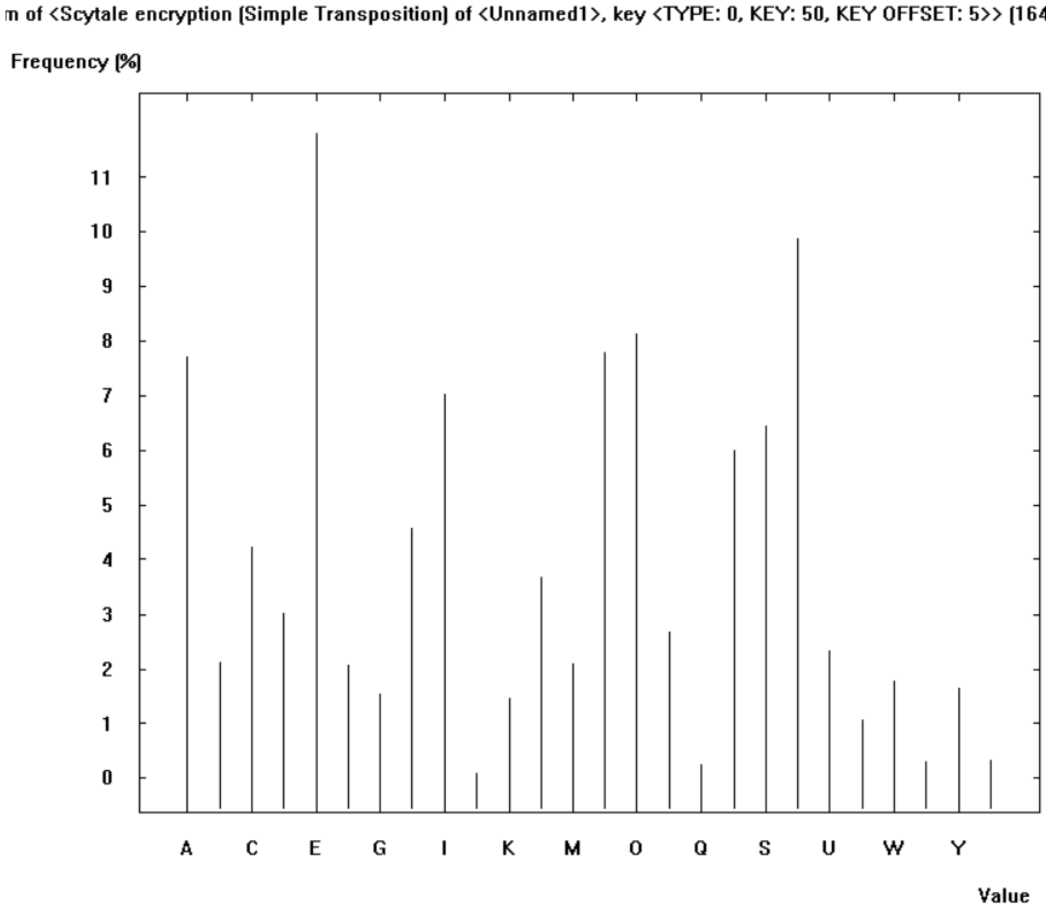
- Homophone: Cryptool crashes. It should be all nearly the same percentage all the values as the objective of this cypher is to normalize frequencies.
- Permutation / Transposition: We still can see patterns (frequencies), but the text is scrambled.



- Solitaire: Random unpredictable pattern



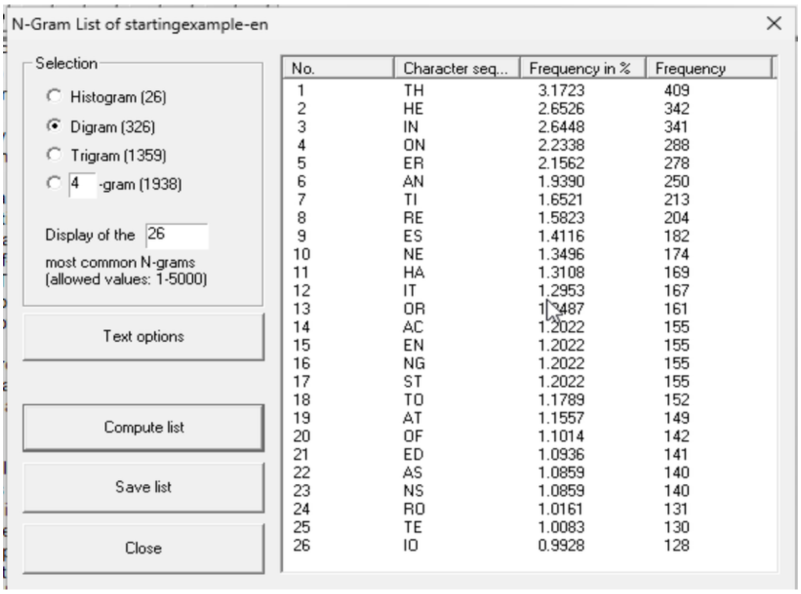
- Scytale / Rail-Fence: The text is the same so same histogram



2.6 <u>Compare n-grams (bi/tri/n) of plaintexts for selected 3 different languages (English, Polish, German, French, Italian, Spanish, ...):</u>

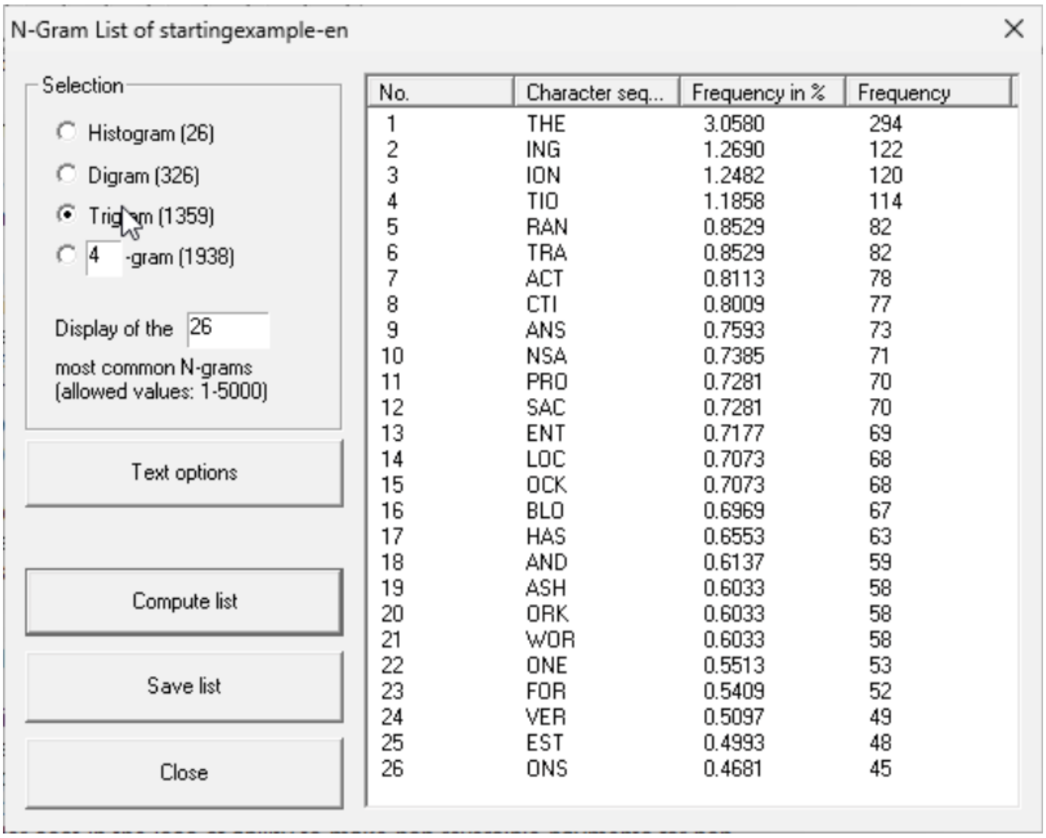
English:

- Digram:



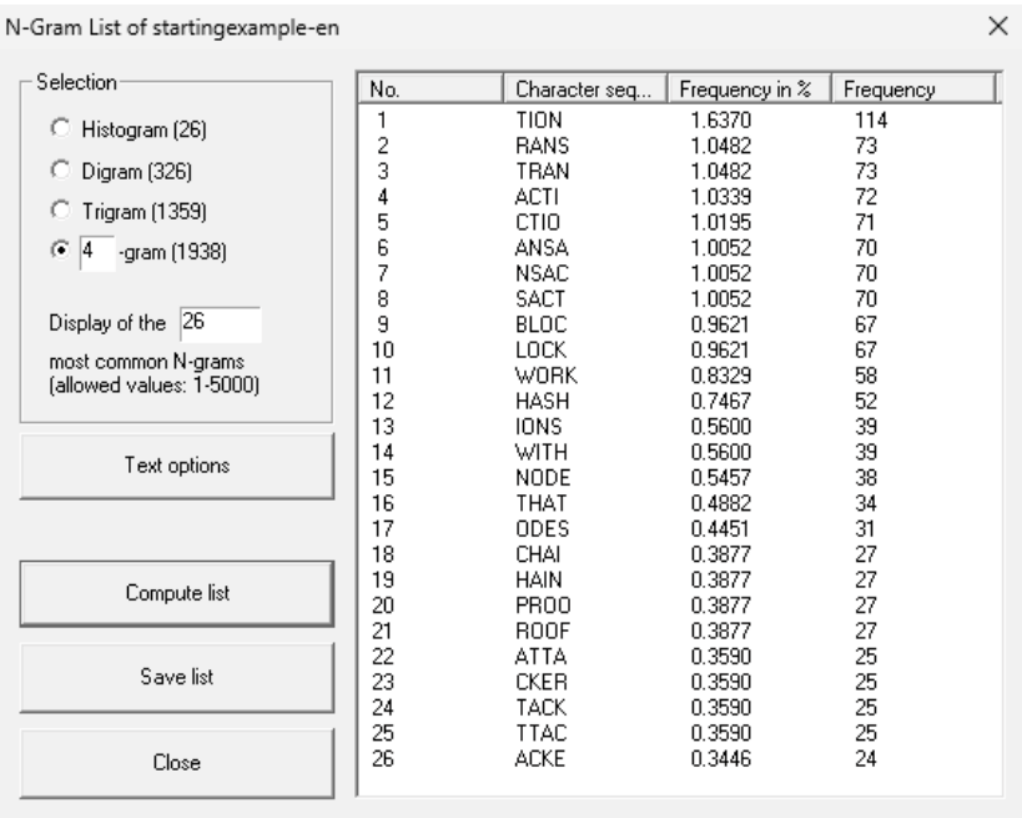
No.	Character seq...	Frequency in %	Frequency
1	TH	3.1723	409
2	HE	2.6526	342
3	IN	2.6448	341
4	ON	2.2338	288
5	ER	2.1562	278
6	AN	1.9390	250
7	TI	1.6521	213
8	RE	1.5823	204
9	ES	1.4116	182
10	NE	1.3496	174
11	HA	1.3108	169
12	IT	1.2953	167
13	OR	1[illegible]487	161
14	AC	1.2022	155
15	EN	1.2022	155
16	NG	1.2022	155
17	ST	1.2022	155
18	TO	1.1789	152
19	AT	1.1557	149
20	OF	1.1014	142
21	ED	1.0936	141
22	AS	1.0859	140
23	NS	1.0859	140
24	RO	1.0161	131
25	TE	1.0083	130
26	IO	0.9928	128

- Trigram:



No.	Character seq...	Frequency in %	Frequency
1	THE	3.0580	294
2	ING	1.2690	122
3	ION	1.2482	120
4	TIO	1.1858	114
5	RAN	0.8529	82
6	TRA	0.8529	82
7	ACT	0.8113	78
8	CTI	0.8009	77
9	ANS	0.7593	73
10	NSA	0.7385	71
11	PRO	0.7281	70
12	SAC	0.7281	70
13	ENT	0.7177	69
14	LOC	0.7073	68
15	OCK	0.7073	68
16	BLO	0.6969	67
17	HAS	0.6553	63
18	AND	0.6137	59
19	ASH	0.6033	58
20	ORK	0.6033	58
21	WOR	0.6033	58
22	ONE	0.5513	53
23	FOR	0.5409	52
24	VER	0.5097	49
25	EST	0.4993	48
26	ONS	0.4681	45

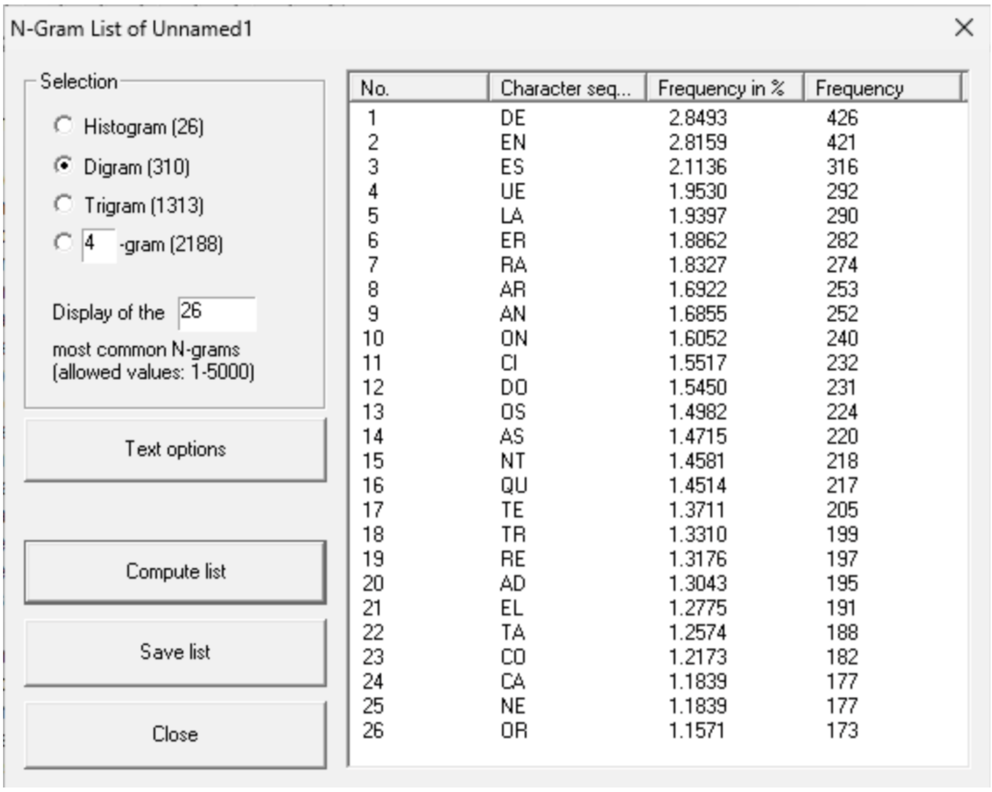
- 4-gram:



No.	Character seq...	Frequency in %	Frequency
1	TION	1.6370	114
2	RANS	1.0482	73
3	TRAN	1.0482	73
4	ACTI	1.0339	72
5	CTIO	1.0195	71
6	ANSA	1.0052	70
7	NSAC	1.0052	70
8	SACT	1.0052	70
9	BLOC	0.9621	67
10	LOCK	0.9621	67
11	WORK	0.8329	58
12	HASH	0.7467	52
13	IONS	0.5600	39
14	WITH	0.5600	39
15	NODE	0.5457	38
16	THAT	0.4882	34
17	ODES	0.4451	31
18	CHAI	0.3877	27
19	HAIN	0.3877	27
20	PROO	0.3877	27
21	ROOF	0.3877	27
22	ATTA	0.3590	25
23	CKER	0.3590	25
24	TACK	0.3590	25
25	TTAC	0.3590	25
26	ACKE	0.3446	24

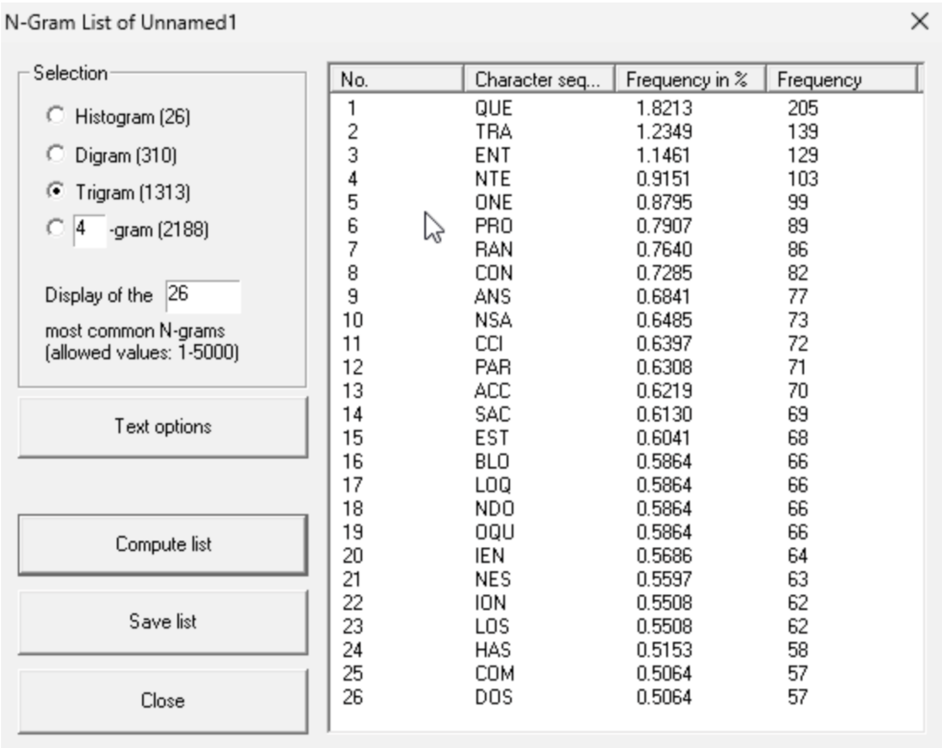
Spanish:

- Digram:



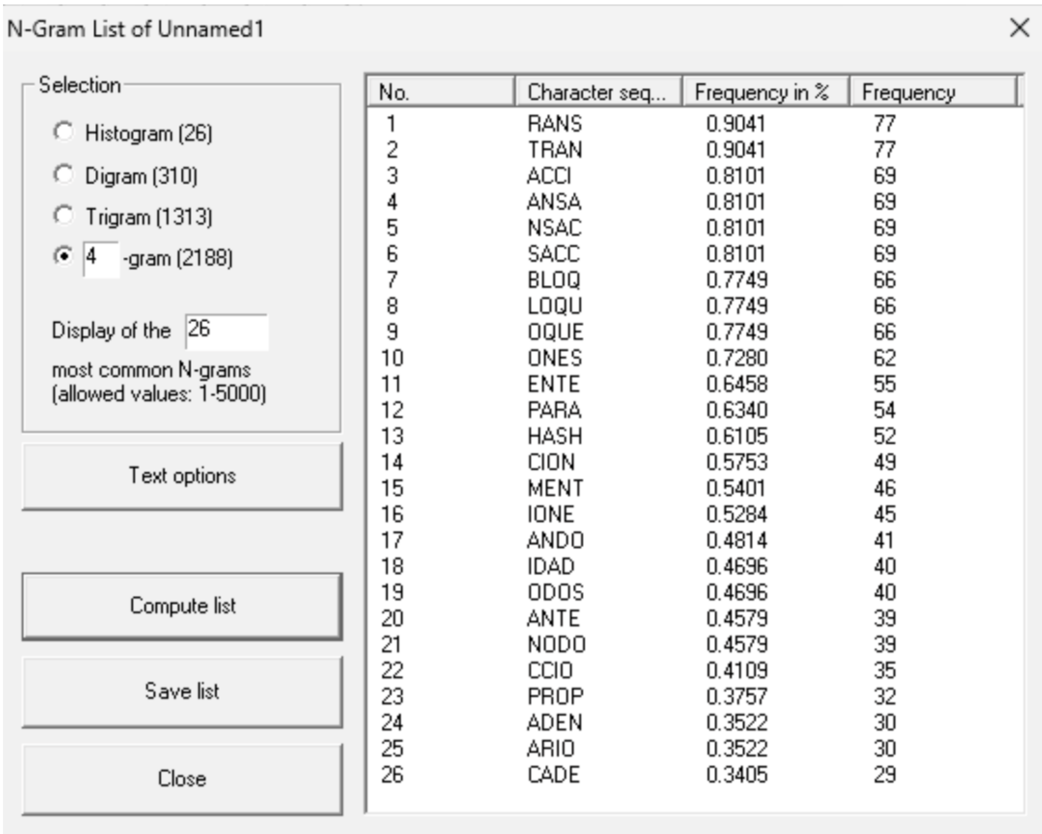
No.	Character seq...	Frequency in %	Frequency
1	DE	2.8493	426
2	EN	2.8159	421
3	ES	2.1136	316
4	UE	1.9530	292
5	LA	1.9397	290
6	ER	1.8862	282
7	RA	1.8327	274
8	AR	1.6922	253
9	AN	1.6855	252
10	ON	1.6052	240
11	CI	1.5517	232
12	DO	1.5450	231
13	OS	1.4982	224
14	AS	1.4715	220
15	NT	1.4581	218
16	QU	1.4514	217
17	TE	1.3711	205
18	TR	1.3310	199
19	RE	1.3176	197
20	AD	1.3043	195
21	EL	1.2775	191
22	TA	1.2574	188
23	CO	1.2173	182
24	CA	1.1839	177
25	NE	1.1839	177
26	OR	1.1571	173

- Trigram:



No.	Character seq...	Frequency in %	Frequency
1	QUE	1.8213	205
2	TRA	1.2349	139
3	ENT	1.1461	129
4	NTE	0.9151	103
5	ONE	0.8795	99
6	PRO	0.7907	89
7	RAN	0.7640	86
8	CON	0.7285	82
9	ANS	0.6841	77
10	NSA	0.6485	73
11	CCI	0.6397	72
12	PAR	0.6308	71
13	ACC	0.6219	70
14	SAC	0.6130	69
15	EST	0.6041	68
16	BLO	0.5864	66
17	LOQ	0.5864	66
18	NDO	0.5864	66
19	OQU	0.5864	66
20	IEN	0.5686	64
21	NES	0.5597	63
22	ION	0.5508	62
23	LOS	0.5508	62
24	HAS	0.5153	58
25	COM	0.5064	57
26	DOS	0.5064	57

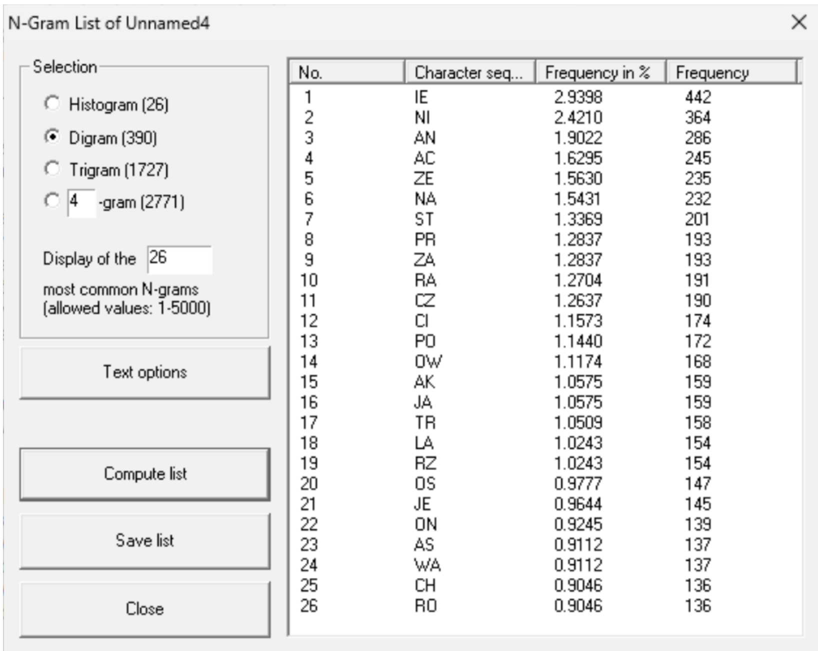
- 4-gram:



No.	Character seq...	Frequency in %	Frequency
1	RANS	0.9041	77
2	TRAN	0.9041	77
3	ACCI	0.8101	69
4	ANSA	0.8101	69
5	NSAC	0.8101	69
6	SACC	0.8101	69
7	BLOQ	0.7749	66
8	LOQU	0.7749	66
9	OQUE	0.7749	66
10	ONES	0.7280	62
11	ENTE	0.6458	55
12	PARA	0.6340	54
13	HASH	0.6105	52
14	CION	0.5753	49
15	MENT	0.5401	46
16	IONE	0.5284	45
17	ANDO	0.4814	41
18	IDAD	0.4696	40
19	ODOS	0.4696	40
20	ANTE	0.4579	39
21	NODO	0.4579	39
22	CCIO	0.4109	35
23	PROP	0.3757	32
24	ADEN	0.3522	30
25	ARIO	0.3522	30
26	CADE	0.3405	29

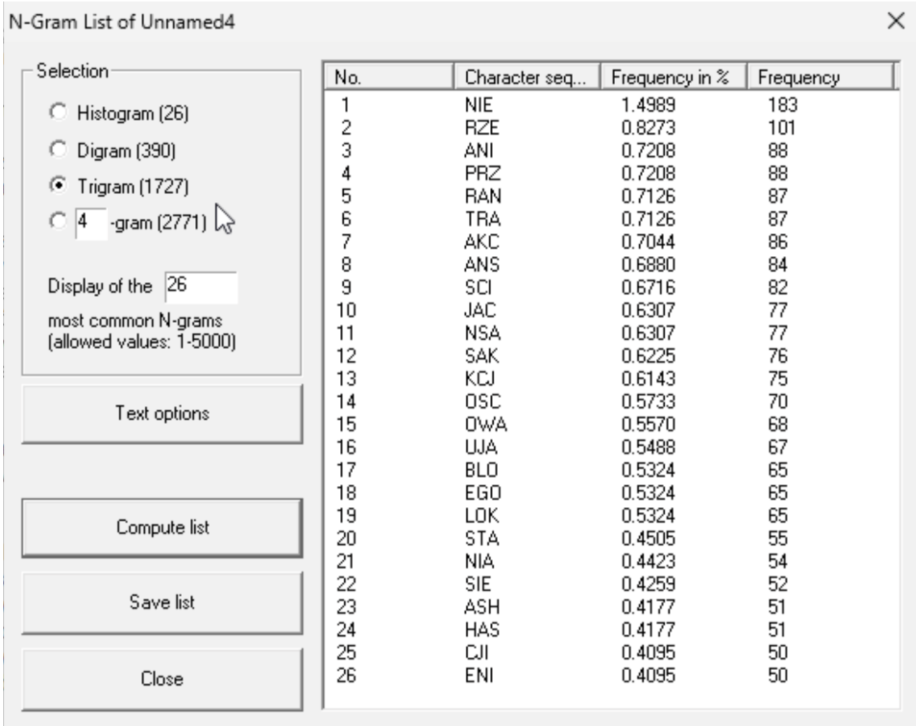
Polish:

- Digram:



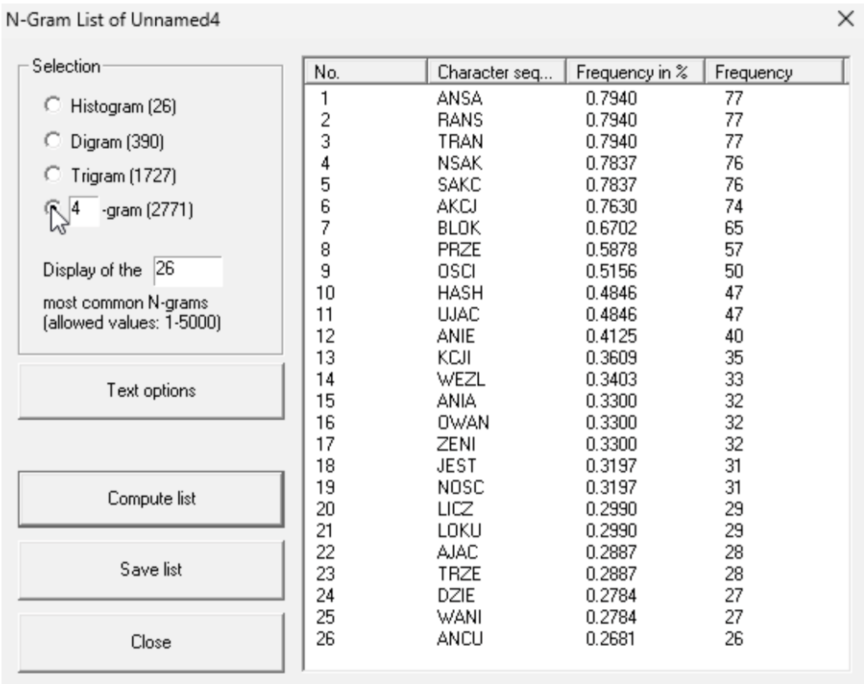
No.	Character seq...	Frequency in %	Frequency
1	IE	2.9398	442
2	NI	2.4210	364
3	AN	1.9022	286
4	AC	1.6295	245
5	ZE	1.5630	235
6	NA	1.5431	232
7	ST	1.3369	201
8	PR	1.2837	193
9	ZA	1.2837	193
10	RA	1.2704	191
11	CZ	1.2637	190
12	CI	1.1573	174
13	PO	1.1440	172
14	OW	1.1174	168
15	AK	1.0575	159
16	JA	1.0575	159
17	TR	1.0509	158
18	LA	1.0243	154
19	RZ	1.0243	154
20	OS	0.9777	147
21	JE	0.9644	145
22	ON	0.9245	139
23	AS	0.9112	137
24	WA	0.9112	137
25	CH	0.9046	136
26	RO	0.9046	136

- Trigram:



No.	Character seq...	Frequency in %	Frequency
1	NIE	1.4989	183
2	RZE	0.8273	101
3	ANI	0.7208	88
4	PRZ	0.7208	88
5	RAN	0.7126	87
6	TRA	0.7126	87
7	AKC	0.7044	86
8	ANS	0.6880	84
9	SCI	0.6716	82
10	JAC	0.6307	77
11	NSA	0.6307	77
12	SAK	0.6225	76
13	KCJ	0.6143	75
14	OSC	0.5733	70
15	OWA	0.5570	68
16	UJA	0.5488	67
17	BLO	0.5324	65
18	EGO	0.5324	65
19	LOK	0.5324	65
20	STA	0.4505	55
21	NIA	0.4423	54
22	SIE	0.4259	52
23	ASH	0.4177	51
24	HAS	0.4177	51
25	CJI	0.4095	50
26	ENI	0.4095	50

- 4-gram:



No.	Character seq...	Frequency in %	Frequency
1	ANSA	0.7940	77
2	RANS	0.7940	77
3	TRAN	0.7940	77
4	NSAK	0.7837	76
5	SAKC	0.7837	76
6	AKCJ	0.7630	74
7	BLOK	0.6702	65
8	PRZE	0.5878	57
9	OSCI	0.5156	50
10	HASH	0.4846	47
11	UJAC	0.4846	47
12	ANIE	0.4125	40
13	KCJI	0.3609	35
14	WEZL	0.3403	33
15	ANIA	0.3300	32
16	OWAN	0.3300	32
17	ZENI	0.3300	32
18	JEST	0.3197	31
19	NOSC	0.3197	31
20	LICZ	0.2990	29
21	LOKU	0.2990	29
22	AJAC	0.2887	28
23	TRZE	0.2887	28
24	DZIE	0.2784	27
25	WANI	0.2784	27
26	ANCU	0.2681	26

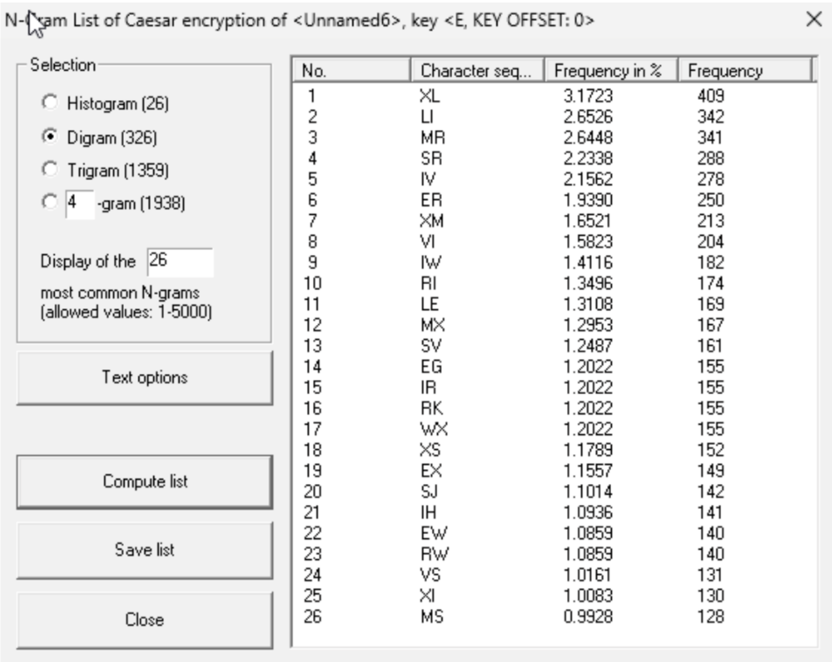
With the n-gram analysis se can get secarral conclusiones. English n-grams show frequent articles and endings as TH, HE, IN, THE,... . Spanish show several combinations as DE, EN, ES, and QUE that shows that Spanish uses a lot of vowels. Finally, Polish reflects the complex of the language, as we can se several combinations of consonantes.

2.7 Compare n-games (bi/tri/n) of plaintext and ciphertext depending on the algorithm. For algorithms such as in point 2:

Caesar: Key: E

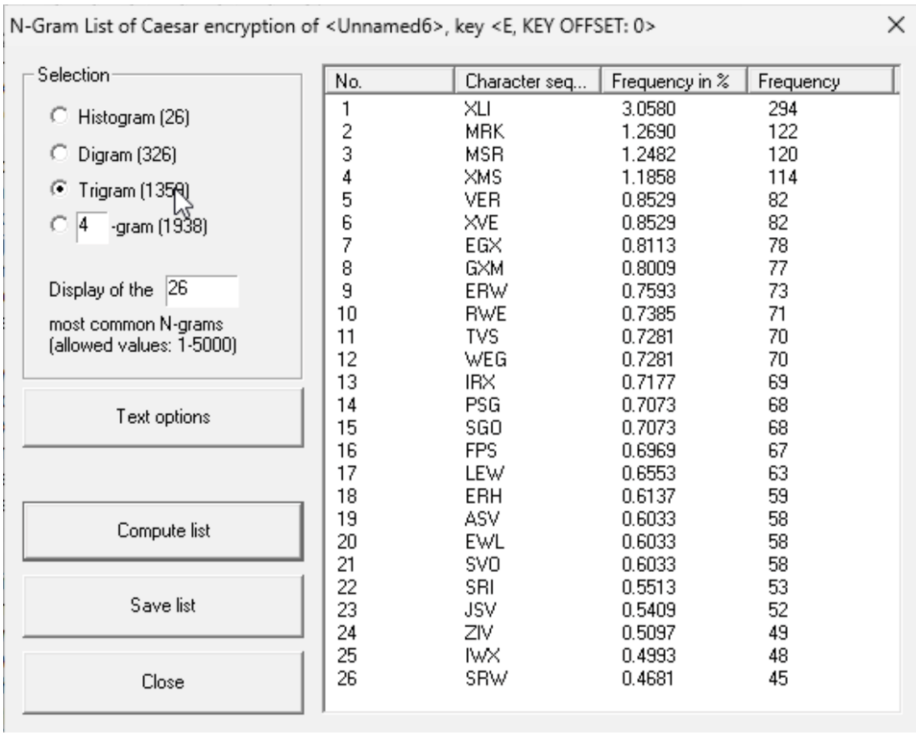
- English:

- Digram:



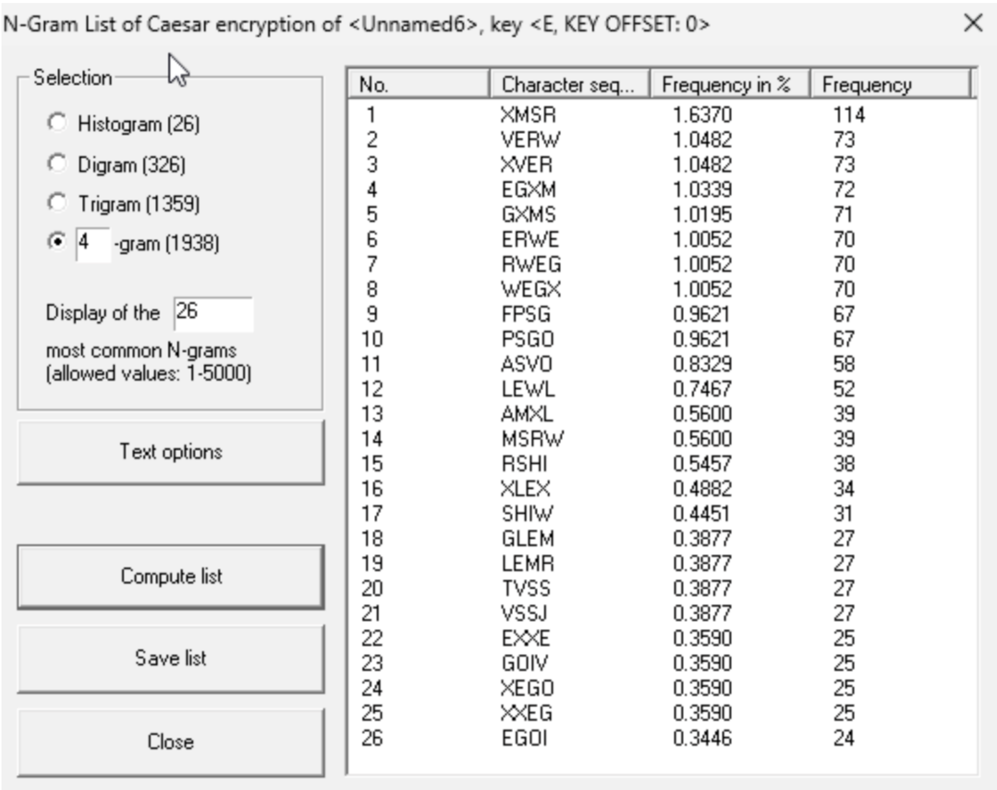
No.	Character seq...	Frequency in %	Frequency
1	XL	3.1723	409
2	LI	2.6526	342
3	MR	2.6448	341
4	SR	2.2338	288
5	IV	2.1562	278
6	ER	1.9390	250
7	XM	1.6521	213
8	VI	1.5823	204
9	IW	1.4116	182
10	RI	1.3496	174
11	LE	1.3108	169
12	MX	1.2953	167
13	SV	1.2487	161
14	EG	1.2022	155
15	IR	1.2022	155
16	RK	1.2022	155
17	WX	1.2022	155
18	XS	1.1789	152
19	EX	1.1557	149
20	SJ	1.1014	142
21	IH	1.0936	141
22	EW	1.0859	140
23	RW	1.0859	140
24	VS	1.0161	131
25	XI	1.0083	130
26	MS	0.9928	128

- Trigram:



No.	Character seq...	Frequency in %	Frequency
1	XLI	3.0580	294
2	MRK	1.2690	122
3	MSR	1.2482	120
4	XMS	1.1858	114
5	VER	0.8529	82
6	XVE	0.8529	82
7	EGX	0.8113	78
8	GXM	0.8009	77
9	ERW	0.7593	73
10	RWE	0.7385	71
11	TVS	0.7281	70
12	WEG	0.7281	70
13	IRX	0.7177	69
14	PSG	0.7073	68
15	SGO	0.7073	68
16	FPS	0.6969	67
17	LEW	0.6553	63
18	ERH	0.6137	59
19	ASV	0.6033	58
20	EWL	0.6033	58
21	SVO	0.6033	58
22	SRI	0.5513	53
23	JSV	0.5409	52
24	ZIV	0.5097	49
25	IWX	0.4993	48
26	SRW	0.4681	45

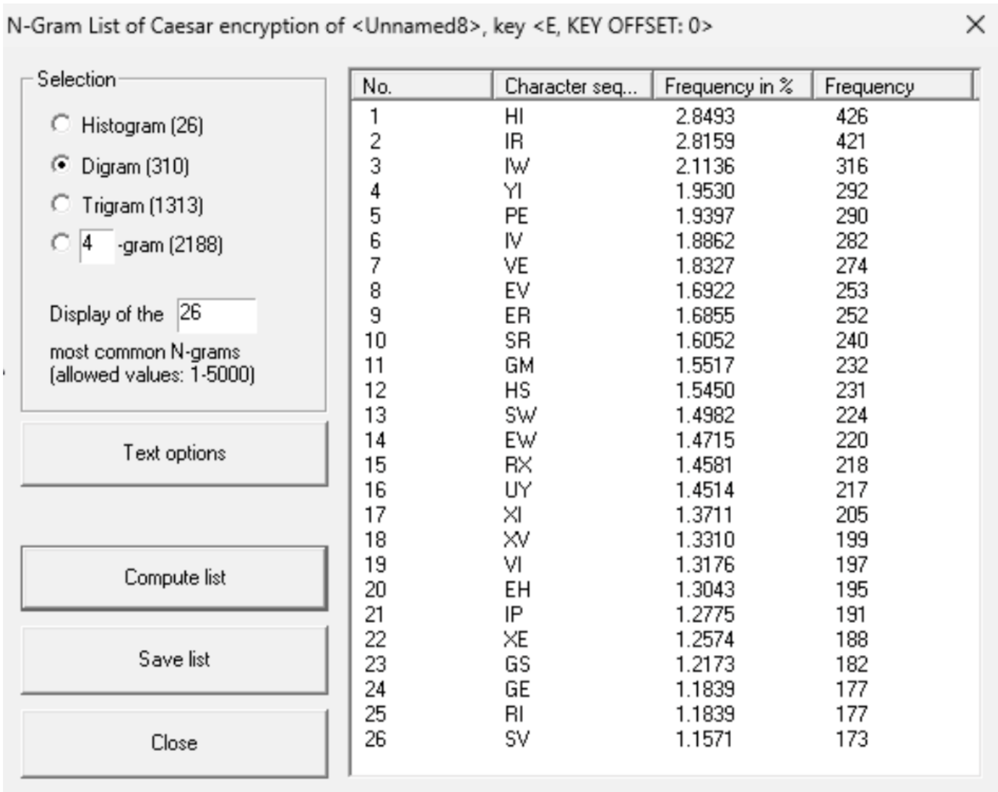
- 4-gram:



No.	Character seq...	Frequency in %	Frequency
1	XMSR	1.6370	114
2	VERW	1.0482	73
3	XVER	1.0482	73
4	EGXM	1.0339	72
5	GXMS	1.0195	71
6	ERWE	1.0052	70
7	RWEG	1.0052	70
8	WEGX	1.0052	70
9	FPSG	0.9621	67
10	PSGO	0.9621	67
11	ASVO	0.8329	58
12	LEWL	0.7467	52
13	AMXL	0.5600	39
14	MSRW	0.5600	39
15	RSHI	0.5457	38
16	XLEX	0.4882	34
17	SHIW	0.4451	31
18	GLEM	0.3877	27
19	LEMR	0.3877	27
20	TVSS	0.3877	27
21	VSSJ	0.3877	27
22	EXXE	0.3590	25
23	GOIV	0.3590	25
24	XEGO	0.3590	25
25	XXEG	0.3590	25
26	EGOI	0.3446	24

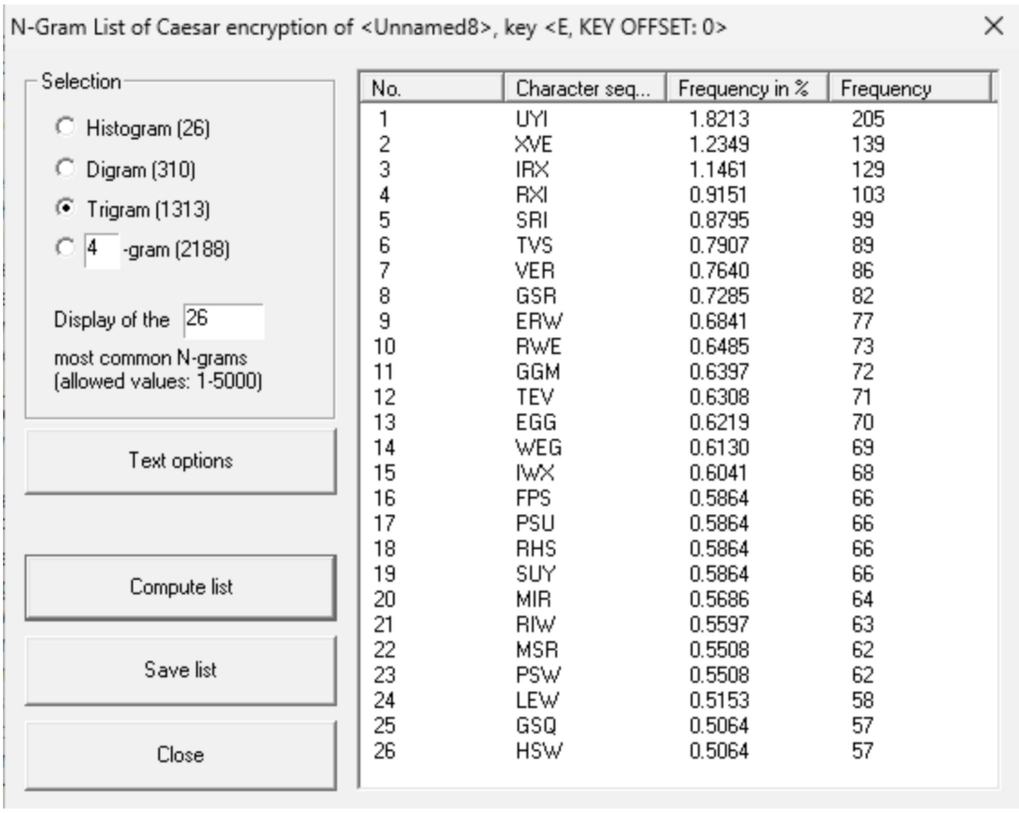
- Spanish:

- Digram:



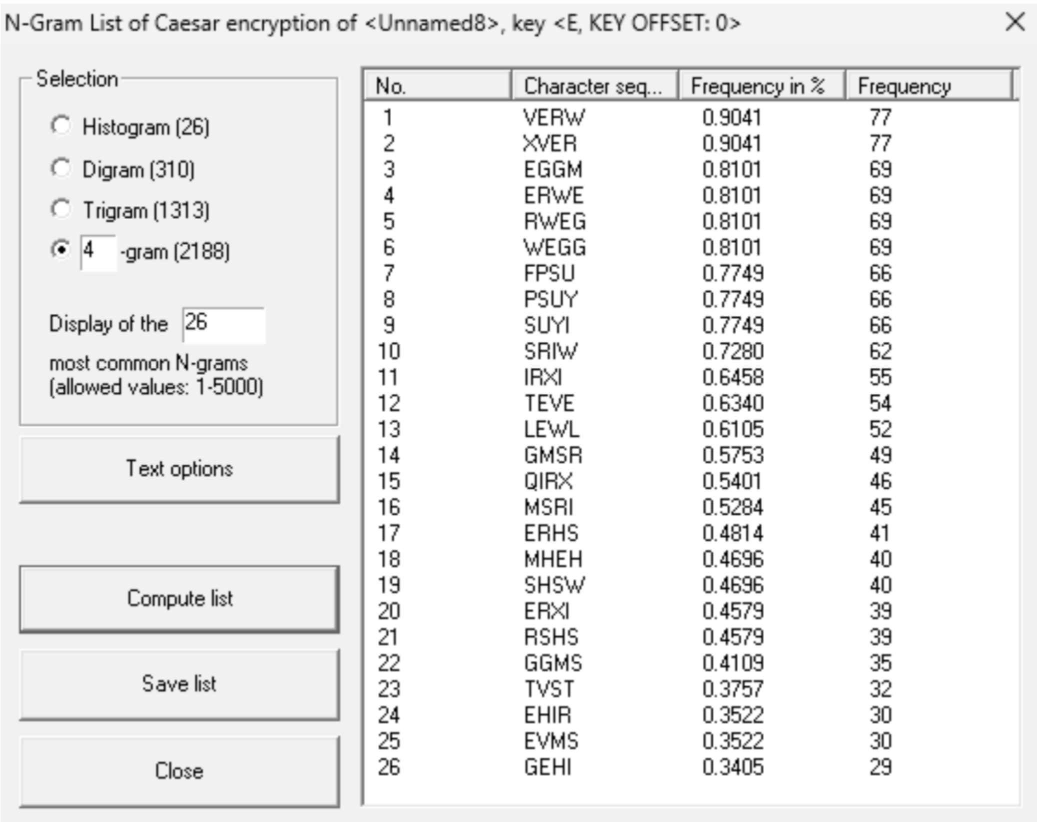
No.	Character seq...	Frequency in %	Frequency
1	HI	2.8493	426
2	IR	2.8159	421
3	IW	2.1136	316
4	YI	1.9530	292
5	PE	1.9397	290
6	IV	1.8862	282
7	VE	1.8327	274
8	EV	1.6922	253
9	ER	1.6855	252
10	SR	1.6052	240
11	GM	1.5517	232
12	HS	1.5450	231
13	SW	1.4982	224
14	EW	1.4715	220
15	RX	1.4581	218
16	UY	1.4514	217
17	XI	1.3711	205
18	XV	1.3310	199
19	VI	1.3176	197
20	EH	1.3043	195
21	IP	1.2775	191
22	XE	1.2574	188
23	GS	1.2173	182
24	GE	1.1839	177
25	RI	1.1839	177
26	SV	1.1571	173

- Trigram:



No.	Character seq...	Frequency in %	Frequency
1	UYI	1.8213	205
2	XVE	1.2349	139
3	IRX	1.1461	129
4	RXI	0.9151	103
5	SRI	0.8795	99
6	TVS	0.7907	89
7	VER	0.7640	86
8	GSR	0.7285	82
9	ERW	0.6841	77
10	RWE	0.6485	73
11	GGM	0.6397	72
12	TEV	0.6308	71
13	EGG	0.6219	70
14	WEG	0.6130	69
15	IWX	0.6041	68
16	FPS	0.5864	66
17	PSU	0.5864	66
18	RHS	0.5864	66
19	SUY	0.5864	66
20	MIR	0.5686	64
21	RIW	0.5597	63
22	MSR	0.5508	62
23	PSW	0.5508	62
24	LEW	0.5153	58
25	GSQ	0.5064	57
26	HSW	0.5064	57

- 4-gram:



No.	Character seq...	Frequency in %	Frequency
1	VERW	0.9041	77
2	XVER	0.9041	77
3	EGGM	0.8101	69
4	ERWE	0.8101	69
5	RWEG	0.8101	69
6	WEGG	0.8101	69
7	FPSU	0.7749	66
8	PSUY	0.7749	66
9	SUYI	0.7749	66
10	SRIW	0.7280	62
11	IRXI	0.6458	55
12	TEVE	0.6340	54
13	LEWL	0.6105	52
14	GMSR	0.5753	49
15	QIRX	0.5401	46
16	MSRI	0.5284	45
17	ERHS	0.4814	41
18	MHEH	0.4696	40
19	SHSW	0.4696	40
20	ERXI	0.4579	39
21	RSHS	0.4579	39
22	GGMS	0.4109	35
23	TVST	0.3757	32
24	EHIR	0.3522	30
25	EVMS	0.3522	30
26	GEHI	0.3405	29

- Polish:

- Digram:

N-Gram List of Caesar encryption of <Unnamed9>, key <E, KEY OFFSET: 0>

Selection
- Histogram (26)
- Digram (390)
- Trigram (1727)
- 4 -gram (2771)

Display of the 26 most common N-grams (allowed values: 1-5000)

Text options | Compute list | Save list | Close

No.	Character seq...	Frequency in %	Frequency
1	MI	2.9398	442
2	RM	2.4210	364
3	ER	1.9022	286
4	EG	1.6295	245
5	DI	1.5630	235
6	RE	1.5431	232
7	WX	1.3369	201
8	DE	1.2837	193
9	TV	1.2837	193
10	VE	1.2704	191
11	GD	1.2637	190
12	GM	1.1573	174
13	TS	1.1440	172
14	SA	1.1174	168
15	EO	1.0575	159
16	NE	1.0575	159
17	XV	1.0509	158
18	PE	1.0243	154
19	VD	1.0243	154
20	SW	0.9777	147
21	NI	0.9644	145
22	SR	0.9245	139
23	AE	0.9112	137
24	EW	0.9112	137
25	GL	0.9046	136
26	VS	0.9046	136

- Trigram:

N-Gram List of Caesar encryption of <Unnamed9>, key <E, KEY OFFSET: 0>

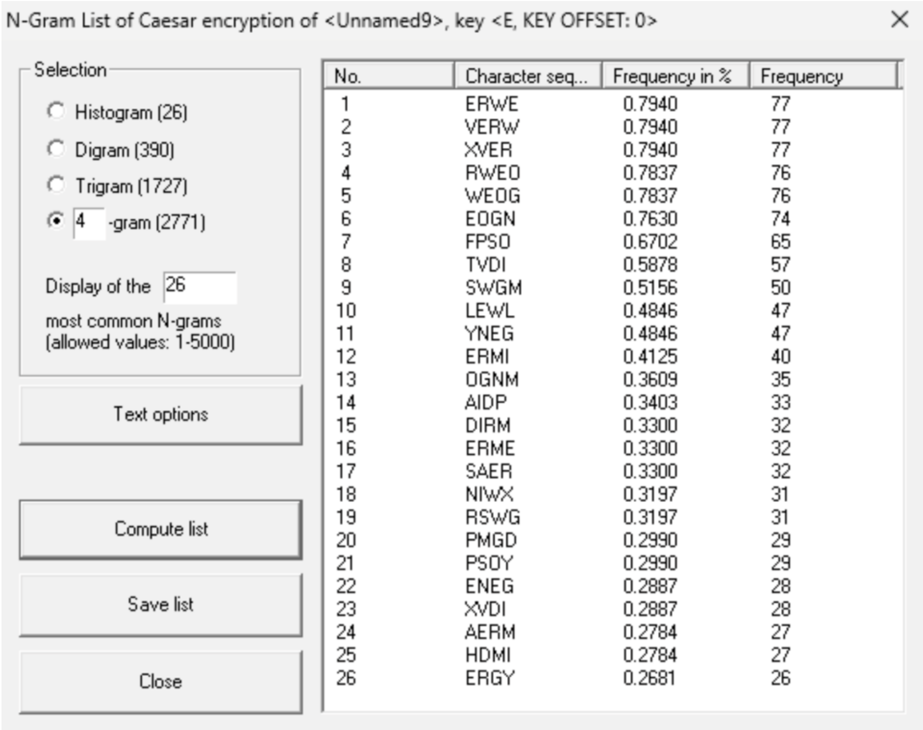
Selection
- Histogram (26)
- Digram (390)
- Trigram (1727)
- 4 -gram (2771)

Display of the 26 most common N-grams (allowed values: 1-5000)

Text options | Compute list | Save list | Close

No.	Character seq...	Frequency in %	Frequency
1	RMI	1.4989	183
2	VDI	0.8273	101
3	ERM	0.7208	88
4	TVD	0.7208	88
5	VER	0.7126	87
6	XVE	0.7126	87
7	EOG	0.7044	86
8	ERW	0.6880	84
9	WGM	0.6716	82
10	NEG	0.6307	77
11	RWE	0.6307	77
12	WEO	0.6225	76
13	OGN	0.6143	75
14	SWG	0.5733	70
15	SAE	0.5570	68
16	YNE	0.5488	67
17	FPS	0.5324	65
18	IKS	0.5324	65
19	PSO	0.5324	65
20	WXE	0.4505	55
21	RME	0.4423	54
22	WMI	0.4259	52
23	EWL	0.4177	51
24	LEW	0.4177	51
25	AER	0.4095	50
26	GNM	0.4095	50

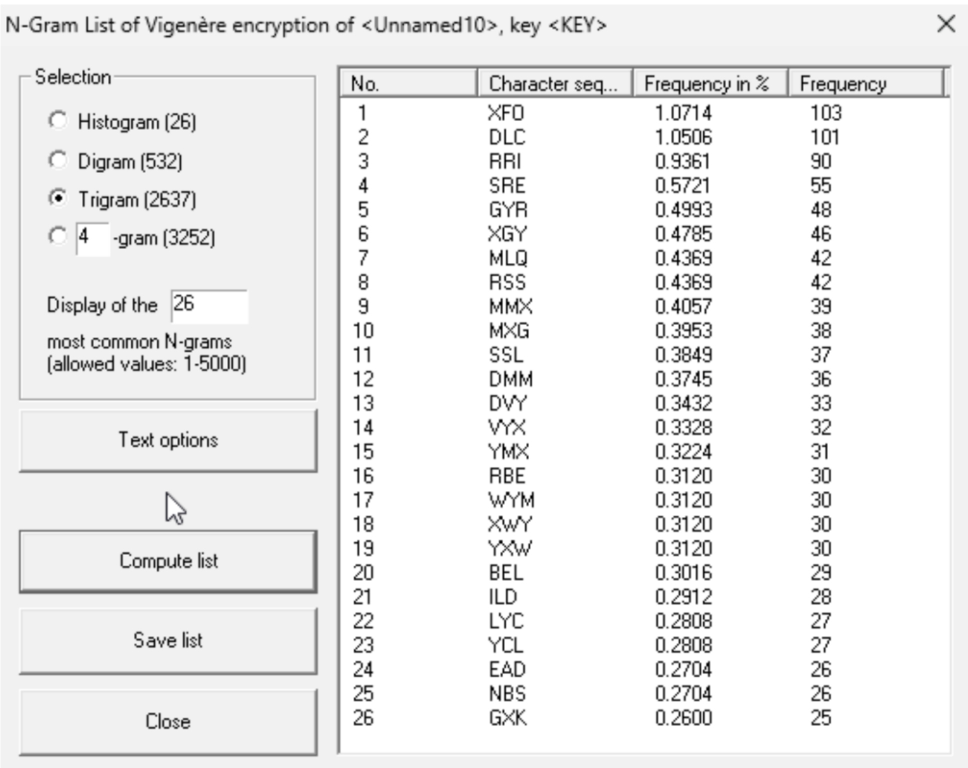
- 4-gram:



No.	Character seq...	Frequency in %	Frequency
1	ERWE	0.7940	77
2	VERW	0.7940	77
3	XVER	0.7940	77
4	RWEO	0.7837	76
5	WEOG	0.7837	76
6	EOGN	0.7630	74
7	FPSO	0.6702	65
8	TVDI	0.5878	57
9	SWGM	0.5156	50
10	LEWL	0.4846	47
11	YNEG	0.4846	47
12	ERMI	0.4125	40
13	OGNM	0.3609	35
14	AIDP	0.3403	33
15	DIRM	0.3300	32
16	ERME	0.3300	32
17	SAER	0.3300	32
18	NIWX	0.3197	31
19	RSWG	0.3197	31
20	PMGD	0.2990	29
21	PSOY	0.2990	29
22	ENEG	0.2887	28
23	XVDI	0.2887	28
24	AERM	0.2784	27
25	HDMI	0.2784	27
26	ERGY	0.2681	26

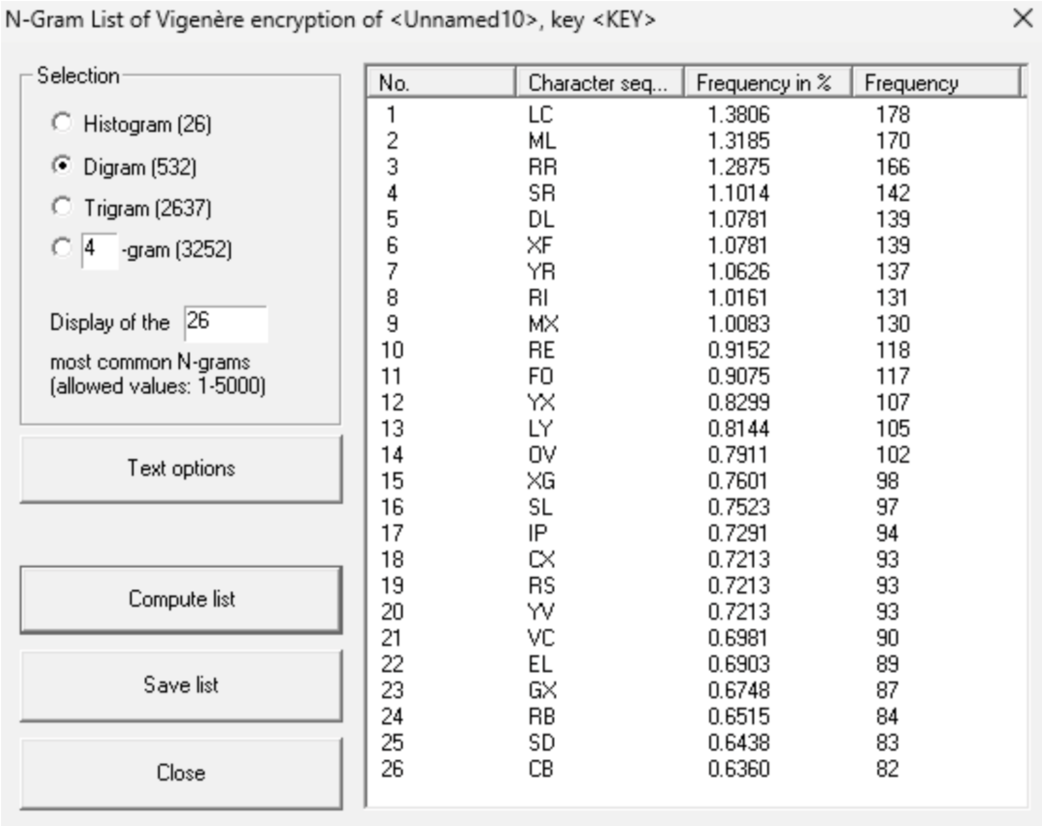
Vigenère: Key:"KEY"

- English:
- Digram:



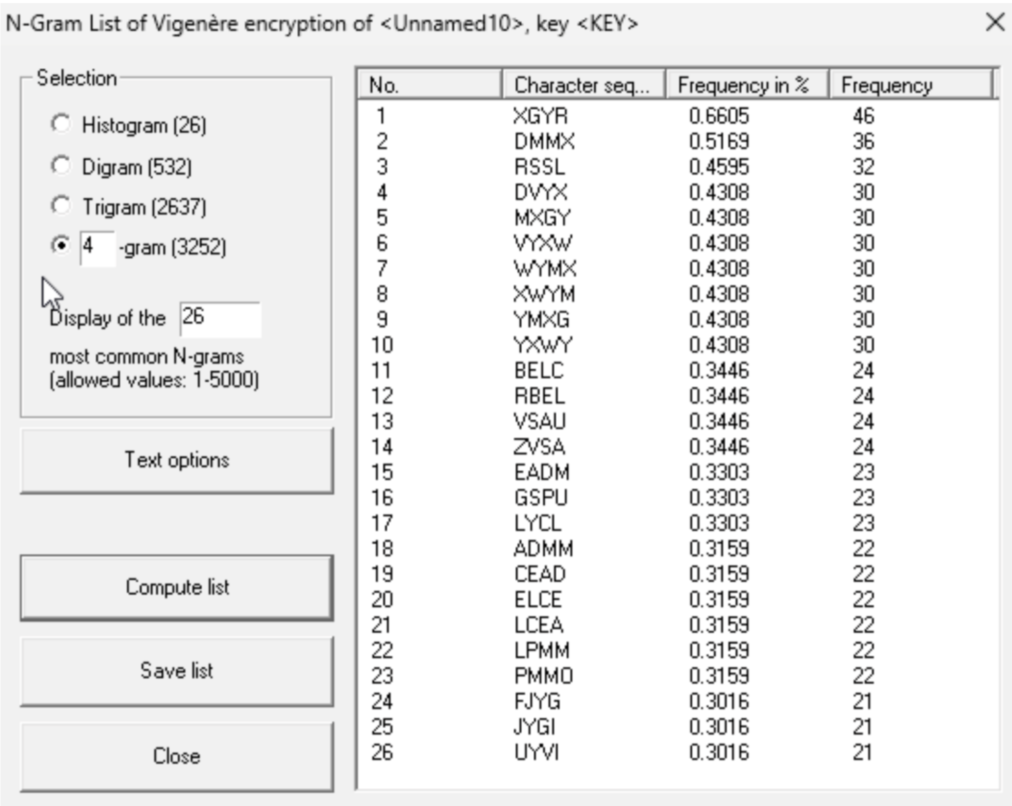
No.	Character seq...	Frequency in %	Frequency
1	XFO	1.0714	103
2	DLC	1.0506	101
3	RRI	0.9361	90
4	SRE	0.5721	55
5	GYR	0.4993	48
6	XGY	0.4785	46
7	MLQ	0.4369	42
8	RSS	0.4369	42
9	MMX	0.4057	39
10	MXG	0.3953	38
11	SSL	0.3849	37
12	DMM	0.3745	36
13	DVY	0.3432	33
14	VYX	0.3328	32
15	YMX	0.3224	31
16	RBE	0.3120	30
17	WYM	0.3120	30
18	XWY	0.3120	30
19	YXW	0.3120	30
20	BEL	0.3016	29
21	ILD	0.2912	28
22	LYC	0.2808	27
23	YCL	0.2808	27
24	EAD	0.2704	26
25	NBS	0.2704	26
26	GXK	0.2600	25

- Trigram:



No.	Character seq...	Frequency in %	Frequency
1	LC	1.3806	178
2	ML	1.3185	170
3	RR	1.2875	166
4	SR	1.1014	142
5	DL	1.0781	139
6	XF	1.0781	139
7	YR	1.0626	137
8	RI	1.0161	131
9	MX	1.0083	130
10	RE	0.9152	118
11	FO	0.9075	117
12	YX	0.8299	107
13	LY	0.8144	105
14	OV	0.7911	102
15	XG	0.7601	98
16	SL	0.7523	97
17	IP	0.7291	94
18	CX	0.7213	93
19	RS	0.7213	93
20	YV	0.7213	93
21	VC	0.6981	90
22	EL	0.6903	89
23	GX	0.6748	87
24	RB	0.6515	84
25	SD	0.6438	83
26	CB	0.6360	82

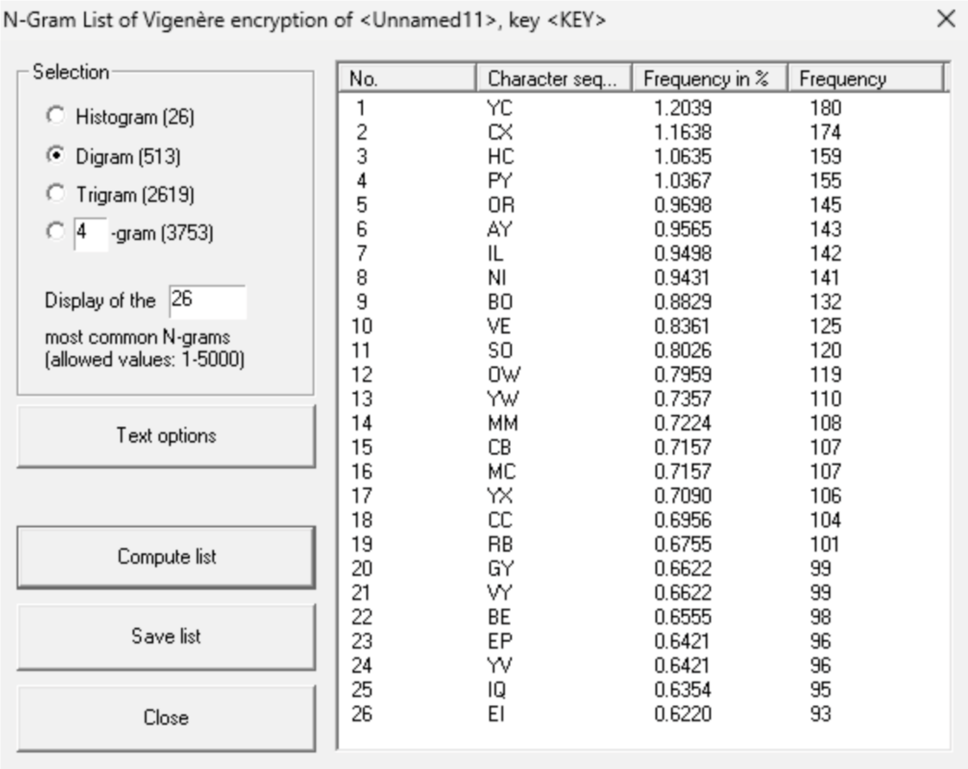
- 4-gram:



No.	Character seq...	Frequency in %	Frequency
1	XGYR	0.6605	46
2	DMMX	0.5169	36
3	RSSL	0.4595	32
4	DVYX	0.4308	30
5	MXGY	0.4308	30
6	VYXW	0.4308	30
7	WYMX	0.4308	30
8	XWYM	0.4308	30
9	YMXG	0.4308	30
10	YXWY	0.4308	30
11	BELC	0.3446	24
12	RBEL	0.3446	24
13	VSAU	0.3446	24
14	ZVSA	0.3446	24
15	EADM	0.3303	23
16	GSPU	0.3303	23
17	LYCL	0.3303	23
18	ADMM	0.3159	22
19	CEAD	0.3159	22
20	ELCE	0.3159	22
21	LCEA	0.3159	22
22	LPMM	0.3159	22
23	PMMO	0.3159	22
24	FJYG	0.3016	21
25	JYGI	0.3016	21
26	UYVI	0.3016	21

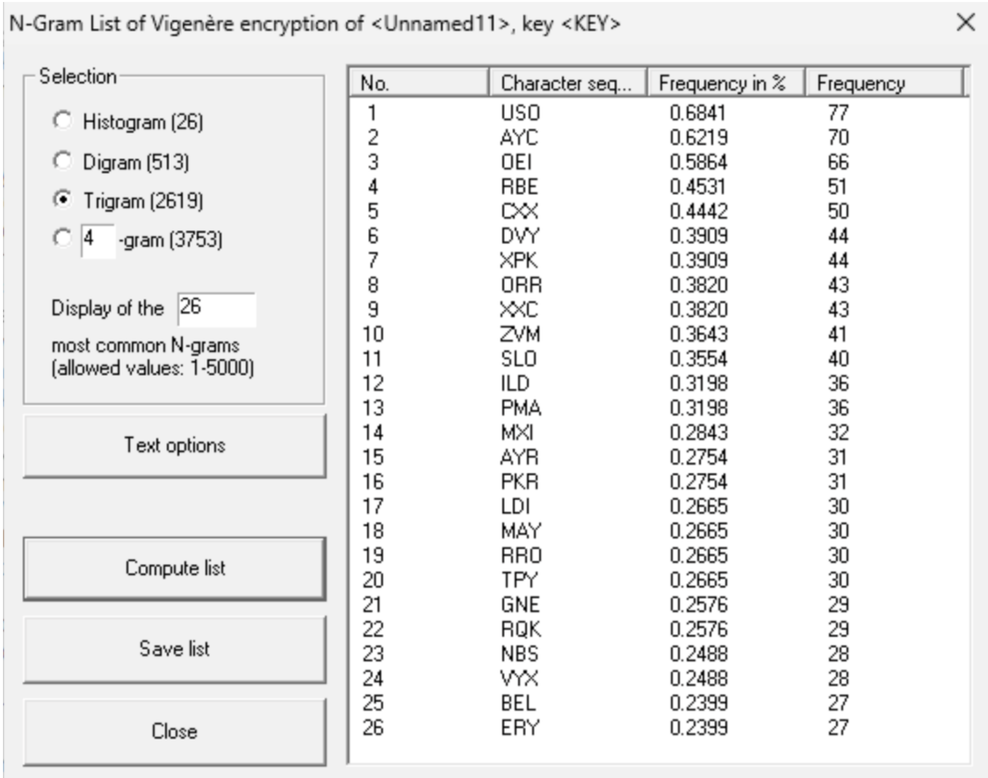
- Spanish:

- Digram:



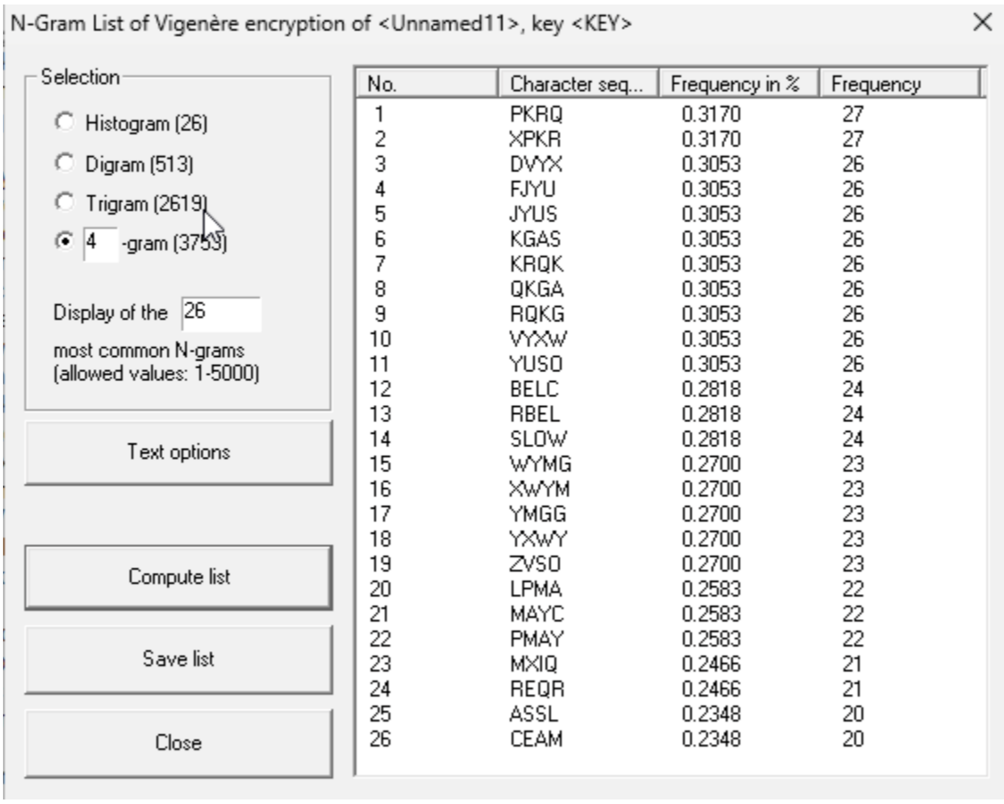
No.	Character seq...	Frequency in %	Frequency
1	YC	1.2039	180
2	CX	1.1638	174
3	HC	1.0635	159
4	PY	1.0367	155
5	OR	0.9698	145
6	AY	0.9565	143
7	IL	0.9498	142
8	NI	0.9431	141
9	BO	0.8829	132
10	VE	0.8361	125
11	SO	0.8026	120
12	OW	0.7959	119
13	YW	0.7357	110
14	MM	0.7224	108
15	CB	0.7157	107
16	MC	0.7157	107
17	YX	0.7090	106
18	CC	0.6956	104
19	RB	0.6755	101
20	GY	0.6622	99
21	VY	0.6622	99
22	BE	0.6555	98
23	EP	0.6421	96
24	YV	0.6421	96
25	IQ	0.6354	95
26	EI	0.6220	93

- Trigram:



No.	Character seq...	Frequency in %	Frequency
1	USO	0.6841	77
2	AYC	0.6219	70
3	OEI	0.5864	66
4	RBE	0.4531	51
5	CXX	0.4442	50
6	DVY	0.3909	44
7	XPK	0.3909	44
8	ORR	0.3820	43
9	XXC	0.3820	43
10	ZVM	0.3643	41
11	SLO	0.3554	40
12	ILD	0.3198	36
13	PMA	0.3198	36
14	MXI	0.2843	32
15	AYR	0.2754	31
16	PKR	0.2754	31
17	LDI	0.2665	30
18	MAY	0.2665	30
19	RRO	0.2665	30
20	TPY	0.2665	30
21	GNE	0.2576	29
22	RQK	0.2576	29
23	NBS	0.2488	28
24	VYX	0.2488	28
25	BEL	0.2399	27
26	ERY	0.2399	27

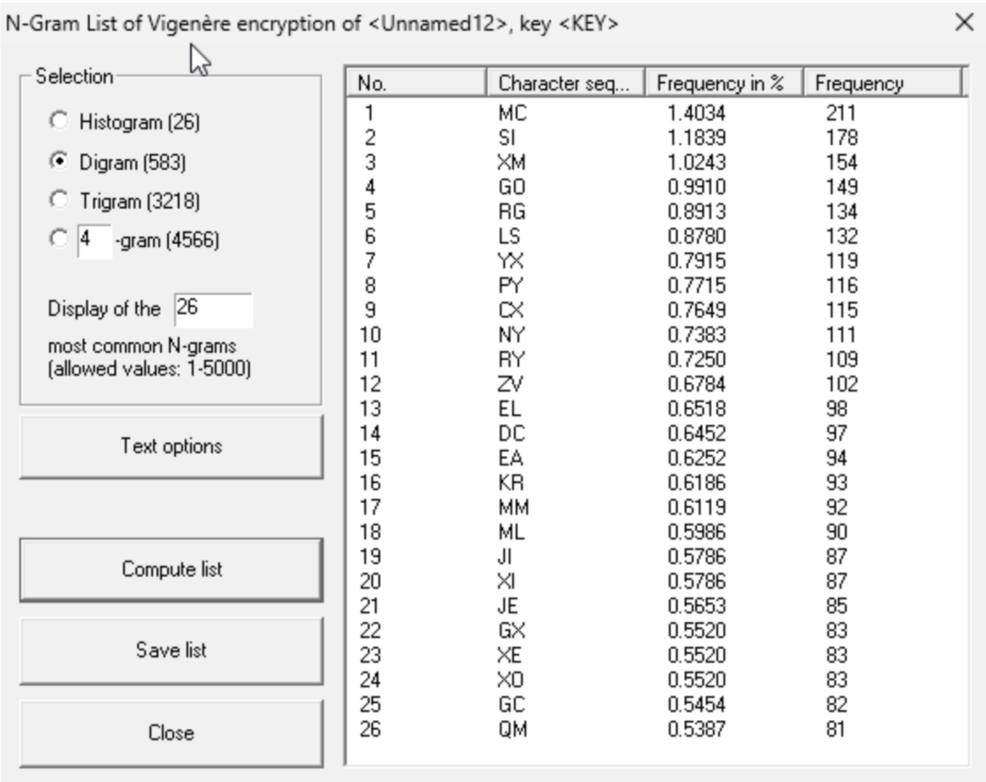
- 4-gram:



No.	Character seq...	Frequency in %	Frequency
1	PKRQ	0.3170	27
2	XPKR	0.3170	27
3	DVYX	0.3053	26
4	FJYU	0.3053	26
5	JYUS	0.3053	26
6	KGAS	0.3053	26
7	KRQK	0.3053	26
8	QKGA	0.3053	26
9	RQKG	0.3053	26
10	VYXW	0.3053	26
11	YUSO	0.3053	26
12	BELC	0.2818	24
13	RBEL	0.2818	24
14	SLOW	0.2818	24
15	WYMG	0.2700	23
16	XWYM	0.2700	23
17	YMGG	0.2700	23
18	YXWY	0.2700	23
19	ZVSO	0.2700	23
20	LPMA	0.2583	22
21	MAYC	0.2583	22
22	PMAY	0.2583	22
23	MXIQ	0.2466	21
24	REQR	0.2466	21
25	ASSL	0.2348	20
26	CEAM	0.2348	20

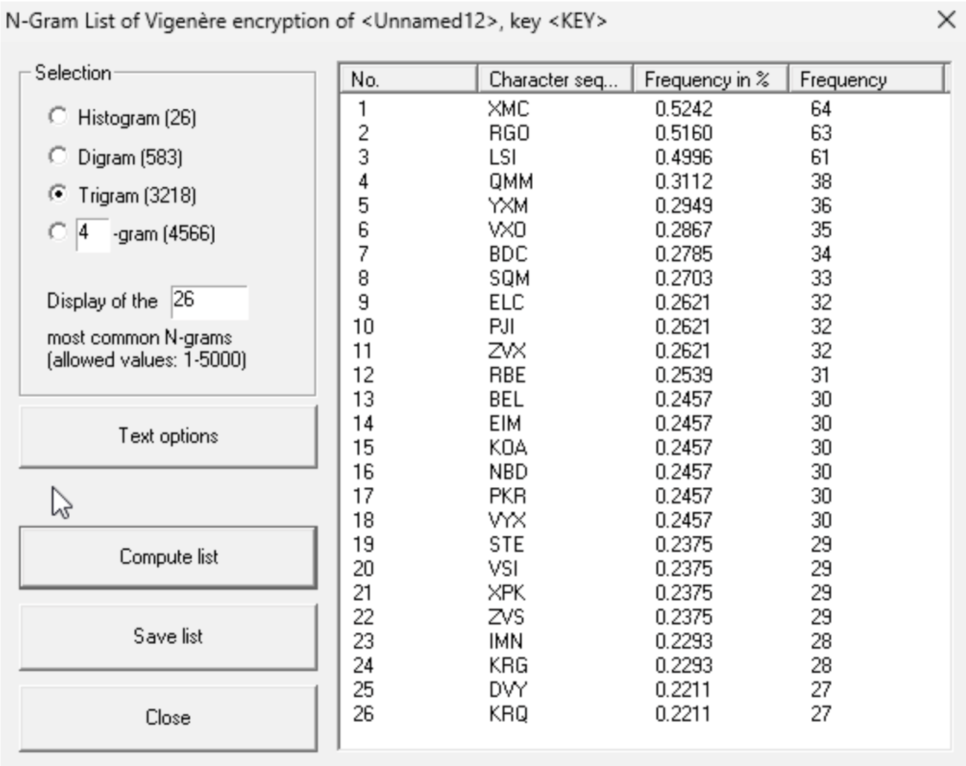
- Polish:

- Digram:



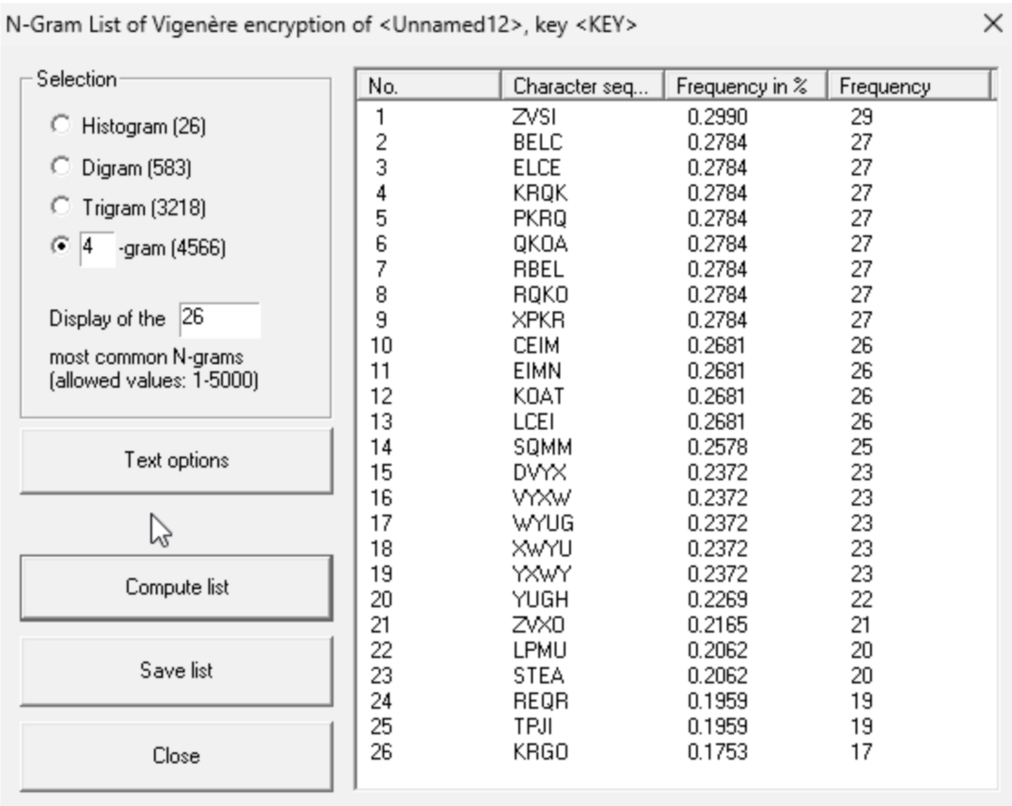
No.	Character seq...	Frequency in %	Frequency
1	MC	1.4034	211
2	SI	1.1839	178
3	XM	1.0243	154
4	GO	0.9910	149
5	RG	0.8913	134
6	LS	0.8780	132
7	YX	0.7915	119
8	PY	0.7715	116
9	CX	0.7649	115
10	NY	0.7383	111
11	RY	0.7250	109
12	ZV	0.6784	102
13	EL	0.6518	98
14	DC	0.6452	97
15	EA	0.6252	94
16	KR	0.6186	93
17	MM	0.6119	92
18	ML	0.5986	90
19	JI	0.5786	87
20	XI	0.5786	87
21	JE	0.5653	85
22	GX	0.5520	83
23	XE	0.5520	83
24	XO	0.5520	83
25	GC	0.5454	82
26	QM	0.5387	81

- Trigram:



No.	Character seq...	Frequency in %	Frequency
1	XMC	0.5242	64
2	RGO	0.5160	63
3	LSI	0.4996	61
4	QMM	0.3112	38
5	YXM	0.2949	36
6	VXO	0.2867	35
7	BDC	0.2785	34
8	SQM	0.2703	33
9	ELC	0.2621	32
10	PJI	0.2621	32
11	ZVX	0.2621	32
12	RBE	0.2539	31
13	BEL	0.2457	30
14	EIM	0.2457	30
15	KOA	0.2457	30
16	NBD	0.2457	30
17	PKR	0.2457	30
18	VYX	0.2457	30
19	STE	0.2375	29
20	VSI	0.2375	29
21	XPK	0.2375	29
22	ZVS	0.2375	29
23	IMN	0.2293	28
24	KRG	0.2293	28
25	DVY	0.2211	27
26	KRQ	0.2211	27

- 4-gram:

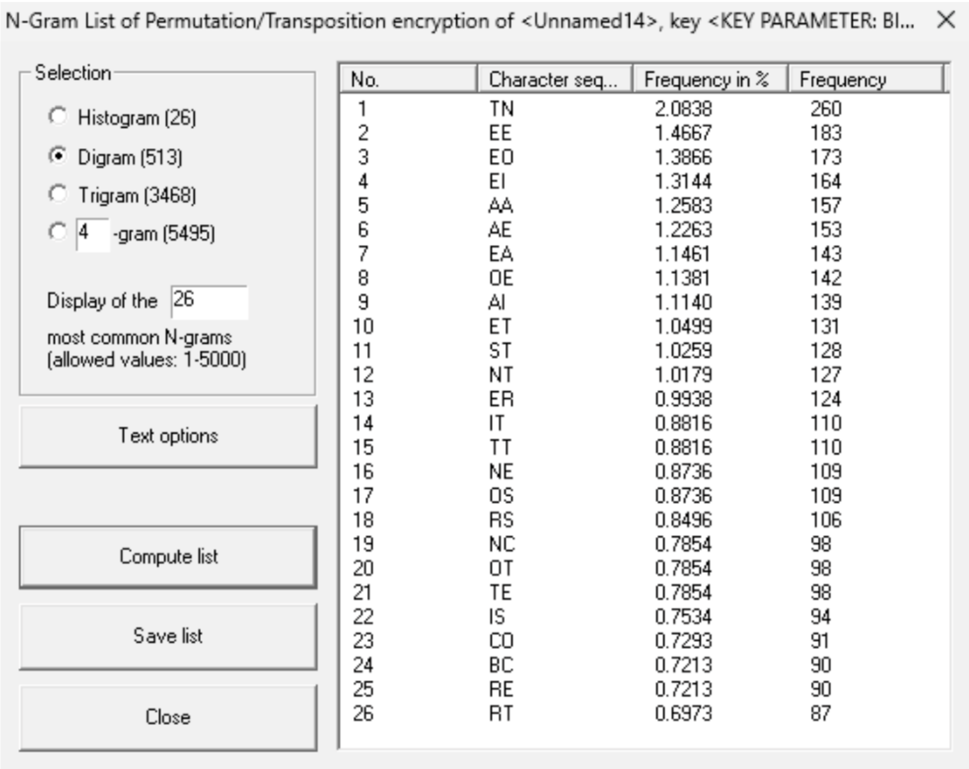


No.	Character seq...	Frequency in %	Frequency
1	ZVSI	0.2990	29
2	BELC	0.2784	27
3	ELCE	0.2784	27
4	KRQK	0.2784	27
5	PKRQ	0.2784	27
6	QKOA	0.2784	27
7	RBEL	0.2784	27
8	RQKO	0.2784	27
9	XPKR	0.2784	27
10	CEIM	0.2681	26
11	EIMN	0.2681	26
12	KOAT	0.2681	26
13	LCEI	0.2681	26
14	SQMM	0.2578	25
15	DVYX	0.2372	23
16	VYXW	0.2372	23
17	WYUG	0.2372	23
18	XWYU	0.2372	23
19	YXWY	0.2372	23
20	YUGH	0.2269	22
21	ZVXO	0.2165	21
22	LPMU	0.2062	20
23	STEA	0.2062	20
24	REQR	0.1959	19
25	TPJI	0.1959	19
26	KRGO	0.1753	17

Permutation: Key: “KEY”

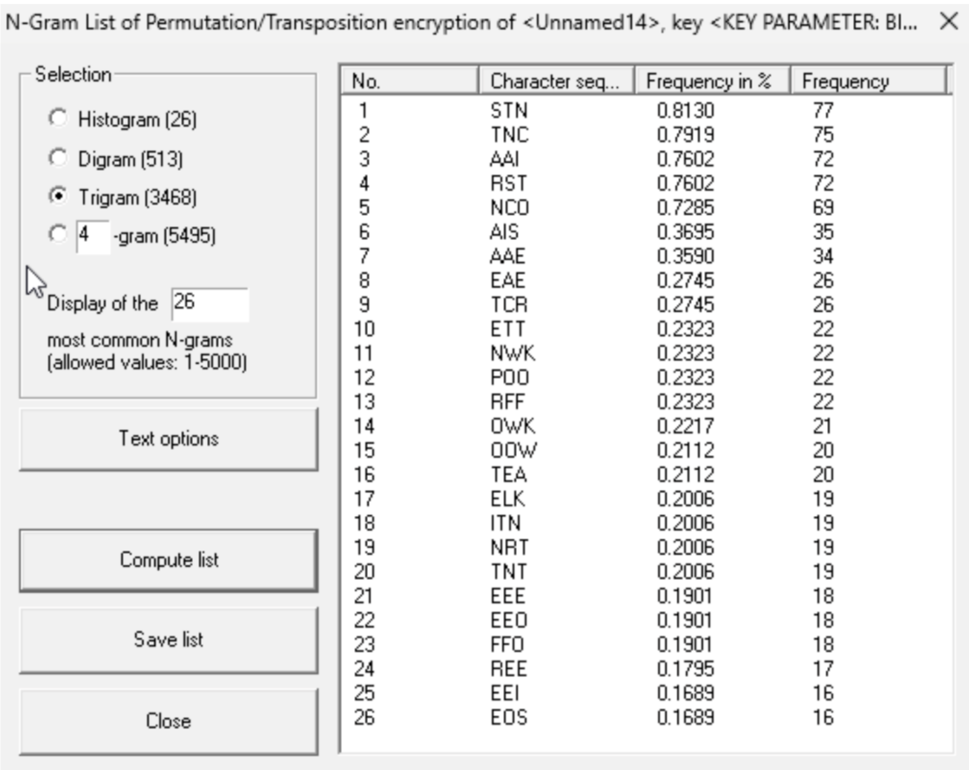
- English:

- Digram:



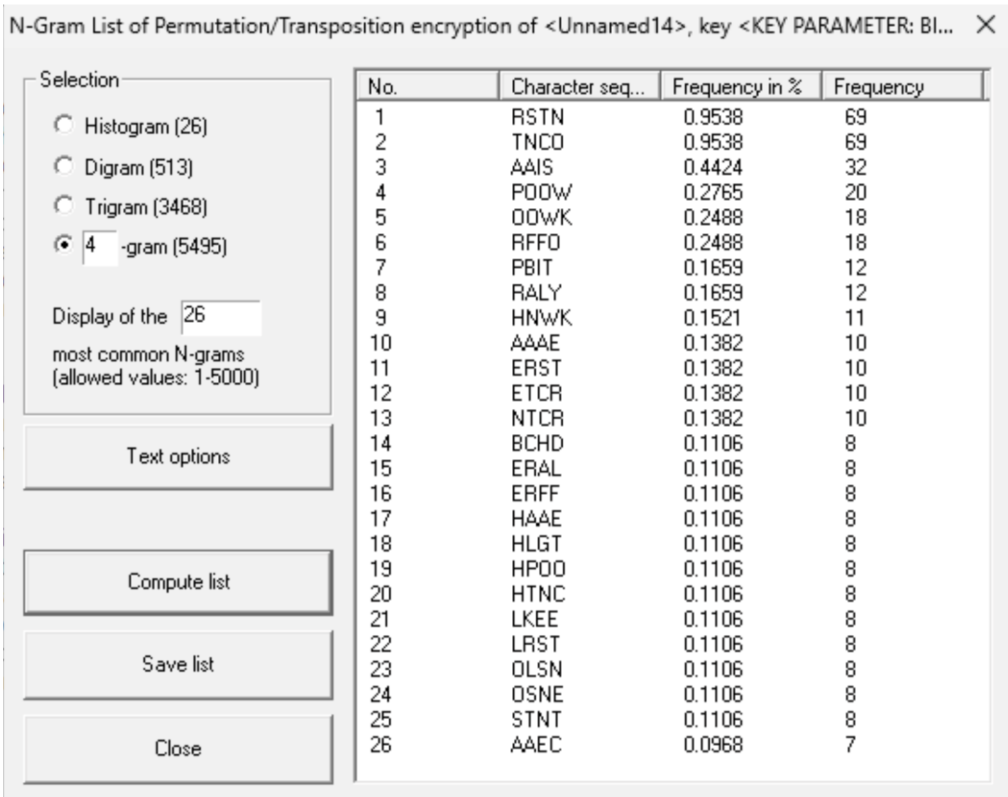
No.	Character seq...	Frequency in %	Frequency
1	TN	2.0838	260
2	EE	1.4667	183
3	EO	1.3866	173
4	EI	1.3144	164
5	AA	1.2583	157
6	AE	1.2263	153
7	EA	1.1461	143
8	OE	1.1381	142
9	AI	1.1140	139
10	ET	1.0499	131
11	ST	1.0259	128
12	NT	1.0179	127
13	ER	0.9938	124
14	IT	0.8816	110
15	TT	0.8816	110
16	NE	0.8736	109
17	OS	0.8736	109
18	RS	0.8496	106
19	NC	0.7854	98
20	OT	0.7854	98
21	TE	0.7854	98
22	IS	0.7534	94
23	CO	0.7293	91
24	BC	0.7213	90
25	RE	0.7213	90
26	RT	0.6973	87

- Trigram:



No.	Character seq...	Frequency in %	Frequency
1	STN	0.8130	77
2	TNC	0.7919	75
3	AAI	0.7602	72
4	RST	0.7602	72
5	NCO	0.7285	69
6	AIS	0.3695	35
7	AAE	0.3590	34
8	EAE	0.2745	26
9	TCR	0.2745	26
10	ETT	0.2323	22
11	NWK	0.2323	22
12	POO	0.2323	22
13	RFF	0.2323	22
14	OWK	0.2217	21
15	OOW	0.2112	20
16	TEA	0.2112	20
17	ELK	0.2006	19
18	ITN	0.2006	19
19	NRT	0.2006	19
20	TNT	0.2006	19
21	EEE	0.1901	18
22	EEO	0.1901	18
23	FFO	0.1901	18
24	REE	0.1795	17
25	EEI	0.1689	16
26	EOS	0.1689	16

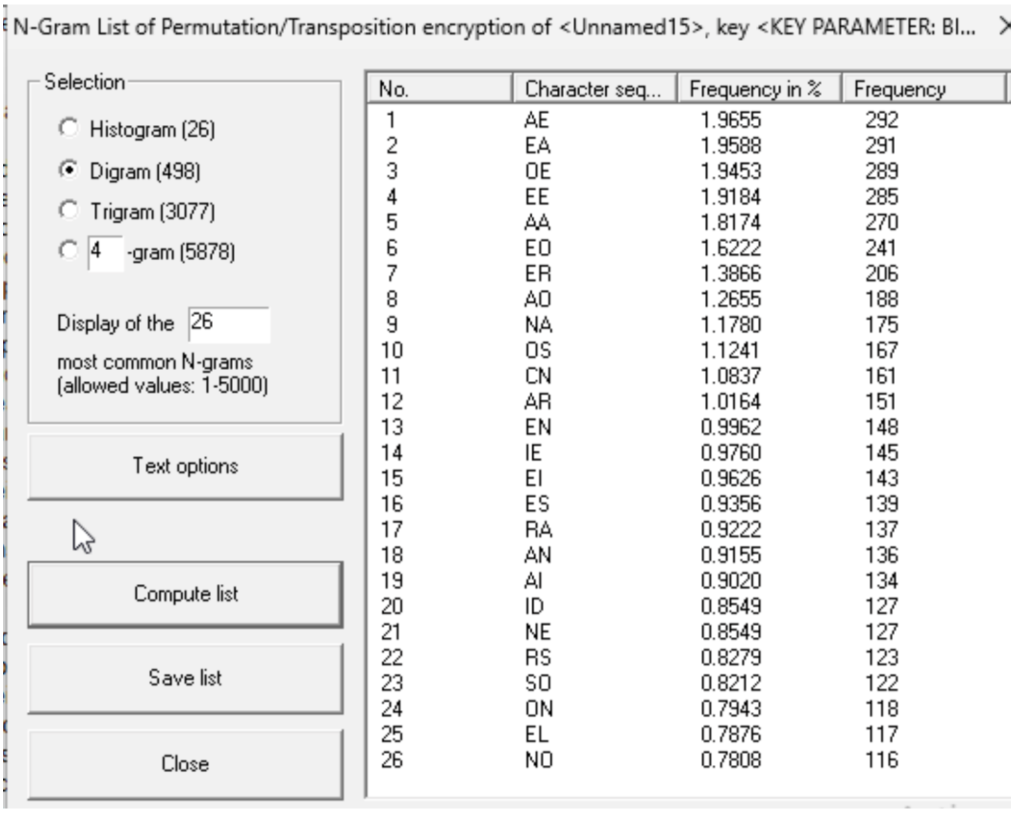
- 4-gram:



No.	Character seq...	Frequency in %	Frequency
1	RSTN	0.9538	69
2	TNCO	0.9538	69
3	AAIS	0.4424	32
4	POOW	0.2765	20
5	OOWK	0.2488	18
6	RFFO	0.2488	18
7	PBIT	0.1659	12
8	RALY	0.1659	12
9	HNWK	0.1521	11
10	AAAE	0.1382	10
11	ERST	0.1382	10
12	ETCR	0.1382	10
13	NTCR	0.1382	10
14	BCHD	0.1106	8
15	ERAL	0.1106	8
16	ERFF	0.1106	8
17	HAAE	0.1106	8
18	HLGT	0.1106	8
19	HPOO	0.1106	8
20	HTNC	0.1106	8
21	LKEE	0.1106	8
22	LRST	0.1106	8
23	OLSN	0.1106	8
24	OSNE	0.1106	8
25	STNT	0.1106	8
26	AAEC	0.0968	7

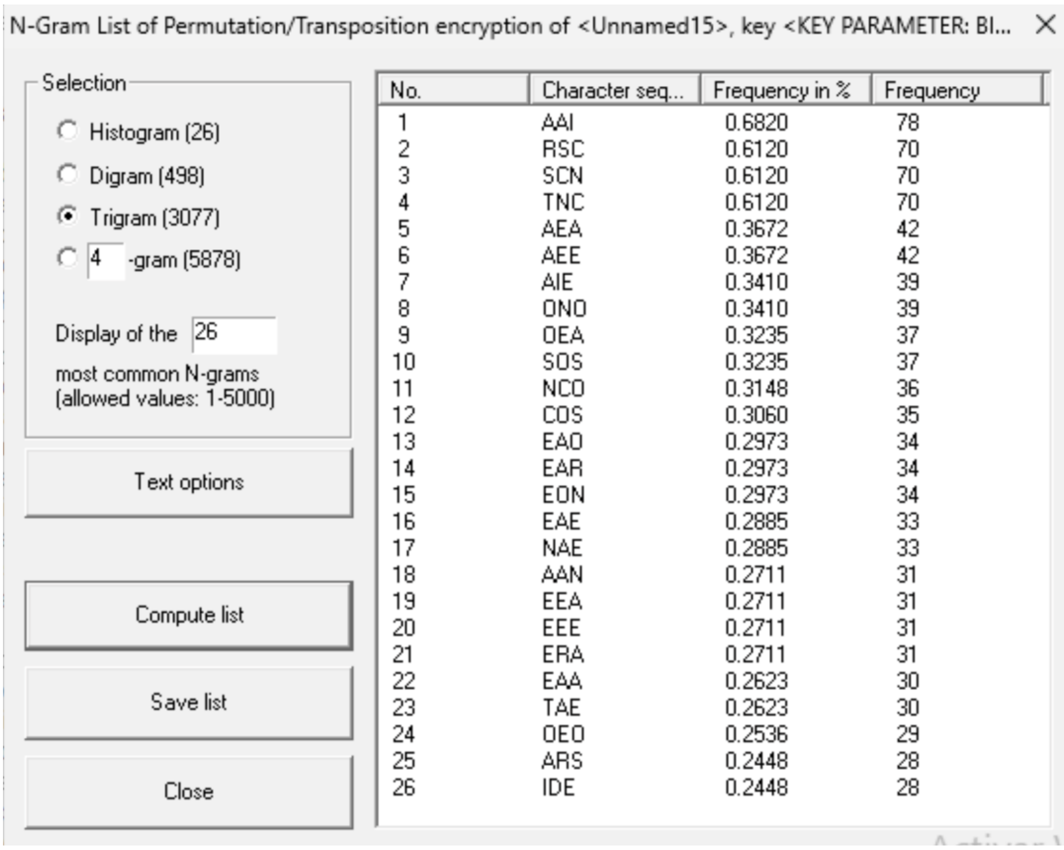
- Spanish:

- Digram:



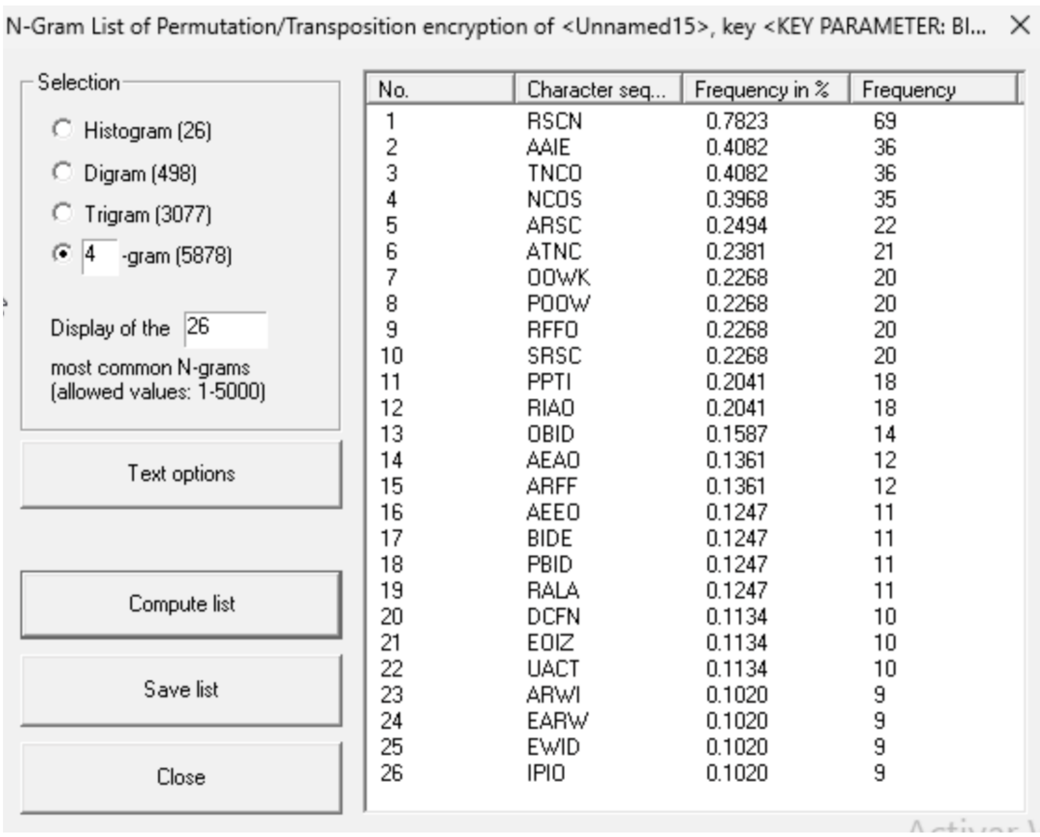
No.	Character seq...	Frequency in %	Frequency
1	AE	1.9655	292
2	EA	1.9588	291
3	OE	1.9453	289
4	EE	1.9184	285
5	AA	1.8174	270
6	EO	1.6222	241
7	ER	1.3866	206
8	AO	1.2655	188
9	NA	1.1780	175
10	OS	1.1241	167
11	CN	1.0837	161
12	AR	1.0164	151
13	EN	0.9962	148
14	IE	0.9760	145
15	EI	0.9626	143
16	ES	0.9356	139
17	RA	0.9222	137
18	AN	0.9155	136
19	AI	0.9020	134
20	ID	0.8549	127
21	NE	0.8549	127
22	RS	0.8279	123
23	SO	0.8212	122
24	ON	0.7943	118
25	EL	0.7876	117
26	NO	0.7808	116

- Trigram:



No.	Character seq...	Frequency in %	Frequency
1	AAI	0.6820	78
2	RSC	0.6120	70
3	SCN	0.6120	70
4	TNC	0.6120	70
5	AEA	0.3672	42
6	AEE	0.3672	42
7	AIE	0.3410	39
8	ONO	0.3410	39
9	OEA	0.3235	37
10	SOS	0.3235	37
11	NCO	0.3148	36
12	COS	0.3060	35
13	EAO	0.2973	34
14	EAR	0.2973	34
15	EON	0.2973	34
16	EAE	0.2885	33
17	NAE	0.2885	33
18	AAN	0.2711	31
19	EEA	0.2711	31
20	EEE	0.2711	31
21	ERA	0.2711	31
22	EAA	0.2623	30
23	TAE	0.2623	30
24	OEO	0.2536	29
25	ARS	0.2448	28
26	IDE	0.2448	28

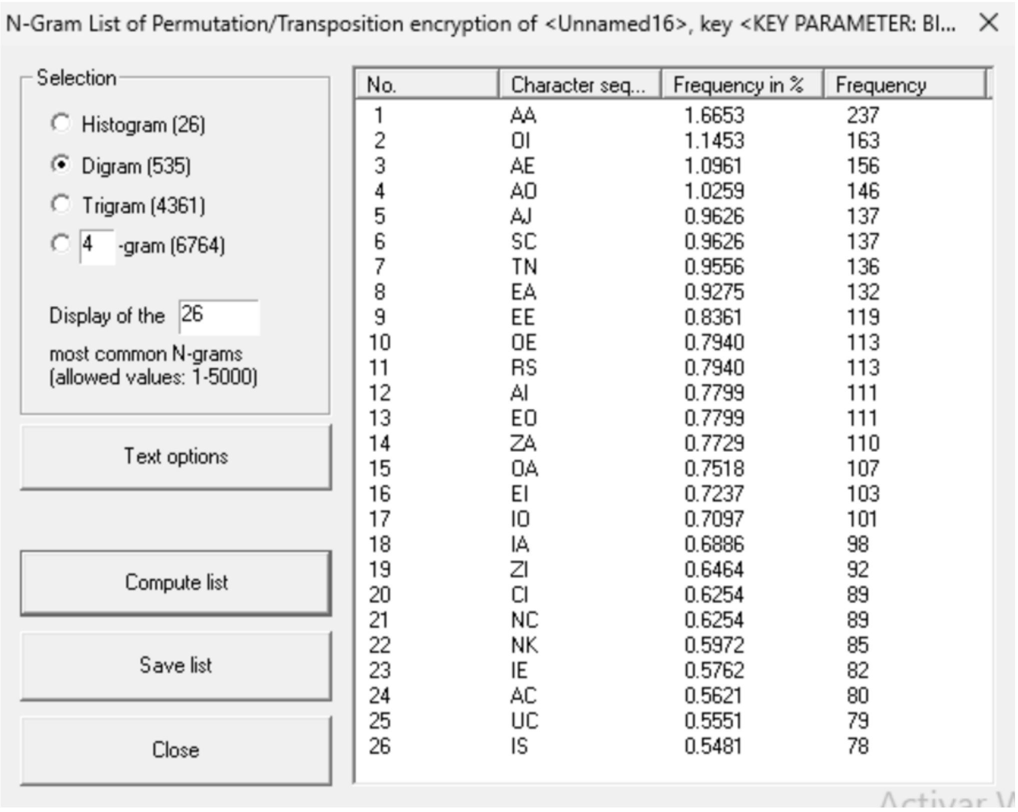
- 4-gram:



No.	Character seq...	Frequency in %	Frequency
1	RSCN	0.7823	69
2	AAIE	0.4082	36
3	TNCO	0.4082	36
4	NCOS	0.3968	35
5	ARSC	0.2494	22
6	ATNC	0.2381	21
7	OOWK	0.2268	20
8	POOW	0.2268	20
9	RFFO	0.2268	20
10	SRSC	0.2268	20
11	PPTI	0.2041	18
12	RIAO	0.2041	18
13	OBID	0.1587	14
14	AEAO	0.1361	12
15	ARFF	0.1361	12
16	AEEO	0.1247	11
17	BIDE	0.1247	11
18	PBID	0.1247	11
19	RALA	0.1247	11
20	DCFN	0.1134	10
21	EOIZ	0.1134	10
22	UACT	0.1134	10
23	ARWI	0.1020	9
24	EARW	0.1020	9
25	EWID	0.1020	9
26	IPIO	0.1020	9

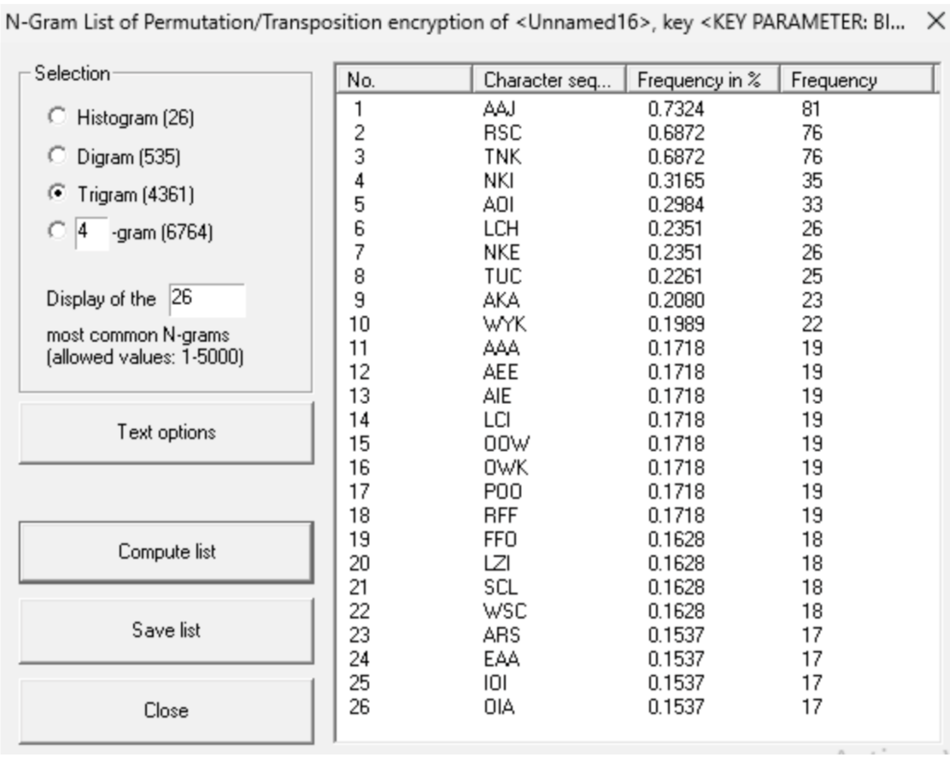
- Polish:

- Digram:



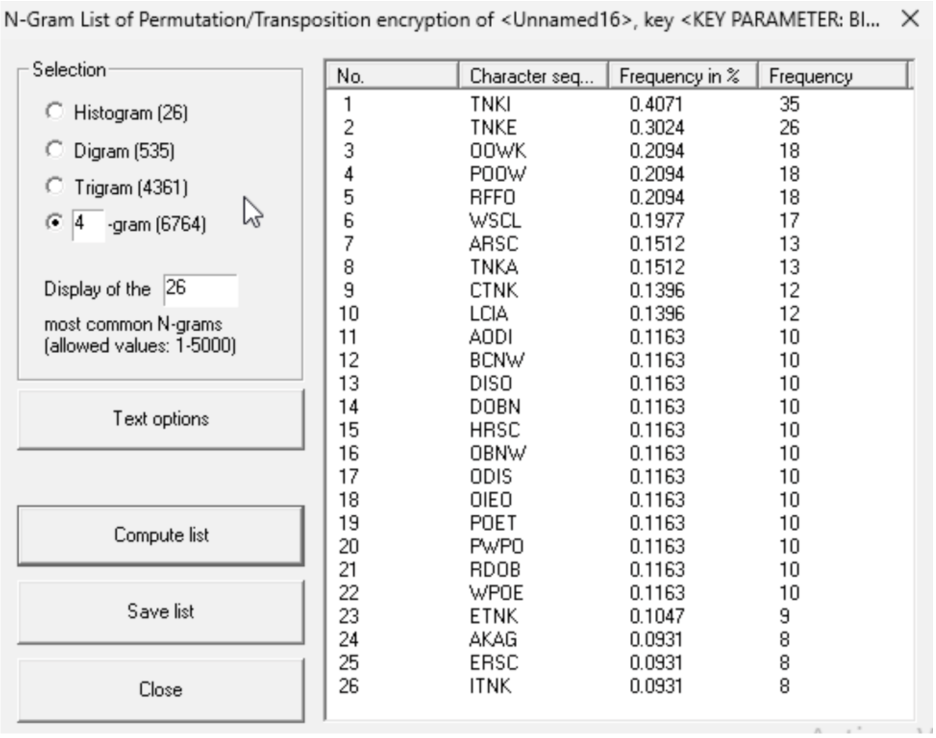
No.	Character seq...	Frequency in %	Frequency
1	AA	1.6653	237
2	OI	1.1453	163
3	AE	1.0961	156
4	AO	1.0259	146
5	AJ	0.9626	137
6	SC	0.9626	137
7	TN	0.9556	136
8	EA	0.9275	132
9	EE	0.8361	119
10	OE	0.7940	113
11	RS	0.7940	113
12	AI	0.7799	111
13	EO	0.7799	111
14	ZA	0.7729	110
15	OA	0.7518	107
16	EI	0.7237	103
17	IO	0.7097	101
18	IA	0.6886	98
19	ZI	0.6464	92
20	CI	0.6254	89
21	NC	0.6254	89
22	NK	0.5972	85
23	IE	0.5762	82
24	AC	0.5621	80
25	UC	0.5551	79
26	IS	0.5481	78

- Trigram:



No.	Character seq...	Frequency in %	Frequency
1	AAJ	0.7324	81
2	RSC	0.6872	76
3	TNK	0.6872	76
4	NKI	0.3165	35
5	AOI	0.2984	33
6	LCH	0.2351	26
7	NKE	0.2351	26
8	TUC	0.2261	25
9	AKA	0.2080	23
10	WYK	0.1989	22
11	AAA	0.1718	19
12	AEE	0.1718	19
13	AIE	0.1718	19
14	LCI	0.1718	19
15	OOW	0.1718	19
16	OWK	0.1718	19
17	POO	0.1718	19
18	RFF	0.1718	19
19	FFO	0.1628	18
20	LZI	0.1628	18
21	SCL	0.1628	18
22	WSC	0.1628	18
23	ARS	0.1537	17
24	EAA	0.1537	17
25	IOI	0.1537	17
26	OIA	0.1537	17

- 4-gram:



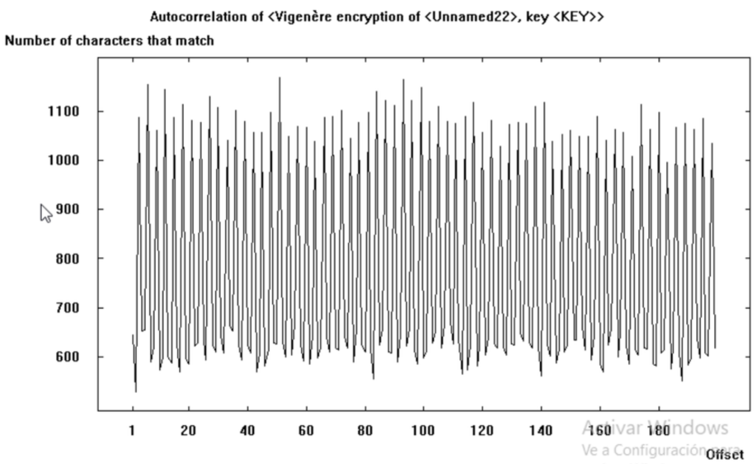
No.	Character seq...	Frequency in %	Frequency
1	TNKI	0.4071	35
2	TNKE	0.3024	26
3	OOWK	0.2094	18
4	POOW	0.2094	18
5	RFFO	0.2094	18
6	WSCL	0.1977	17
7	ARSC	0.1512	13
8	TNKA	0.1512	13
9	CTNK	0.1396	12
10	LCIA	0.1396	12
11	AODI	0.1163	10
12	BCNW	0.1163	10
13	DISO	0.1163	10
14	DOBN	0.1163	10
15	HRSC	0.1163	10
16	OBNW	0.1163	10
17	ODIS	0.1163	10
18	OIEO	0.1163	10
19	POET	0.1163	10
20	PWPO	0.1163	10
21	RDOB	0.1163	10
22	WPOE	0.1163	10
23	ETNK	0.1047	9
24	AKAG	0.0931	8
25	ERSC	0.0931	8
26	ITNK	0.0931	8

In the Caesar cipher, bigram and trigram distributions remained almost identical to those of the plaintext as it is a mono alphabetic substitution and the frequencies remain the same.
For Vigenère encryption, as larger is the Key, the frequencies will have a smaller difference between the.
The permutation cipher reduced the frequencies, as its transposition destroyed the positional correlations.

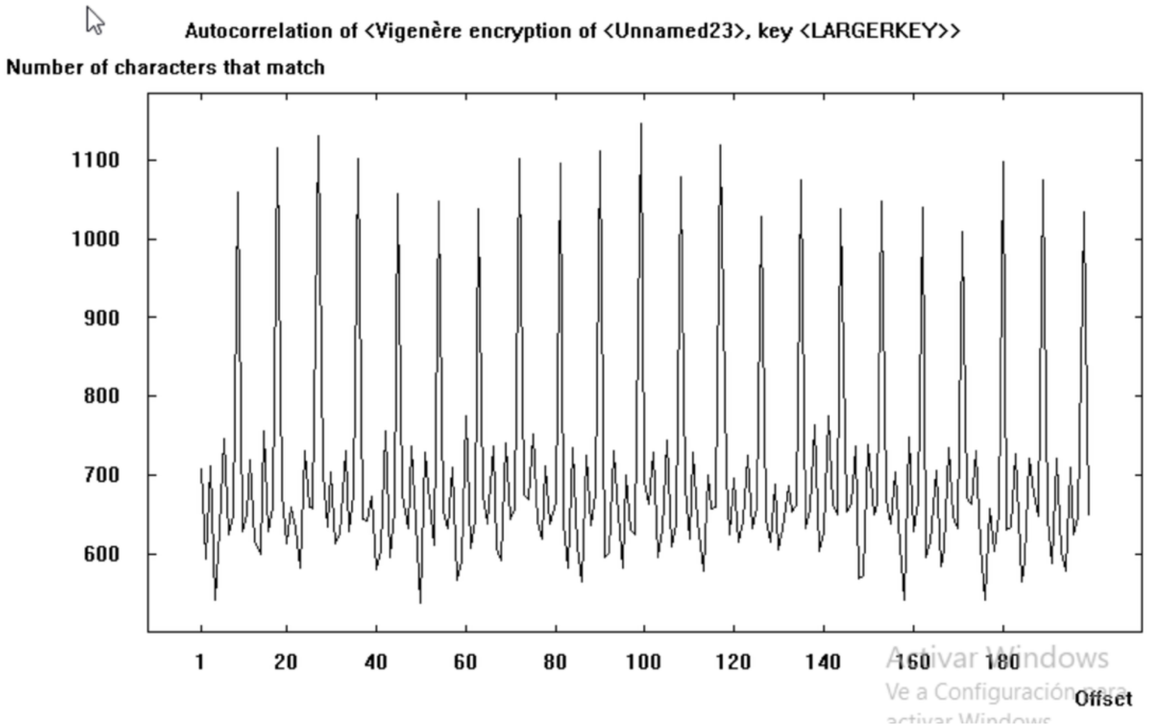
2.8 Analyze the autocorrelation values of the ciphertext obtained by encryption XOR and Vigenère algorithms for different password lengths:

I am using the English Plain text.

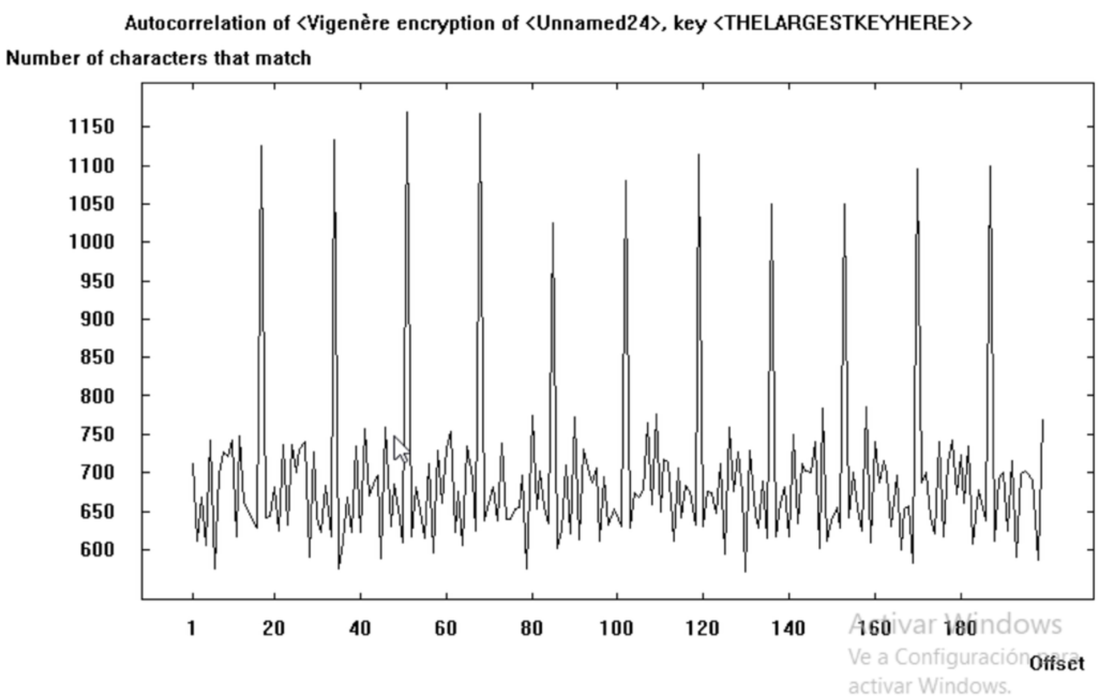
- Vigenère: Key:"KEY"



- Vigenère: Key:"LARGERKEY"

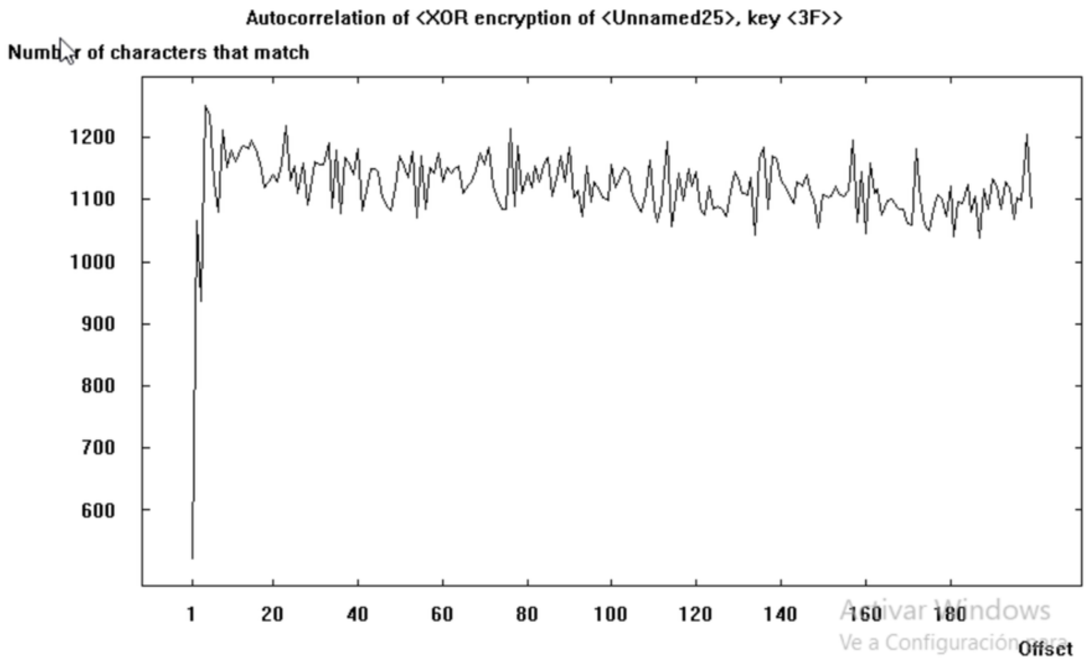


- Vigenère: Key: "THELARGESTKEYHERE"

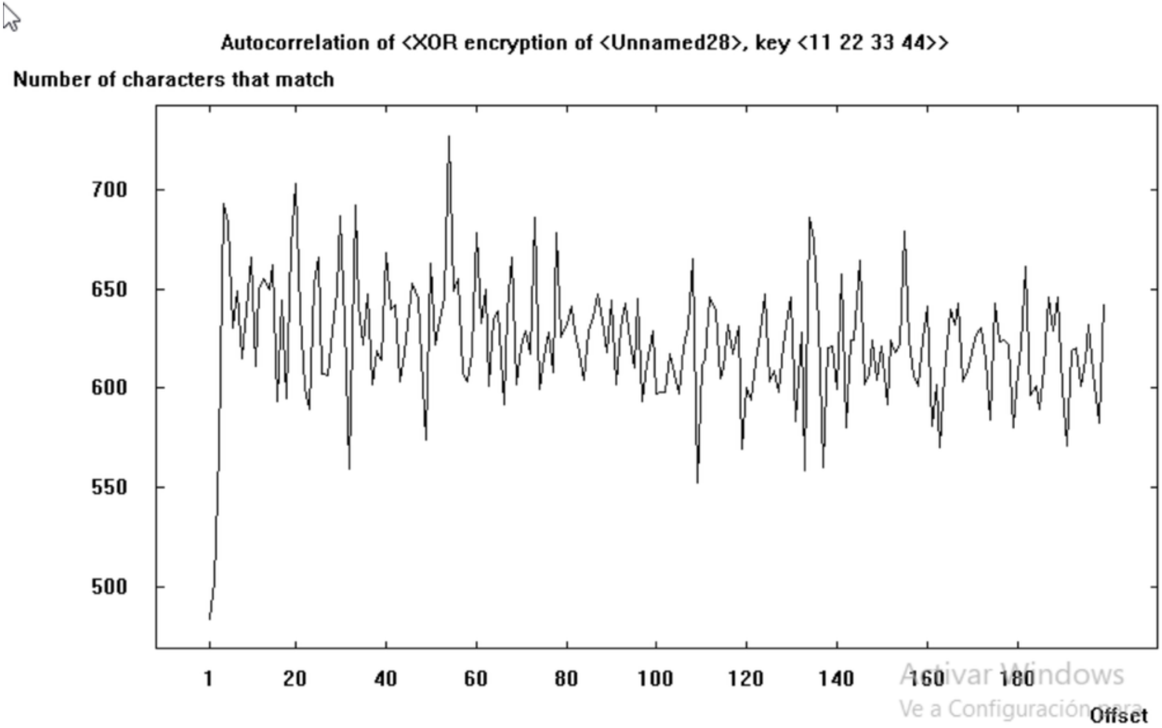


You can see for the Vigenère cypher a pattern between the length of the key and the separation between spikes. For short keys the spikes have a higher frequency and for larger ones a lower frequency of spikes.

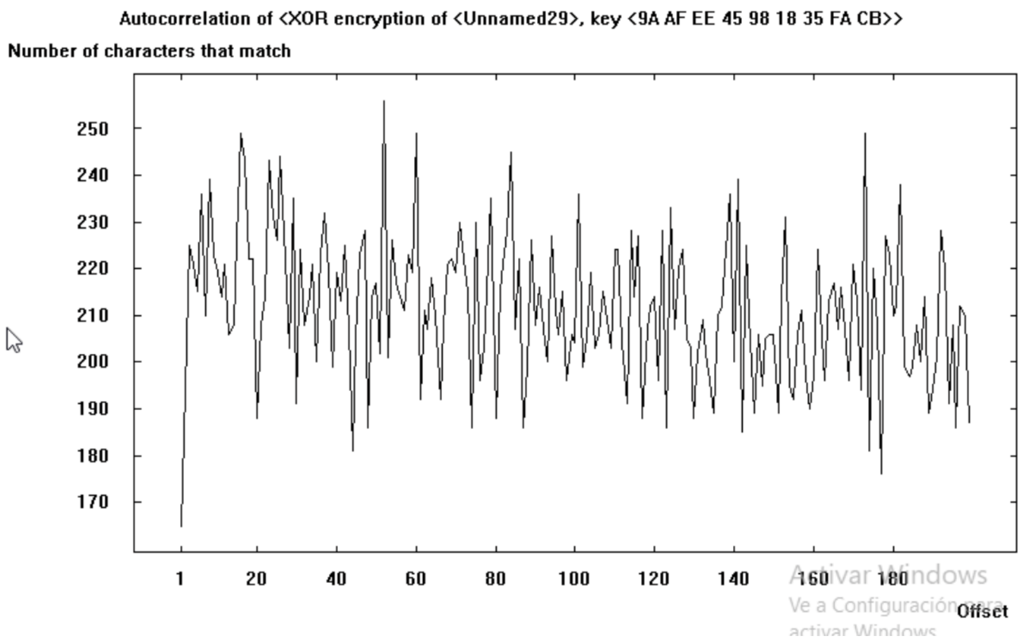
- XOR: Key: 3F



- XOR: Key: 11223344



- XOR: Key: 9AAFEE45981835FACB



With the XOR algorithm we can see a negative correlation between the length of the Key and the number of characters that match in the autocorrection.

2.9 How do the observed parameters change?

As the encryption key length increases or as the cipher becomes more complex, all statistical parameters progressively approach those of a random sequence.

The entropy rises, meaning that the information content and unpredictability of the ciphertext increase. Histograms become flatter, showing that all symbols appear with nearly equal probability. N-gram distributions lose the characteristic patterns of natural language; common letter pairs or triplets vanish, and no linguistic structure remains. Finally, the autocorrelation curve becomes flat, indicating the absence of periodicity or repetition.

In general, higher key lengths and stronger algorithms destroy the statistical regularities of the plaintext, producing ciphertexts that are increasingly random and therefore more resistant to classical cryptanalysis.

2.10 How can the text analysis tools available in CrypTool be used to determine the encryption algorithm for a given ciphertext?

Each tool in Cryptool provides us with different information, such as:

- Histogram: Shows letter frequencies. A shifted histogram suggests a Caesar/shift or simple monoalphabetic substitution.

- N-grams (bigrams/trigrams): Frequent sequences point to common words/particles in the language and help recover substitution mappings or cribs.
- Autocorrelation: Periodic peaks reveal repeating-key ciphers (Vigenère, repeating XOR); the peak spacing estimates key length.
- Index of Coincidence / Kasiski / Friedman tests: Distinguish monoalphabetic vs polyalphabetic ciphers and give key-length estimates.
- Ciphertext-only solvers: Once you classify the family, run automated solvers (Caesar, Vigenère, substitution, columnar, ADFGVX, XOR, etc.) to recover keys/plaintext.

2.11 How can the text analysis tools available in the CrypTool program be used to determine the password used for encryption?

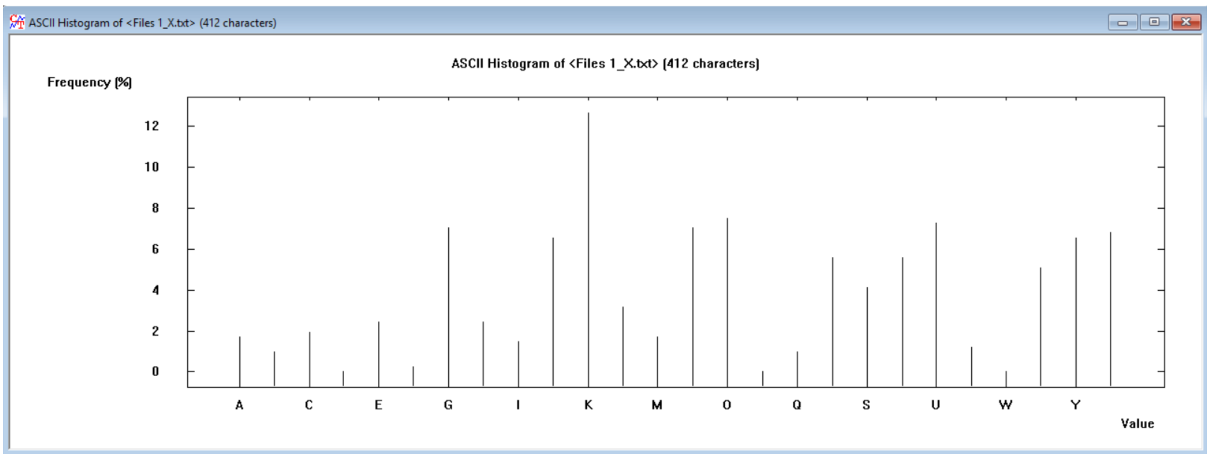
The general workflow has 4 steps:

1. Classify the cipher using Histogram, IC, Autocorrelation, N-grams.
2. Estimate key length / structure (Friedman, Kasiski, autocorrelation).
3. Run targeted solver for that family (CrypTool has ciphertext-only modules).
4. Validate candidate keys by decrypting and scoring the plaintext.

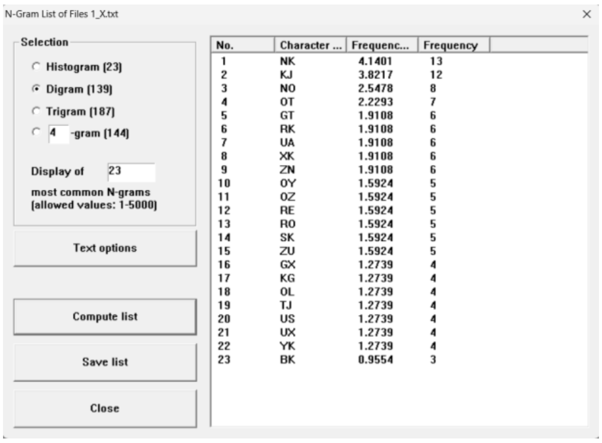
3 Analysis of delivered files:

3.1 Try to recognize which encryption algorithm has been used based on an analysis of the ciphertext program only. All cryptograms were created based on the same plaintext. Files 1 X.txt,..,:

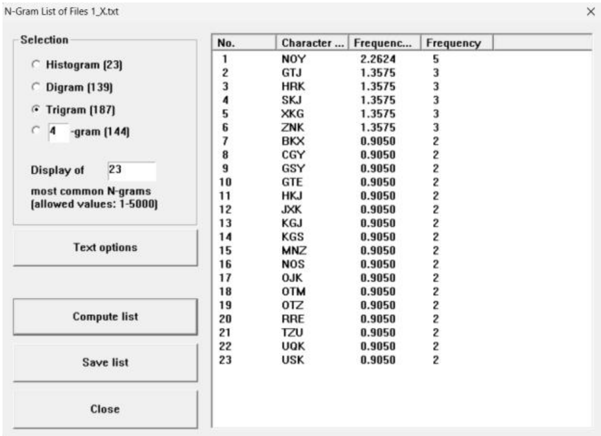
The file 1 can be classified as a monoalphabetic substitution as the histogram shows a normal frequency in English language:



We can see also high frequencies in the diagram and trigram, meaning that the letters are substituted but not permutated:

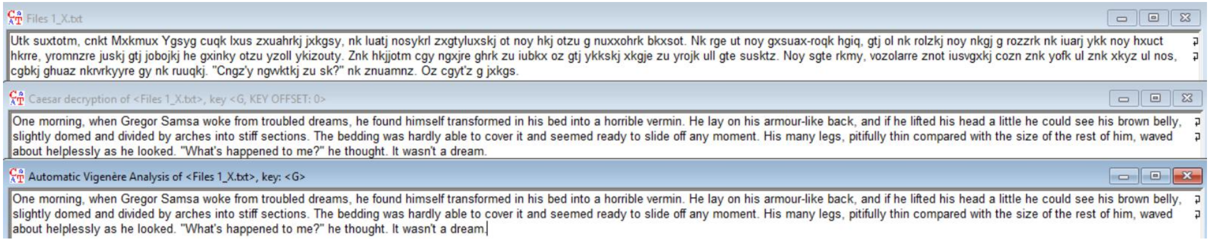


No.	Character ...	Frequenc...	Frequency
1	NK	4.1401	13
2	KJ	3.8217	12
3	NO	2.5478	8
4	OT	2.2293	7
5	GT	1.9108	6
6	RK	1.9108	6
7	UA	1.9108	6
8	XK	1.9108	6
9	ZN	1.9108	6
10	OY	1.5924	5
11	OZ	1.5924	5
12	RE	1.5924	5
13	RO	1.5924	5
14	SK	1.5924	5
15	ZU	1.5924	5
16	GX	1.2739	4
17	KG	1.2739	4
18	OL	1.2739	4
19	TJ	1.2739	4
20	US	1.2739	4
21	UX	1.2739	4
22	YK	1.2739	4
23	BK	0.9554	3



No.	Character ...	Frequenc...	Frequency
1	NOY	2.2624	5
2	GTJ	1.3575	3
3	HRK	1.3575	3
4	SKJ	1.3575	3
5	XKG	1.3575	3
6	ZNK	1.3575	3
7	BKX	0.9050	2
8	CGY	0.9050	2
9	GSY	0.9050	2
10	GTE	0.9050	2
11	HKJ	0.9050	2
12	JXK	0.9050	2
13	KGJ	0.9050	2
14	KGS	0.9050	2
15	MNZ	0.9050	2
16	NOS	0.9050	2
17	OJK	0.9050	2
18	OTM	0.9050	2
19	OTZ	0.9050	2
20	RRE	0.9050	2
21	TZU	0.9050	2
22	UQK	0.9050	2
23	USK	0.9050	2

So, we try with deciphering it with Cesar/Vigenère (Key length 1) and the key is “G”:

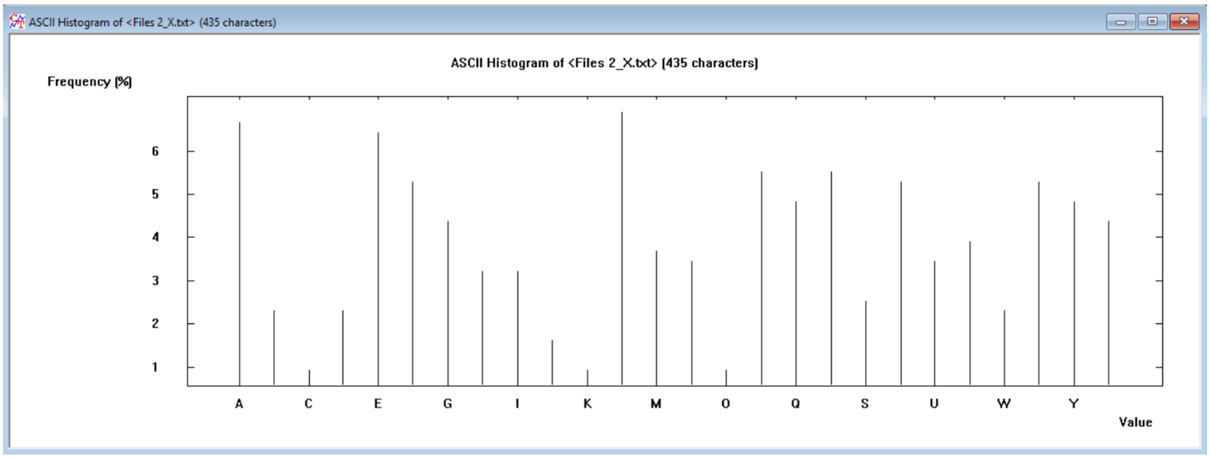


The decrypted text is: “One morning, when Gregor Samsa woke from troubled dreams, he found himself transformed in his bed into a horrible vermin. He lay on his armour-like back, and if he lifted his head a little he could see his brown belly, slightly domed and divided by arches into stiff sections. The bedding was hardly able to cover it and seemed ready to slide off any moment. His many legs, pitifully thin compared with the size of the rest of him, waved about helplessly as he looked. "What's happened to me?" he thought. It wasn't a dream.”

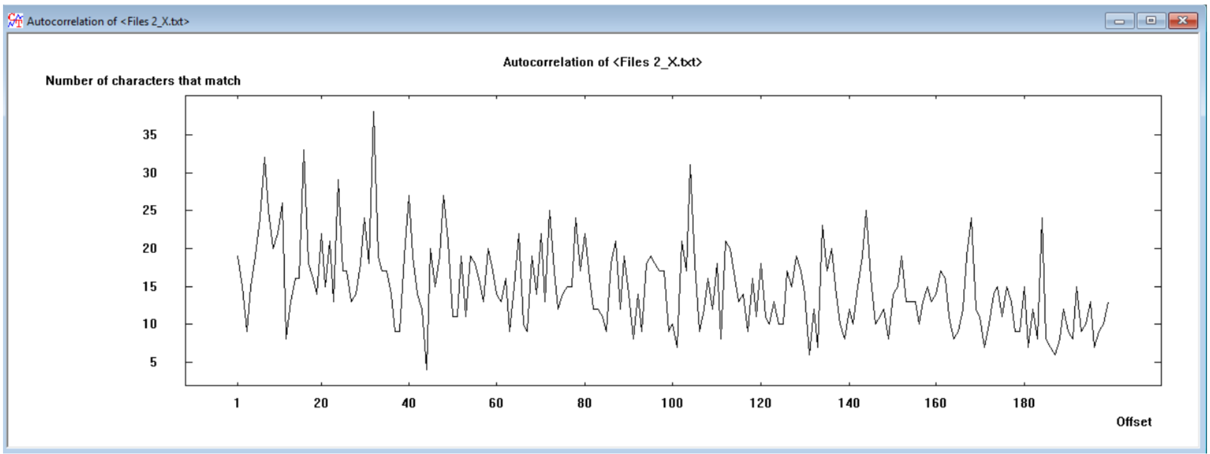
3.2 Try to decrypt by hand (using premises such as type algorithm, histogram, and other tools) or by using the automatic cipher-breaking option (Analysis - > Symmetric Encryption (Classic) -> Ciphertext-only) encryption programs placed in files 2_x.txt, ...:

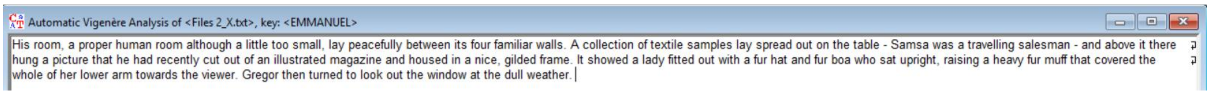
The automatic Vigenère analysis showed that the key length was 7, if you increased by 1 as there is an small error, you could see that finally the text is decrypted with the key EMMANUEL.

The histogram more randomness than in file 1:



In the autocorrelation we don't see a pattern discarding some algorithms:




Automatic Vigenère Analysis of <Files 2_X.txt>, key: <EMMANUEL>

His room, a proper human room although a little too small, lay peacefully between its four familiar walls. A collection of textile samples lay spread out on the table - Samsa was a travelling salesman - and above it there hung a picture that he had recently cut out of an illustrated magazine and housed in a nice, gilded frame. It showed a lady fitted out with a fur hat and fur boa who sat upright, raising a heavy fur muff that covered the whole of her lower arm towards the viewer. Gregor then turned to look out the window at the dull weather.

The decrypted text is:

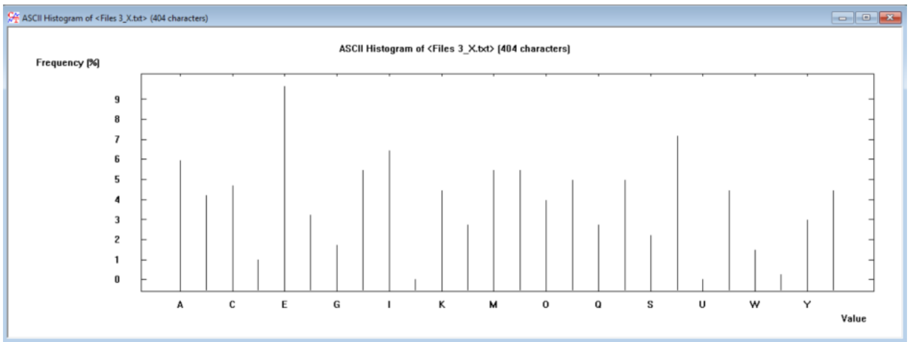
"His room, a proper human room although a little too small, lay peacefully between its four familiar walls. A collection of textile samples lay spread out on the table - Samsa was a travelling salesman - and above it there hung a picture that he had recently cut out of an illustrated magazine and housed in a nice, gilded frame. It showed a lady fitted out with a fur hat and fur boa who sat upright, raising a heavy fur muff that covered the whole of her lower arm towards the viewer. Gregor then turned to look out the window at the dull weather."

3.3 Recognize what encryption algorithm was used, and then try to decrypt the provided ciphertext programs. Files: 3_x.txt, ...:

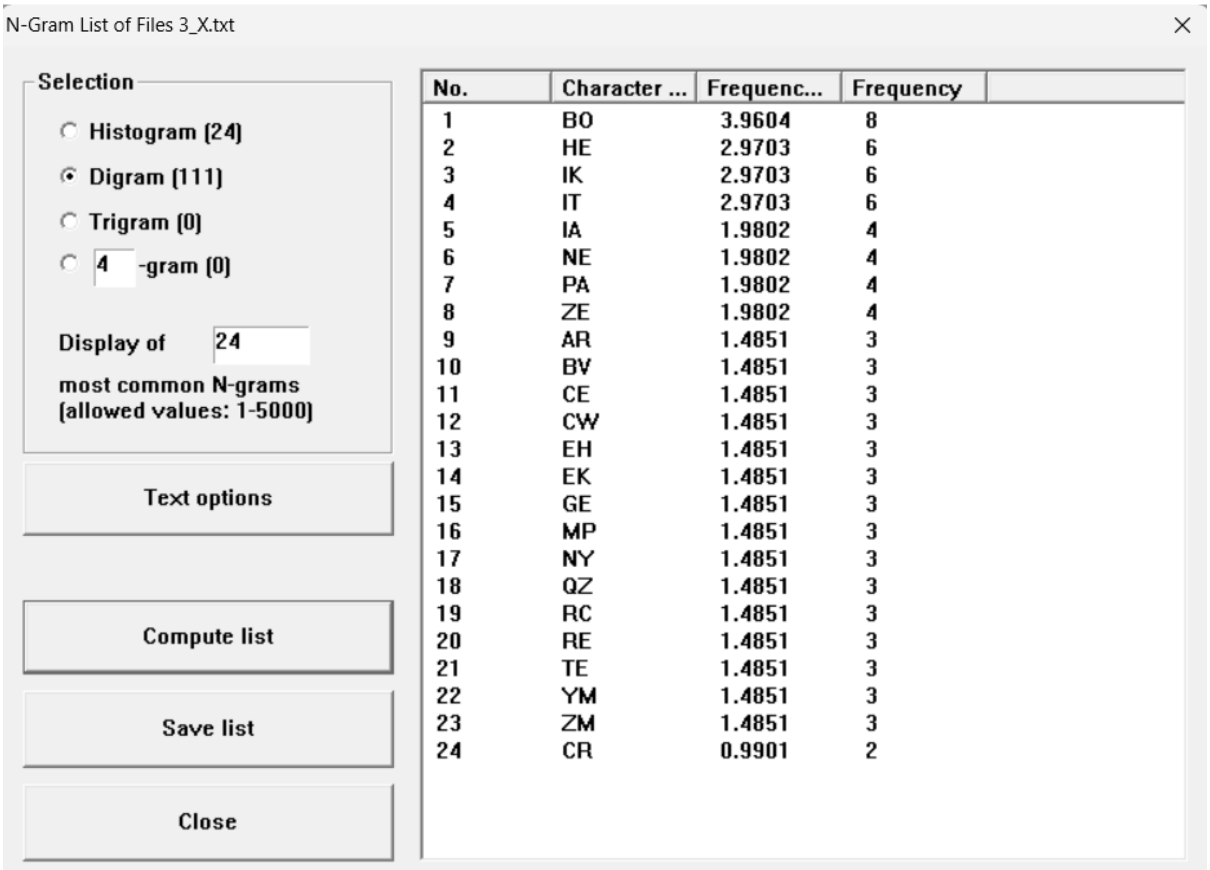
This file 3 is clearly encrypted with playfair as in it histogram there is no appearance of the letter J also the division of the text in 2 character blocks, meaning it was encrypted with 5x5 matrix.

It must be a text from: Franz Kafka's The Metamorphosis (Die Verwandlung), originally written in German (first published 1915). As the rest of the texts.

Histogram:



Digram:



No.	Character ...	Frequenc...	Frequency
1	BO	3.9604	8
2	HE	2.9703	6
3	IK	2.9703	6
4	IT	2.9703	6
5	IA	1.9802	4
6	NE	1.9802	4
7	PA	1.9802	4
8	ZE	1.9802	4
9	AR	1.4851	3
10	BV	1.4851	3
11	CE	1.4851	3
12	CW	1.4851	3
13	EH	1.4851	3
14	EK	1.4851	3
15	GE	1.4851	3
16	MP	1.4851	3
17	NY	1.4851	3
18	QZ	1.4851	3
19	RC	1.4851	3
20	RE	1.4851	3
21	TE	1.4851	3
22	YM	1.4851	3
23	ZM	1.4851	3
24	CR	0.9901	2

3.4 What does the strength of the algorithm depend on?

The strength of an algorithm depends on:

- **Key secrecy (known plaintext/ciphertext):** even if an **attacker knows the algorithm** and sees example message pairs, they **still can't** realistically **figure out your secret key.**
- **Avalanche (confusion + diffusion):** a tiny change in the key or the message makes the entire encrypted output change wildly and **unpredictably**.

- **All-or-nothing:** a **wrong key doesn't reveal bits of the message**; it just produces useless, random-looking data.
- **(Asymmetric) Key-pair infeasibility:** having one key (e.g., the public key) and the algorithm shouldn't allow anyone to compute the matching private key.

3.5 How can the encryption strength be increased for known ciphers?

Some general rules for increasing strength in a cypher are:

- Use **long, truly random keys and never reuse them**.
- Multi cypher: **combine substitution + transposition** and repeat with different keys.
- **Flatten frequencies**: polyalphabetic keys, homophones, padding.
- For stream/XOR/byte-addition -> derive keystream from a **secure PRNG (Pseudorandom Number Generator) and never reuse it**.
- **OTP is perfect** only if key = message length, truly random, secret, and never reused.
- Increase block/matrix size for Hill, use random key tables for Playfair and treat **Caesar/Atbash as obsolete**.